

EuroSkills Test Project

IT Network Systems Administration (39)

Module B – Microsoft Environment

Submitted by:

Gen Lee EE

José Alves PT

Silvio Papic HR

Valentin Creton FR

INDEX

INTRODUCTION	4
LOGIN	4
SYSTEM CONFIGURATION	4
GENERAL CONFIGURATIONS.....	4
DESCRIPTION OF PROJECT AND TASKS.....	18
INET	18
INET - Task1: DNS server	18
INET - Task2: SMTP server	31
INET - Task3: web server	41
RTR-CPH.....	51
RTR-CPH - Task1: Site-to-site VPN	51
RTR-CPH - Task2: RRAS FW	98
RTR-CPH - Task3: Port forwarding.....	103
RTR-CPH - Task4: Reverse proxy.....	103
RTR-CPH - Task5: NAT (masquerade).....	103
RTR-AAL.....	109
RTR-AAL - Task1: Site-to-site VPN	109
RTR-AAL - Task2: NAT (masquerade).....	110
DC.....	110
Active Directory & Identity Services	110
DC AD - Task1: Domain controller	110
DC AD - Task2: Configure sites	124
DC AD - Task3: Create OU structure	127
DC AD - Task4: Password policies	135
DC AD - Task5: ADFS.....	135
DC AD - Task6: Import users.....	135
Group Policies	135
DC GPO - Task1: Baseline Security Policy for Servers.....	135
DC GPO - Task2: Baseline Security Policy for Desktops.....	136
DC GPO - Task3: Administrative accounts	136
DC GPO - Task4: Users – Restrict Tools	136
DC GPO - Task5: Contractors – Customization	136
DC GPO - Task6: Folder Redirection	137
DC GPO - Task7: Disk Encryption	137
Name Resolution	137
DC DNS - Task1: A & AAAA records	137
DC DNS - Task2: Reverse lookup zones	138
DC DNS - Task3: Authority delegation	138
DC DNS - Task4: DNS forwarding.....	139
Certificate services	139
DC Certs - Task1: PKI and Root CA.....	139

DC Certs - Task2: CDP	163
DC Certs - Task3: AIA	163
DC Certs - Task4: OCSP	163
DC Certs - Task5: CDP and AIA on SRV2	164
DC Certs - Task6: certificate templates	164
Part 6: Enable Autoenrollment via Group Policy	172
Backups	172
DC Backups - Task1: Backup script	172
RODC	173
RODC- Task1: Promote as RODC	173
SRV2	190
SRV2 - Task1: public web server	190
SRV2 - Task2: intranet	190
SRV2 - Task3: SAML based auth app	191
SRV2 - Task4: web server certificates	191
SRV2 - Task5: iSCSI Target	191
SRV2 - Task6: Windows Event Forwarding	191
SRV1 & SRV2	192
SRV1 & SRV2 - Task1: SMB encryption	192
SRV1 & SRV2 - Task2: DFS shares	192
SRV1 & SRV2 - Task3: Bitlocker encryption	192
SRV1 & SRV2 - Task4: Prevent writing	192
SRV1 & SRV2 - Task5: Quota for roaming profiles	193
SRV1 & SRV2 - Task6: DFS replication	193
SRV1 & SRV2 - Task7: HA DHCP	193
SRV1 & SRV2 - Task8: DHCP reservation	193
DEV-PC	193
Ansible - Task1: hostname	193
Ansible - Task2: ADDS	194
Ansible - Task3: Users	194
Ansible - Task4: Web server	194
Ansible - Task5: Shares	194
Appendix A: Configuration table	196
Appendix B: Network topology	197
Appendix C: OU Structure	197
Appendix D: Password Policies	197
Appendix E: INET DNS records	198
Appendix F: DNS Authoritative Servers	198
Appendix G: ADFS successful integration	198

INTRODUCTION

The competition has a fixed start and finish time. You must decide how to best divide your time.

Please **carefully read** the following instructions!

This Test Project consists of the following documentation/files:

1. ES2025_TP39_ModuleB.docx
2. ES2025_TP39_ModuleB_Users_Skillsnet.csv
3. ES2025_TP39_ModuleB_Users_Skillsdev.json
4. ES2025_TP39_ModuleB_Shares.yaml
5. ES2025_TP39_ModuleB_SSO_App.zip

When the competition time ends, please leave your station in a running state. The assessment will be done in the state as it is. No reboot will be initiated as well as powered off machines will not be powered on!

LOGIN

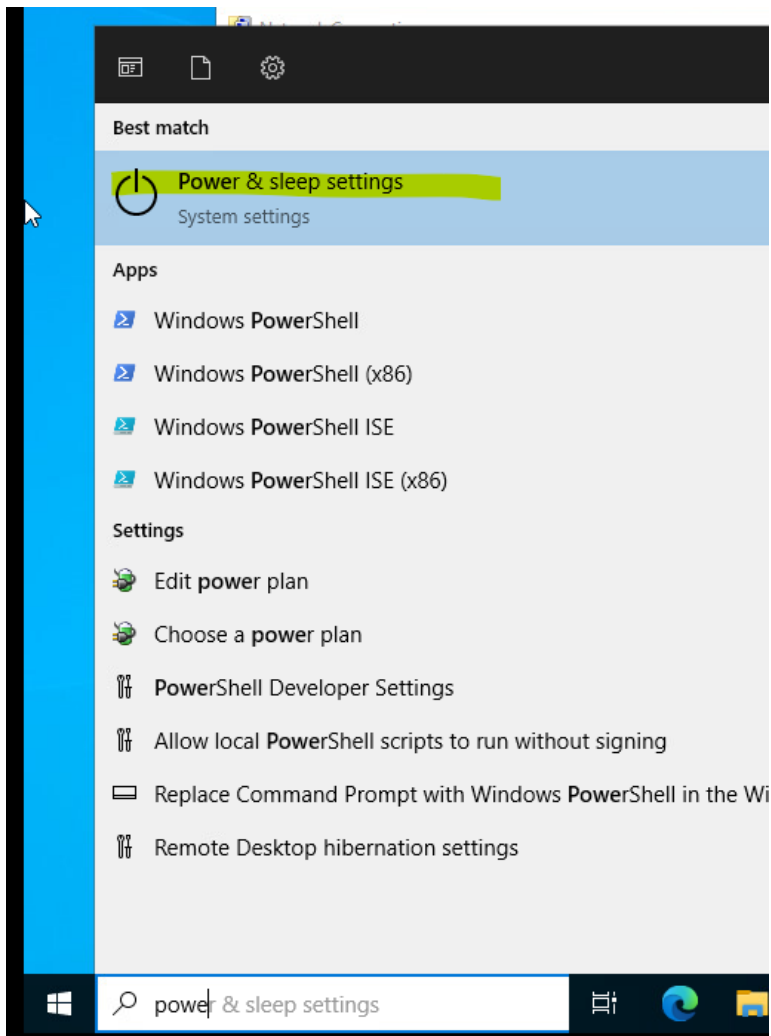
Username: Administrator
Password: Passw0rd!

SYSTEM CONFIGURATION

Language: English US
Time zone: Europe/Copenhagen
Key Map: English US

GENERAL CONFIGURATIONS

AVOID SCREEN LOCKING



Power & sleep

Screen

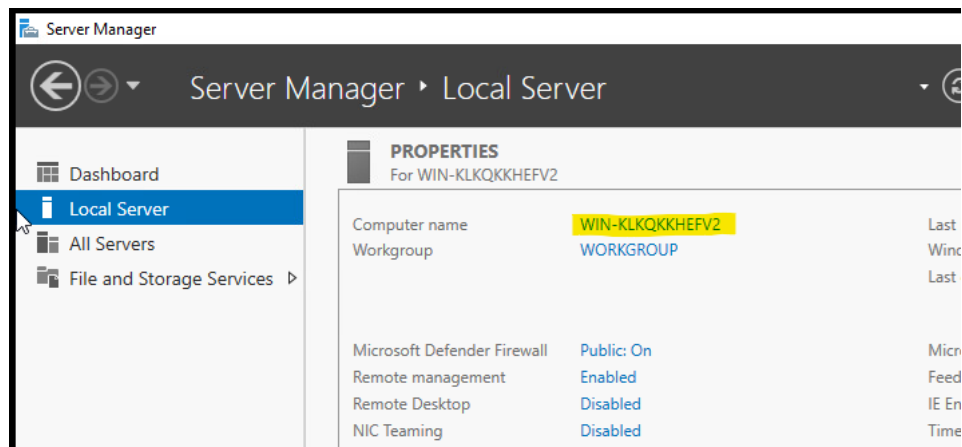
When plugged in, turn off after

Never

Related settings

[Additional power settings](#)

HOSTNAME:



The screenshot shows the Windows Server Manager interface. The left-hand navigation pane is open, displaying a tree view with 'Local Server' selected. The main pane on the right is titled 'PROPERTIES For WIN-KLKQKKHEFV2'. It contains a table of system properties:

Property Name	Value	Category
Computer name	WIN-KLKQKKHEFV2	Last
Workgroup	WORKGROUP	Winc
Microsoft Defender Firewall	Public: On	Micr
Remote management	Enabled	Feed
Remote Desktop	Disabled	IE En
NIC Teaming	Disabled	Time

System Properties

Computer Name Hardware Advanced Remote

Windows uses the following information to identify your computer on the network.

Computer description:

For example: "IIS Production Server" or "Accounting Server".

Full computer name: WIN-KLKQKKHEFV2

Workgroup: WORKGROUP

To rename this computer or change its domain or workgroup, click Change.

Change...

OK Cancel Apply

Computer Name/Domain Changes

You can change the name and the membership of this computer. Changes might affect access to network resources.

Computer name:

Full computer name: RTR-CPH

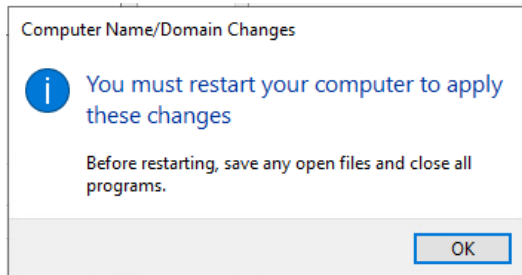
More...

Member of

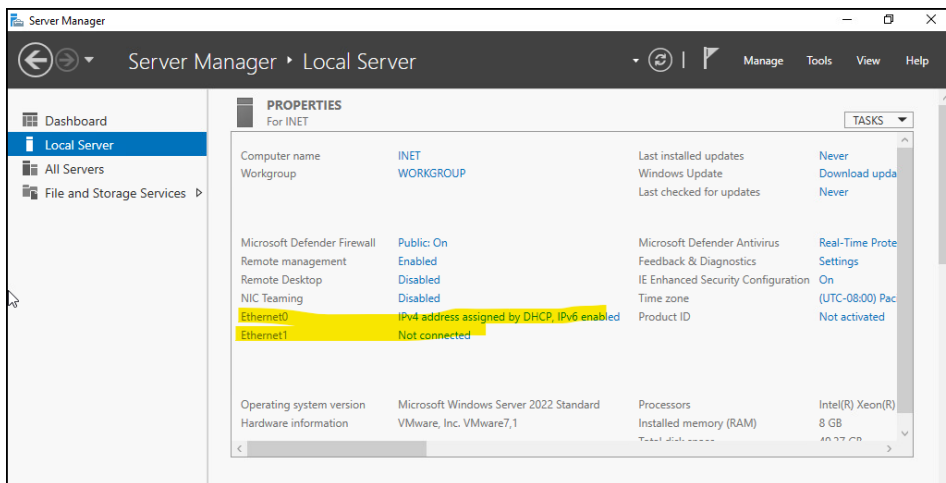
☐ Domain:

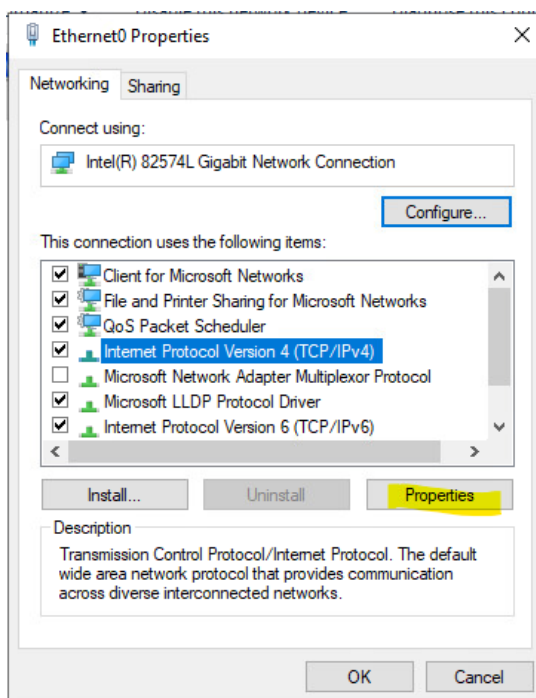
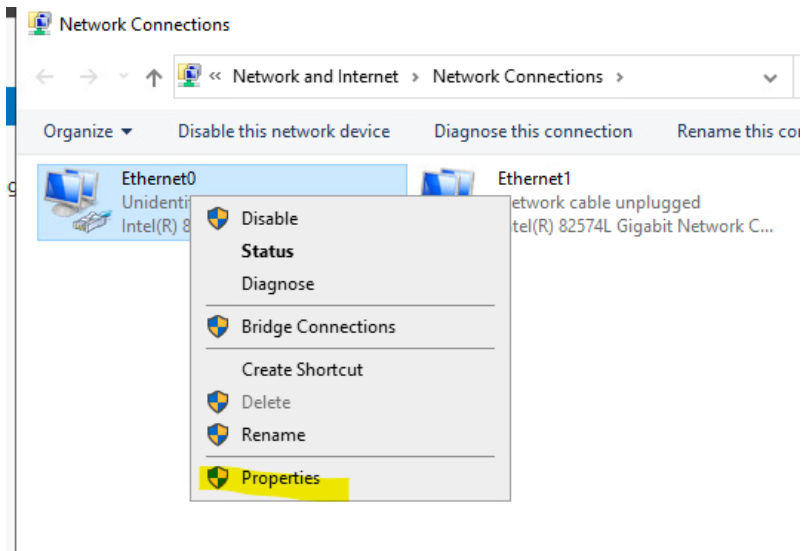
☒ Workgroup:

OK Cancel



NETWORK CONFIGURATION - GUI





Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 198 . 51 . 100 . 1

Subnet mask: 255 . 255 . 255 . 0

Default gateway: . . .

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

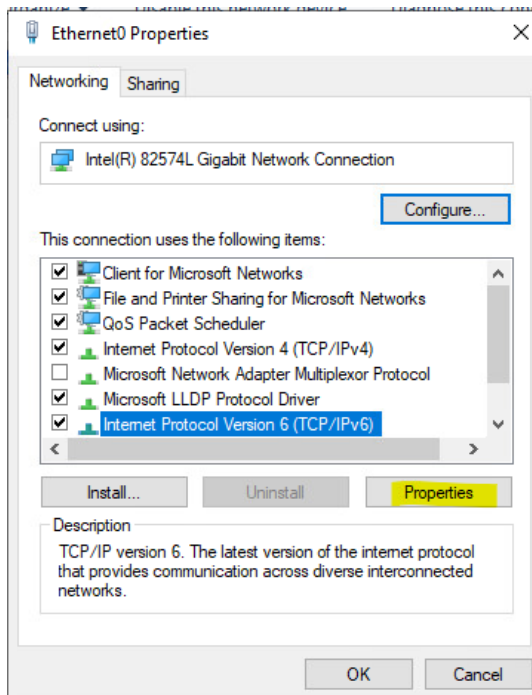
Preferred DNS server: 127 . 0 . 0 . 1

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel



Internet Protocol Version 6 (TCP/IPv6) Properties

General

You can get IPv6 settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IPv6 settings.

☐ Obtain an IPv6 address automatically

☒ Use the following IPv6 address:

IPv6 address: 2001:db8:100::1

Subnet prefix length: 64

Default gateway:

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: ::1

Alternate DNS server:

☐ Validate settings upon exit

Advanced...

OK Cancel

```

Administrator: Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

Host Name . . . . . : INET
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) 82574L Gigabit Network Connection #2
Physical Address. . . . . : 00-0C-29-D5-4B-94
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-D5-4B-8A
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:db8:100::1(Preferred)
Link-local IPv6 Address . . . . : fe80::d4a4:88b5:3632:9bf0%4(Preferred)
IPv4 Address. . . . . : 198.51.100.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-D2-7A-2E-00-0C-29-D5-4B-8A
DNS Servers . . . . . : ::1
                        127.0.0.1
NetBIOS over Tcpip. . . . . : Enabled
PS C:\Users\Administrator>

```

NETWORK CONFIGURATION - CORE

IPv4 via menu

IPv6 via Powershell

New-NetIPAddress -InterfaceAlias Ethernet0 -AddressFamily IPv6 -IPAddress fd01:2:1::1 -PrefixLength 64 -DefaultGateway fd01:2:1::254

```
PS C:\Users\Administrator> New-NetIPAddress -InterfaceAlias Ethernet0 -AddressFamily IPv6 -IPAddress fd01:2:1::1 -PrefixLength 64 -DefaultGateway fd01:2:1::254
```

```
PS C:\Users\Administrator> New-NetIPAddress -InterfaceAlias Ethernet0 -AddressFamily IPv6 -IPAddress fd01:2:1::1 -PrefixLength 64 -DefaultGateway fd01:2:1::254

IPAddress      : fd01:2:1::1
InterfaceIndex : 4
InterfaceAlias : Ethernet0
AddressFamily  : IPv6
Type           : Unicast
PrefixLength   : 64
PrefixOrigin   : Manual
SuffixOrigin   : Manual
AddressState   : Tentative
ValidLifetime  : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource   : False
PolicyStore    : ActiveStore
```

Set-DnsClientServerAddress -InterfaceAlias Ethernet0 -ServerAddresses ::1

```
PS C:\Users\Administrator> Set-DnsClientServerAddress -InterfaceAlias Ethernet0 -ServerAddresses ::1
PS C:\Users\Administrator> Get-DnsClientServerAddress

InterfaceAlias      Interface Address ServerAddresses
-----
Index      Family
-----
Ethernet0      4 IPv4 {127.0.0.1}
Ethernet0      4 IPv6 {::1}
Loopback Pseudo-Interface 1 1 IPv4 {}
Loopback Pseudo-Interface 1 1 IPv6 {fec0:0:0:ffff::1, fec0:0:0:ffff::2, fec0:0:0:ffff::3}
```

checking

ipconfig /all

```
PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

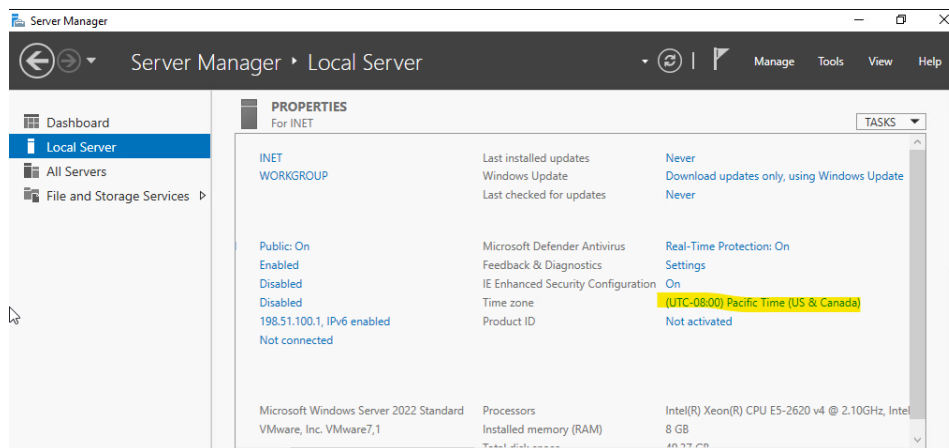
Host Name . . . . . : RODC
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-BB-F8-CA
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : fd01:2::1(Preferred)
Link-local IPv6 Address . . . . . : fe80::7532:aa9b:b425:af0a%4(Preferred)
IPv4 Address. . . . . : 10.2.1.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fd01:2::254
                          10.2.1.254
DHCPv6 IAID . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-D2-77-FC-00-0C-29-BB-F8-CA
DNS Servers . . . . . : ::1
                          127.0.0.1
NetBIOS over Tcpip. . . . . : Enabled

PS C:\Users\Administrator>
```

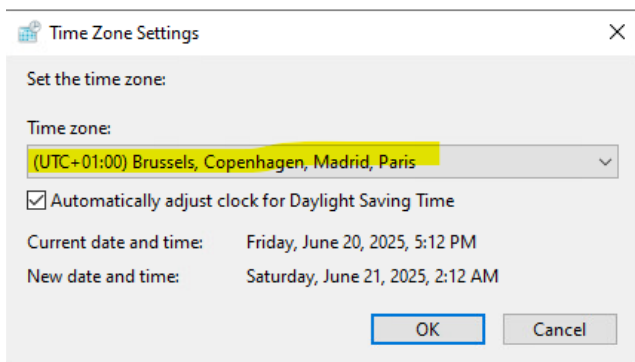
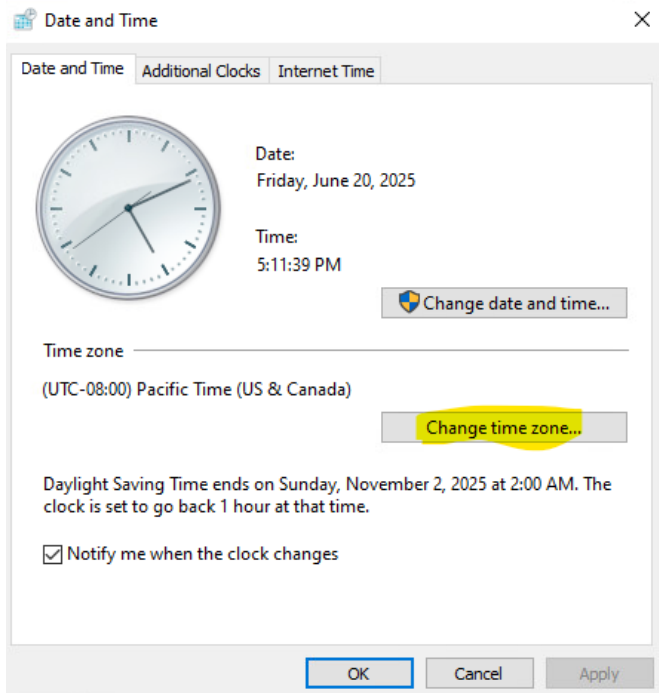
TIMEZONE



Server Manager - Local Server

PROPERTIES For INET

INET WORKGROUP	Last installed updates	Never
	Windows Update	Download updates only, using Windows Update
	Last checked for updates	Never
Public: On	Microsoft Defender Antivirus	Real-Time Protection: On
Enabled	Feedback & Diagnostics	Settings
Disabled	IE Enhanced Security Configuration	On
Disabled	Time zone	(UTC-08:00) Pacific Time (US & Canada)
198.51.100.1, IPv6 enabled	Product ID	Not activated
Not connected		
Microsoft Windows Server 2022 Standard	Processors	Intel(R) Xeon(R) CPU E5-2620 v4 @ 2.10GHz, Intel
VMware, Inc. VMware7,1	Installed memory (RAM)	8 GB
	Total disk space	40.37 GB



DESCRIPTION OF PROJECT AND TASKS

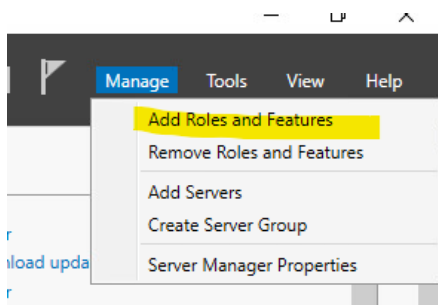
This project simulates a Microsoft-based enterprise network with two independent domains. You are tasked with deploying services such as Active Directory, DNS, certificate services, Group Policies, backups, automation via Ansible, and secure connectivity between sites. The environment must follow best practices and reflect real-world enterprise infrastructure.

INET

Role: Acts as external ISP – provides DNS, SMTP.
Server is preconfigured:

INET - Task1: DNS server

1. Authoritative Nameserver for **nordicbackup.net** and **skillspublic.dk** domain with records according to **Appendix E**



Add Roles and Features Wizard

Select server roles

DESTINATION SERVER
INET

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select one or more roles to install on the selected server.

Roles	Description
☐ Active Directory Certificate Services	
☐ Active Directory Domain Services	
☐ Active Directory Federation Services	
☐ Active Directory Lightweight Directory Services	
☐ Active Directory Rights Management Services	
☐ Device Health Attestation	
☐ DHCP Server	
☐ DNS Server	Domain Name System (DNS) Server provides name resolution for TCP/IP networks. DNS Server is easier to manage when it is installed on the same server as Active Directory Domain Services. If you select the Active Directory Domain Services role, you can install and configure DNS Server and Active Directory Domain Services to work together.
☐ Fax Server	
☒ File and Storage Services (1 of 12 installed)	
☐ Host Guardian Service	
☐ Hyper-V	
☐ Network Policy and Access Services	
☐ Print and Document Services	
☐ Remote Access	
☐ Remote Desktop Services	
☐ Volume Activation Services	
☐ Web Server (IIS)	
☐ Windows Deployment Services	
☐ Windows Server Update Services	

< Previous Next > Install Cancel

Add Roles and Features Wizard

Add features that are required for DNS Server?

The following tools are required to manage this feature, but do not have to be installed on the same server.

- Remote Server Administration Tools
 - Role Administration Tools
 - [Tools] DNS Server Tools

☒ Include management tools (if applicable)

Add Features Cancel

Add Roles and Features Wizard

Select server roles

DESTINATION SERVER
INET

Before You Begin
Installation Type
Server Selection
Server Roles
Features
DNS Server
Confirmation
Results

Select one or more roles to install on the selected server.

Roles	Description
☐ Active Directory Certificate Services	Domain Name System (DNS) Server provides name resolution for TCP/IP networks. DNS Server is easier to manage when it is installed on the same server as Active Directory Domain Services. If you select the Active Directory Domain Services role, you can install and configure DNS Server and Active Directory Domain Services to work together.
☐ Active Directory Domain Services	
☐ Active Directory Federation Services	
☐ Active Directory Lightweight Directory Services	
☐ Active Directory Rights Management Services	
☐ Device Health Attestation	
☐ DHCP Server	
☒ DNS Server	
☐ Fax Server	
☒ File and Storage Services (1 of 12 installed)	
☐ Host Guardian Service	
☐ Hyper-V	
☐ Network Policy and Access Services	
☐ Print and Document Services	
☐ Remote Access	
☐ Remote Desktop Services	
☐ Volume Activation Services	
☐ Web Server (IIS)	
☐ Windows Deployment Services	
☐ Windows Server Update Services	

< Previous Next >

Activate Windows
Go to Settings to activate Windows.

<https://extratruco.com/informatica-tecla-suprimir>

Add Roles and Features Wizard

Confirm installation selections

DESTINATION SERVER
INET

Before You Begin
Installation Type
Server Selection
Server Roles
Features
DNS Server
Confirmation
Results

To install the following roles, role services, or features on selected server, click Install.

☐ Restart the destination server automatically if required

Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.

DNS Server

Remote Server Administration Tools

Role Administration Tools

DNS Server Tools

[Export configuration settings](#)
[Specify an alternate source path](#)

< Previous Next > **Install** Cancel


Add Roles and Features Wizard

Installation progress

- Before You Begin
- Installation Type
- Server Selection
- Server Roles
- Features
- DNS Server
- Confirmation
- Results**

View installation progress


Feature installation

Installation succeeded on INET.

DNS Server

Remote Server Administration Tools

Role Administration Tools

DNS Server Tools



You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.

[Export configuration settings](#)

< Previous

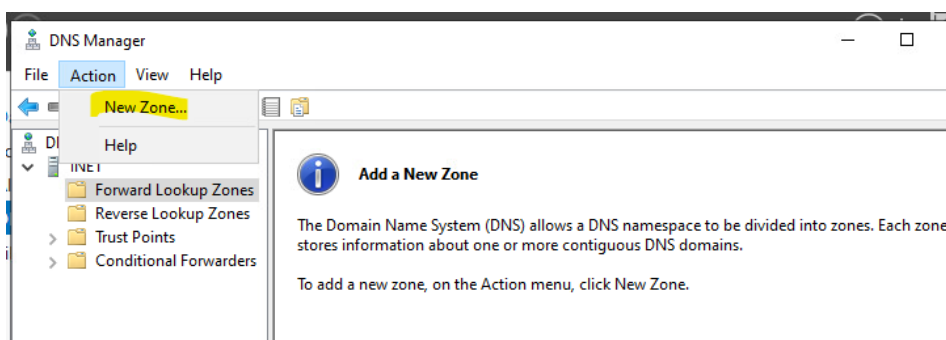
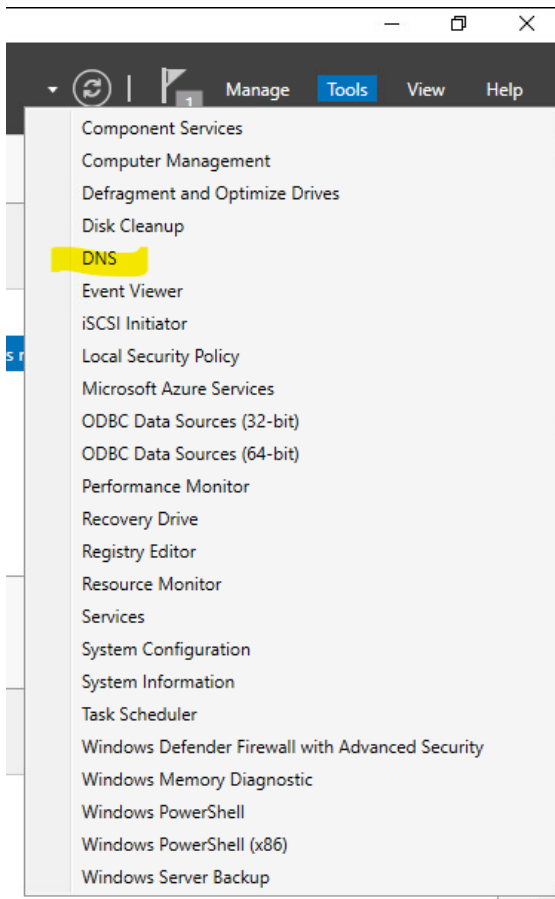
Next >

Close

Cancel

Activate Windows

Go to Settings to activate



<https://extratruocos.com/informatica-tecla-suprimir>

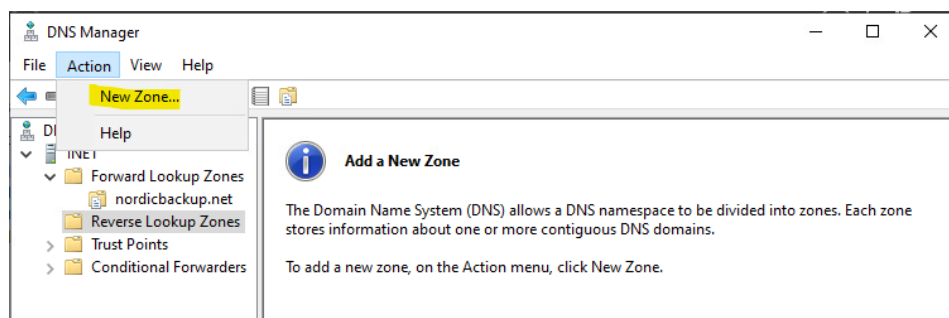
New Zone Wizard

Zone Name
What is the name of the new zone?

The zone name specifies the portion of the DNS namespace for which this server is authoritative. It might be your organization's domain name (for example, microsoft.com) or a portion of the domain name (for example, newzone.microsoft.com). The zone name is not the name of the DNS server.

Zone name:
nordicbackup.net

< Back Next > Cancel



New Zone Wizard

Reverse Lookup Zone Name

A reverse lookup zone translates IP addresses into DNS names.



Choose whether you want to create a reverse lookup zone for IPv4 addresses or IPv6 addresses.

☒ IPv4 Reverse Lookup Zone

☐ IPv6 Reverse Lookup Zone

< Back

Next >

Cancel

New Zone Wizard

Reverse Lookup Zone Name

A reverse lookup zone translates IP addresses into DNS names.



To identify the reverse lookup zone, type the network ID or the name of the zone.

☒ Network ID:

 The network ID is the portion of the IP addresses that belongs to this zone. Enter the network ID in its normal (not reversed) order.

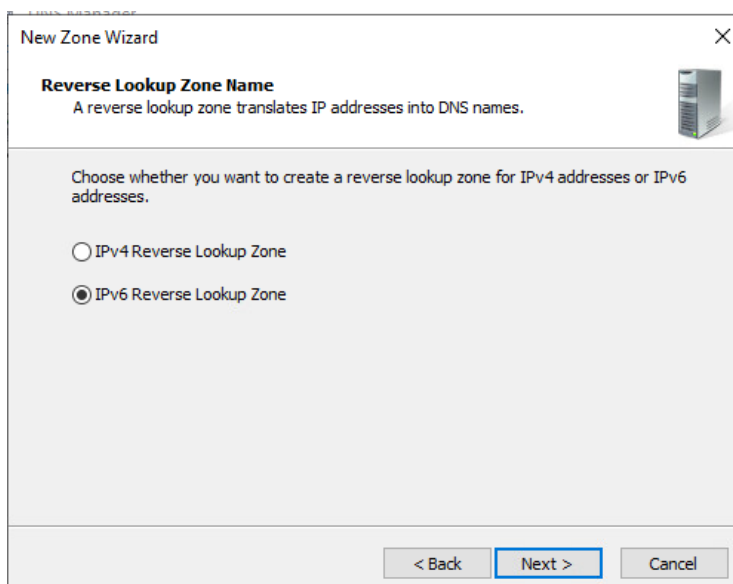
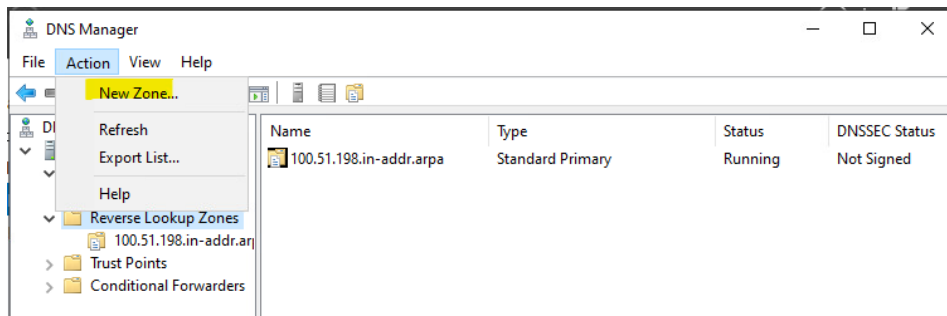
 If you use a zero in the network ID, it will appear in the zone name. For example, network ID 10 would create zone 10.in-addr.arpa, and network ID 10.0 would create zone 0.10.in-addr.arpa.

☐ Reverse lookup zone name:

< Back

Next >

Cancel



New Zone Wizard

Reverse Lookup Zone Name

A reverse lookup zone translates IP addresses into DNS names.

To name the reverse lookup zone, enter an IPv6 address prefix to auto generate the zone name(s). Depending on the prefix you enter, up to 8 zones may be created.

IPv6 Address Prefix:

2001:db8:100::/64

Reverse Lookup Zones

☒ 0.0.0.0.0.0.1.0.8.b.d.0.1.0.0.2.ip6.arpa

< Back

Next >

Cancel

DNS Manager

File Action View Help

DNS

INET

Forward Lookup Zones

nordicbackup.net

Reverse Lookup Zones

100.51.198.in-addr.arpa

0.0.0.0.0.1.0.8.b.d.0.

Trust Points

Conditional Forwarders

Name	Type	Data
(same as parent folder)	Start of Authority (SOA)	[1], inet, hostmaster.
(same as parent folder)	Name Server (NS)	inet.

Update Server Data File

Reload

New Host (A or AAAA)...

New Alias (CNAME)...

New Mail Exchanger (MX)...

New Domain...

New Delegation...

ES2025_TP_IT Network Systems Administration_39_ModuleB

Version: 1.0
Date: 07.06.25

27 of 198

New Host

Name (uses parent domain name if blank):
mail

Fully qualified domain name (FQDN):
mail.nordicbackup.net.

IP address:
198.51.100.1

☒ Create associated pointer (PTR) record

Add Host Cancel

DNS

 The host record mail.nordicbackup.net was successfully created.

OK

New Host

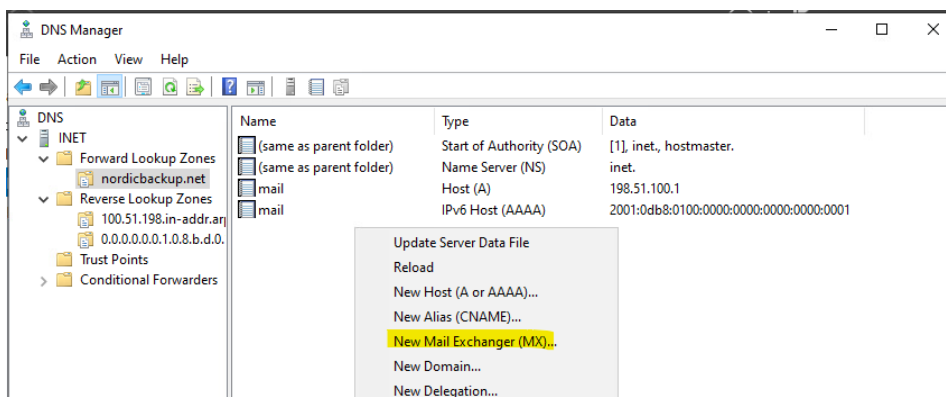
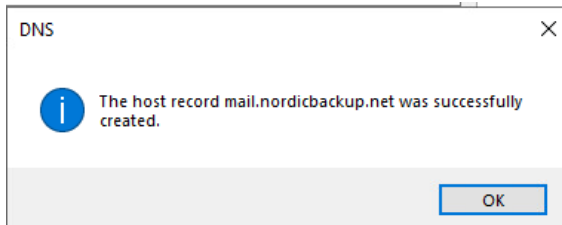
Name (uses parent domain name if blank):
mail

Fully qualified domain name (FQDN):
mail.nordicbackup.net.

IP address:
2001:db8:100::1

☒ Create associated pointer (PTR) record

Add Host Done



New Resource Record

Mail Exchanger (MX)

Host or child domain:

By default, DNS uses the parent domain name when creating a Mail Exchange record. You can specify a host or child name, but in most deployments, the above field is left blank.

Fully qualified domain name (FQDN):

nordicbackup.net.

Fully qualified domain name (FQDN) of mail server:

mail.nordicbackup.net Browse...

Mail server priority:

10

OK Cancel Help

DNS Manager

File Action View Help

Name	Type	Data
(same as parent folder)	Start of Authority (SOA)	[1] inet., hostmaster.
(same as parent folder)	Name Server (NS)	inet.
mail	Host (A)	198.51.100.1
mail	IPv6 Host (AAAA)	2001:0db8:0100:0000:0000:0000:0001
(same as parent folder)	Mail Exchanger (MX)	[10] mail.nordicbackup.net

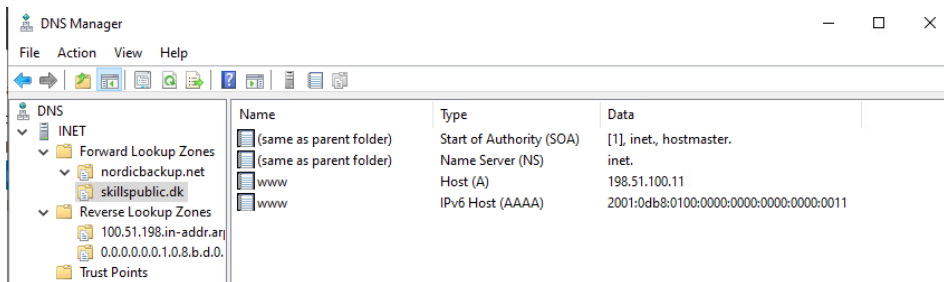
```

PS C:\Users\Administrator> nslookup mail.nordicbackup.net
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

Name:   mail.nordicbackup.net
Addresses: 2001:db8:100::1
           198.51.100.1

PS C:\Users\Administrator>
  
```

REPEAT THE PROCESS WITH skillspublic.dk domain - Here is the result:

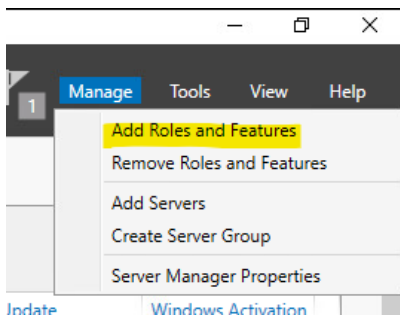


```
PS C:\Users\Administrator> nslookup www.skillspublic.dk
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

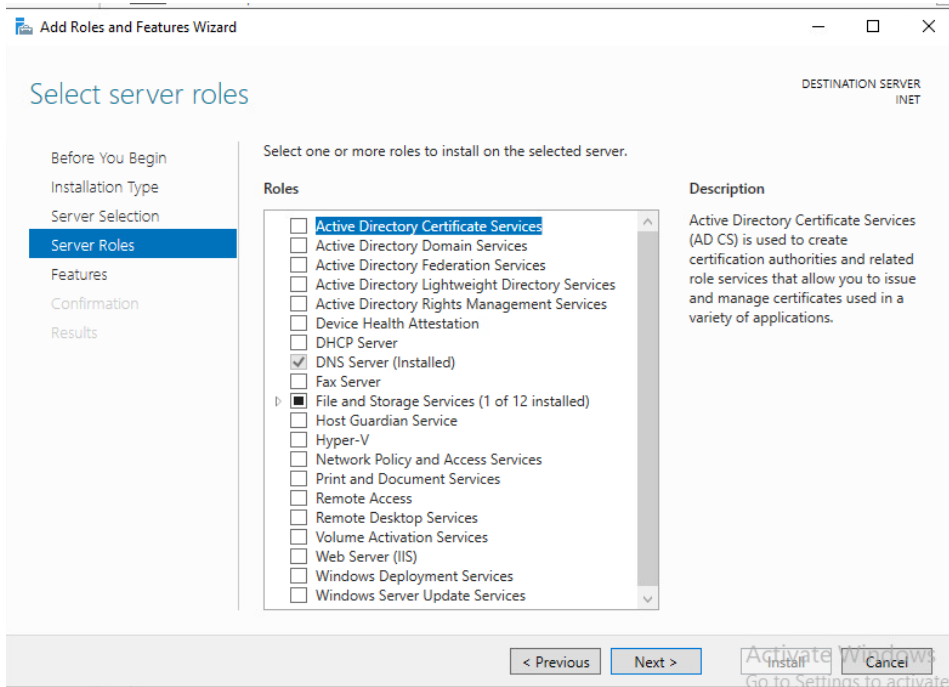
Name:    www.skillspublic.dk
Addresses: 2001:db8:100::11
           198.51.100.11
PS C:\Users\Administrator>
```

INET - Task2: SMTP server

2. SMTP server for notifications



Let "Select server roles" selection as it is, DO NOT CHOOSE ANYTHING



In “Features”, choose “SMTP Server”:

Add Roles and Features Wizard

Select features

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select one or more features to install on the selected server.

DESTINATION SERVER
INET

Features	Description
☐ Peer Name Resolution Protocol	
☐ Quality Windows Audio Video Experience	
☐ RAS Connection Manager Administration Kit (CMA)	
☐ Remote Assistance	
☐ Remote Differential Compression	
☒ Remote Server Administration Tools (1 of 43 installed)	
☐ RPC over HTTP Proxy	
☐ Setup and Boot Event Collection	
☐ Simple TCP/IP Services	
☐ SMB 1.0/CIFS File Sharing Support	
☐ SMB Bandwidth Limit	
☒ SMTP Server	SMTP Server supports the transfer of e-mail messages between e-mail systems.
☐ SNMP Service	
☐ Storage Migration Service	
☐ Storage Migration Service Proxy	
☐ Storage Replica	
☒ System Data Archiver (Installed)	
☐ System Insights	
☐ Telnet Client	

< Previous Next > **Install** **Cancel**

Activate Windows
Go to Settings to activate Windows.

Add Roles and Features Wizard

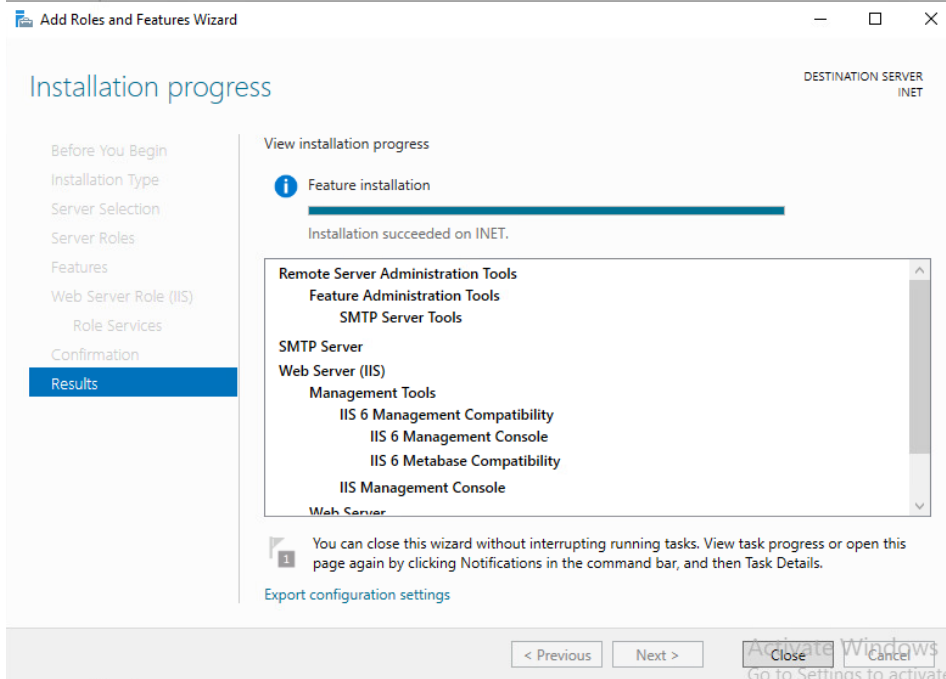
Add features that are required for SMTP Server?

You cannot install SMTP Server unless the following role services or features are also installed.

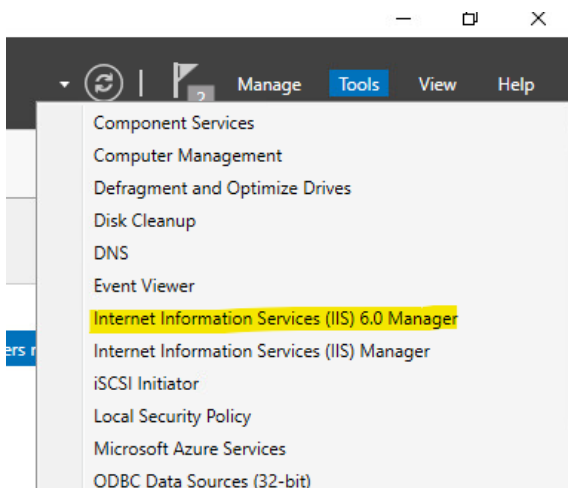
- ▲ Remote Server Administration Tools
 - ▲ Feature Administration Tools
 - SMTP Server Tools
- ▲ Web Server (IIS)
 - ▲ Management Tools
 - ▲ IIS 6 Management Compatibility
 - IIS 6 Management Console
 - IIS 6 Metabase Compatibility
 - [Tools] IIS Management Console
 - ▲ Web Server
 - ▲ Health and Diagnostics
 - ODBC Logging

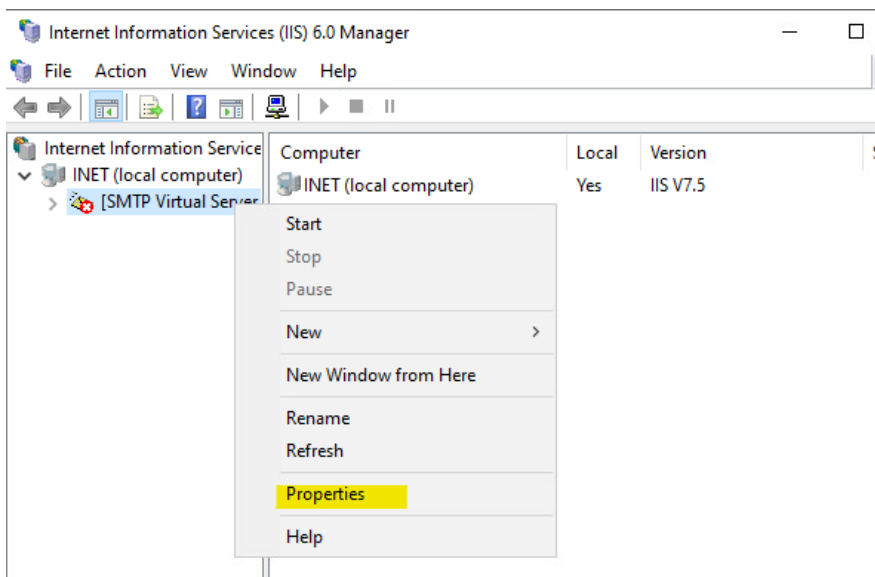
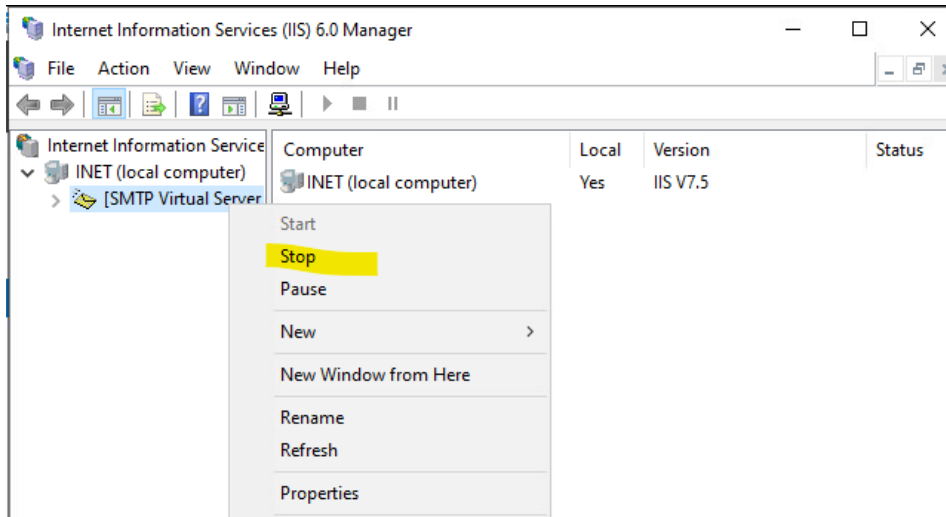
☒ Include management tools (if applicable)

Add Features **Cancel**



CONFIGURE



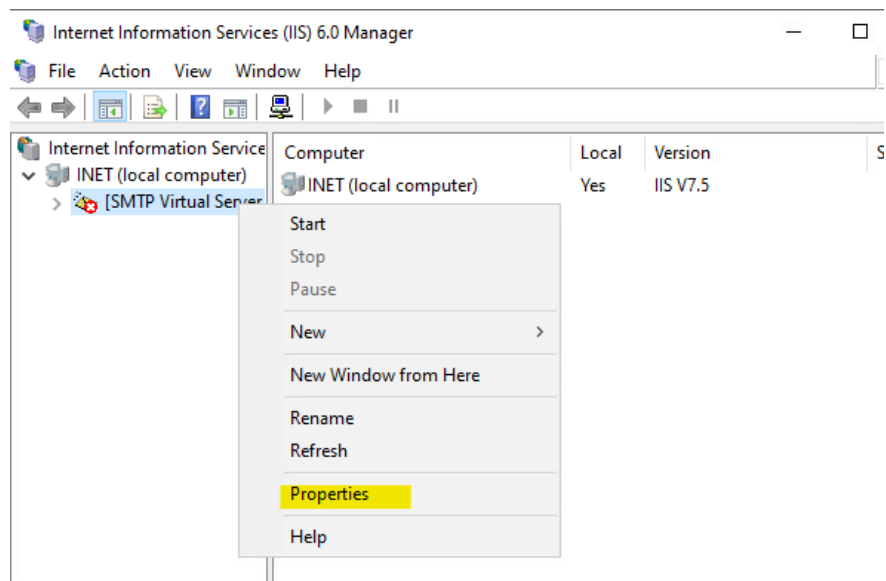


If it gives a MMC error, do this:

```
Open services.msc
Stop SMTPSVC service [Display Name: Simple Mail Transfer Protocol (SMTP)]
Stop IISADMIN service [Display name: IIS Admin Service]
Edit "C:\Windows\System32\inetsrv\MetaBase.xml"
Find: <IISMtpServer Location ="/LM/SmtpSvc/1"
Add (Settings are alphabetical): RelayIpList=""
```

Save file
Start IISAdmin Service
Start SMTPSVC service

Try again:



[SMTP Virtual Server #1] Properties

General Access Messages Delivery LDAP Routing Security

Fully-qualified domain name:
INET

IP address:
(All Unassigned) Advanced...

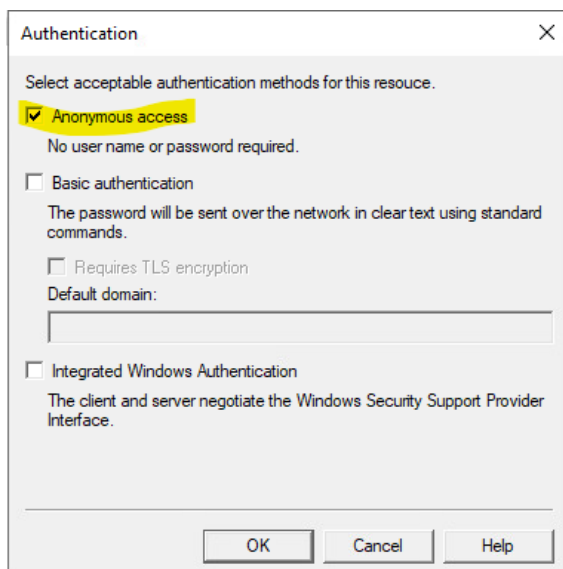
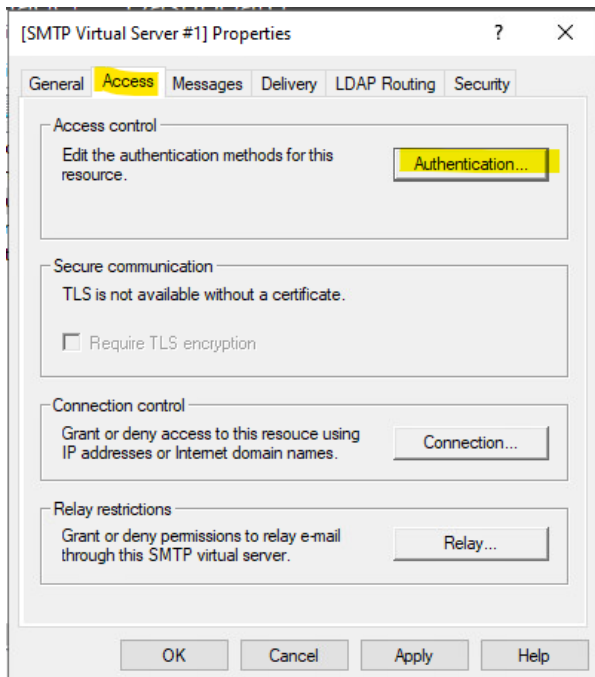
☐ Limit number of connections to:

Connection time-out (minutes):

☒ Enable logging

Active log format:
W3C Extended Log File Format Properties...

OK Cancel Apply Help



[SMTP Virtual Server #1] Properties

General Access Messages **Delivery** LDAP Routing Security

Outbound

First retry interval (minutes): 15

Second retry interval (minutes): 30

Third retry interval (minutes): 60

Subsequent retry interval (minutes): 240

Delay notification: 12 Hours

Expiration timeout: 2 Days

Local

Delay notification: 12 Hours

Expiration timeout: 2 Days

Outbound Security... Outbound connections... **Advanced...**

OK Cancel Apply Help

[SMTP Virtual Server #1] Properties

Advanced Delivery

Maximum hop count:
15

Masquerade domain:

Fully-qualified domain name:
mail.nordicbackup.net Check DNS

Smart host:

☐ Attempt direct delivery before sending to smart host

☐ Perform reverse DNS lookup on incoming messages

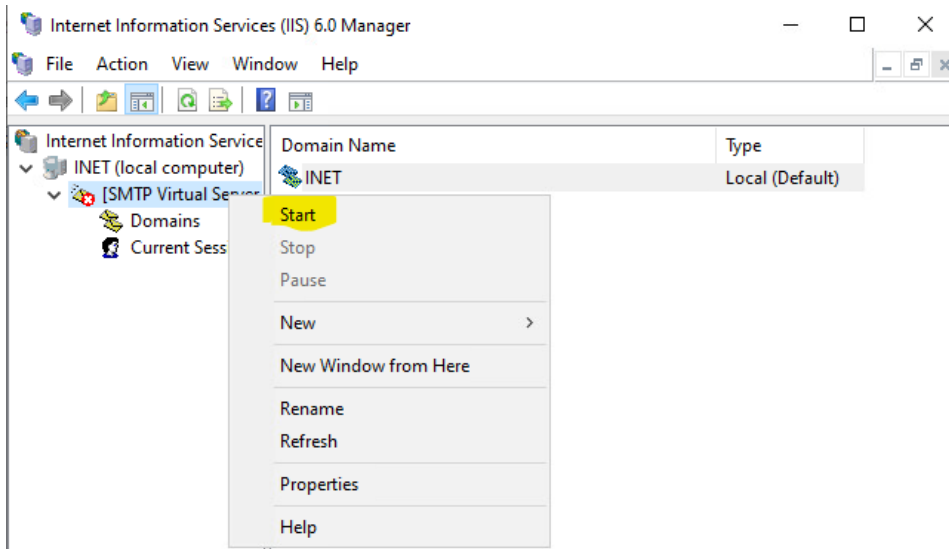
OK Cancel Help

OK Cancel Apply Help

SMTP Configuration

 The domain name is valid.

OK



INET - Task3: web server

3. Web server running on ports 80 and 8080

Main Server Role already installed with SMTP. But you must install some Common HTTP Features for the IIS to work:

Add Roles and Features Wizard

Select server roles

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select one or more roles to install on the selected server.

Roles

- ☐ Network Policy and Access Services
- ☐ Print and Document Services
- ☐ Remote Access
- ☐ Remote Desktop Services
- ☐ Volume Activation Services
- ☒ Web Server (IIS) (8 of 43 installed)
 - ☒ Web Server (5 of 34 installed)
 - ☒ **Common HTTP Features (4 of 6 installed)**
 - ☒ Default Document (Installed)
 - ☒ Directory Browsing (Installed)
 - ☒ HTTP Errors (Installed)
 - ☒ Static Content (Installed)
 - ☐ HTTP Redirection
 - ☐ WebDAV Publishing
 - ☒ Health and Diagnostics (1 of 6 installed)
 - ☐ Performance
 - ☐ Security
 - ☐ Application Development
 - ☐ FTP Server
 - ☒ Management Tools (3 of 7 installed)

Description

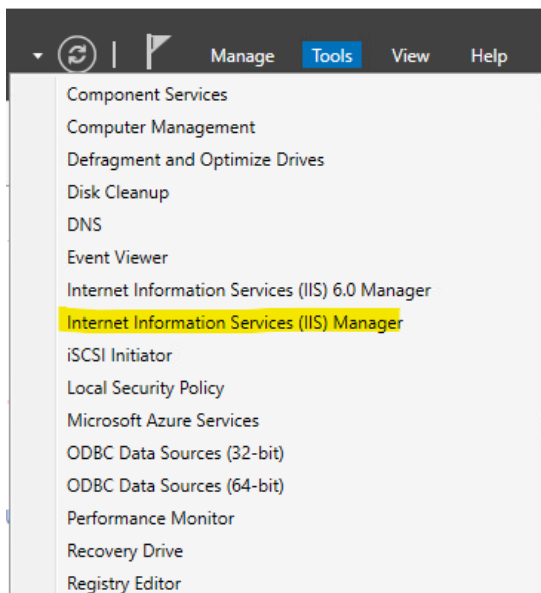
Common HTTP Features supports basic HTTP functionality, such as delivering standard file formats and configuring custom server properties. Use Common HTTP Features to create custom error messages, to configure how the server responds to requests that do not specify a document, or to automatically redirect some requests to a different location.

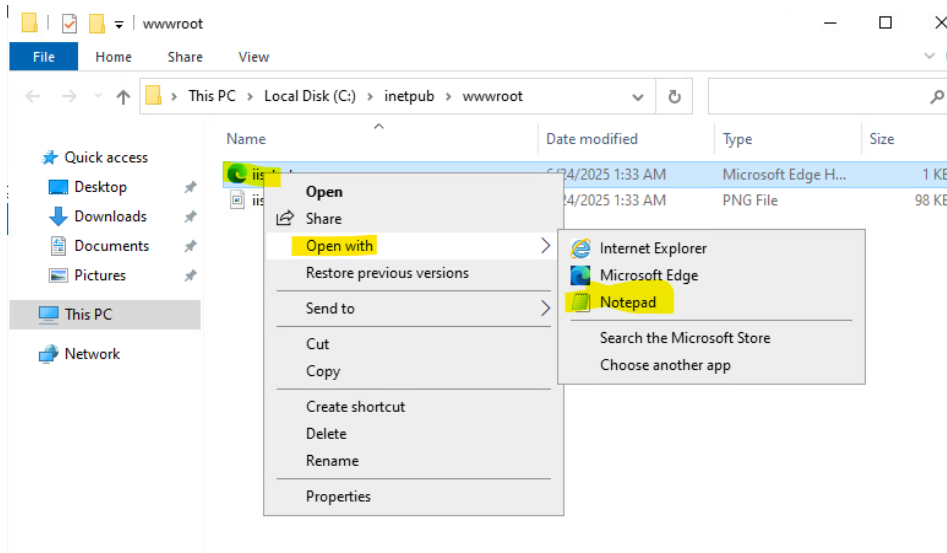
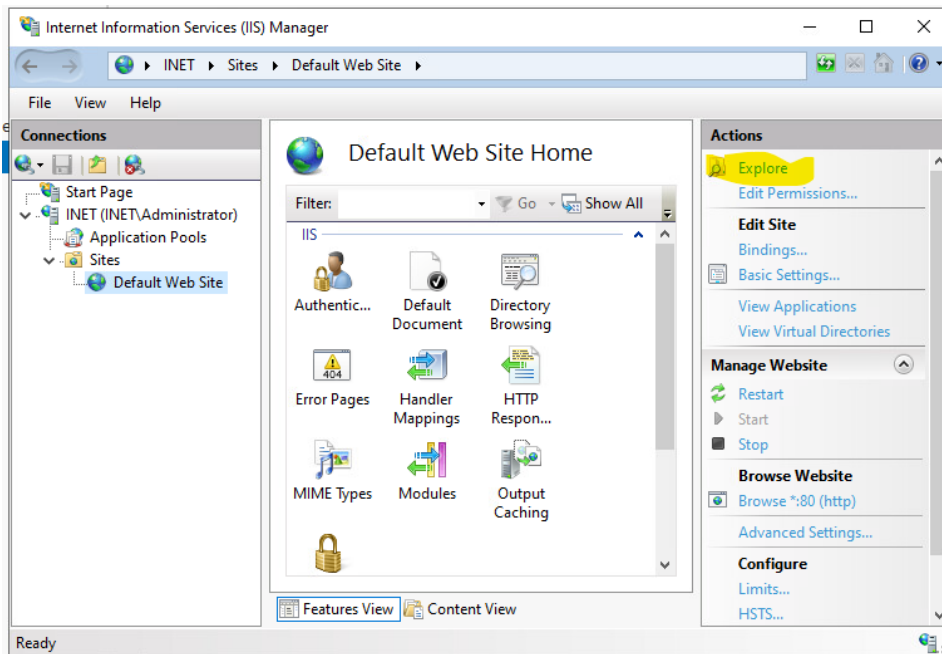
< Previous

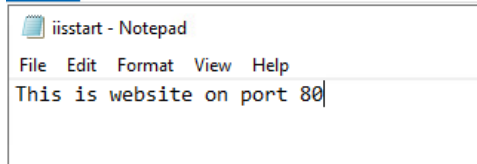
Next >

Install

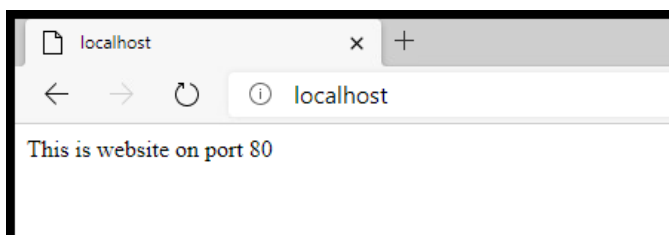
Cancel



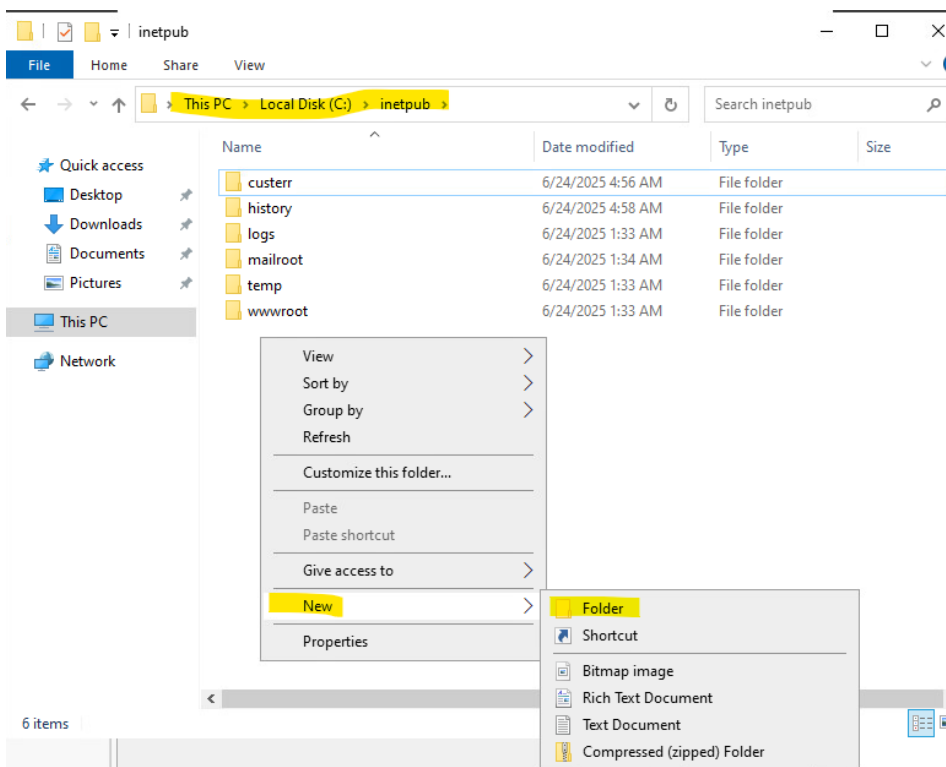




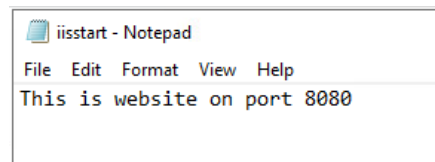
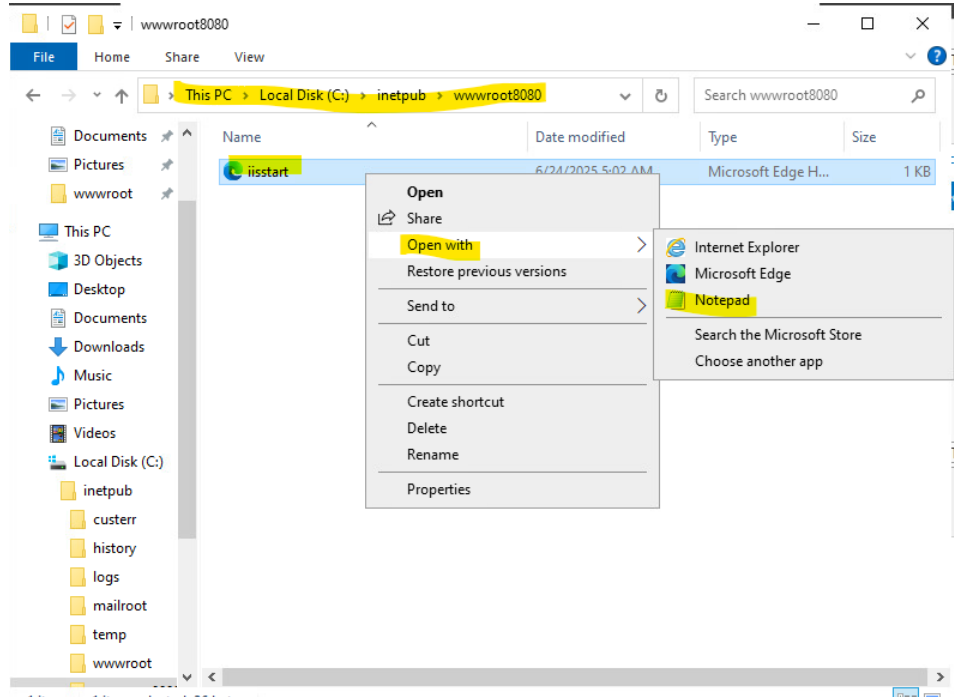
Web browser:

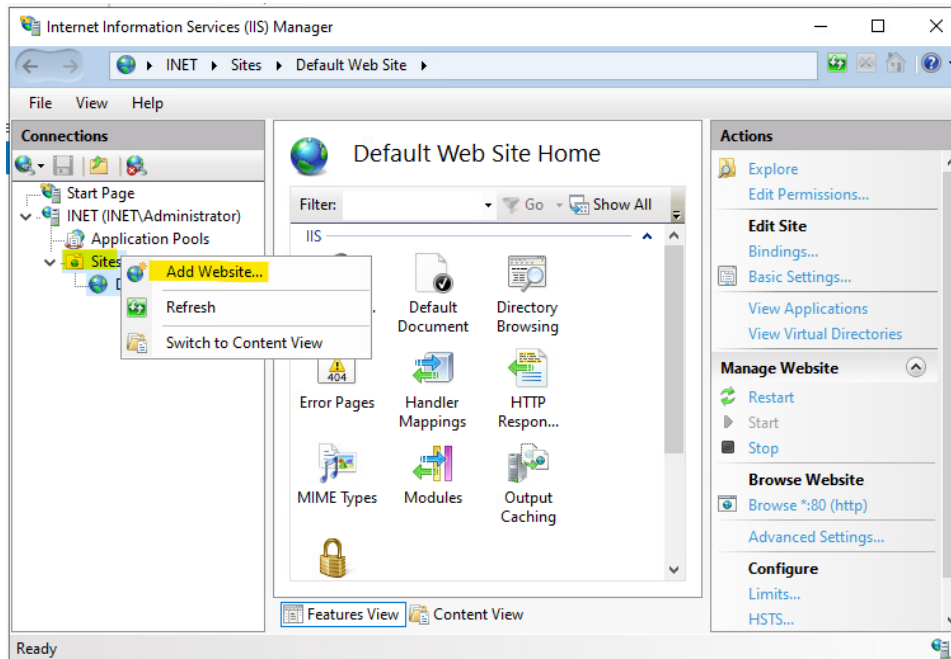


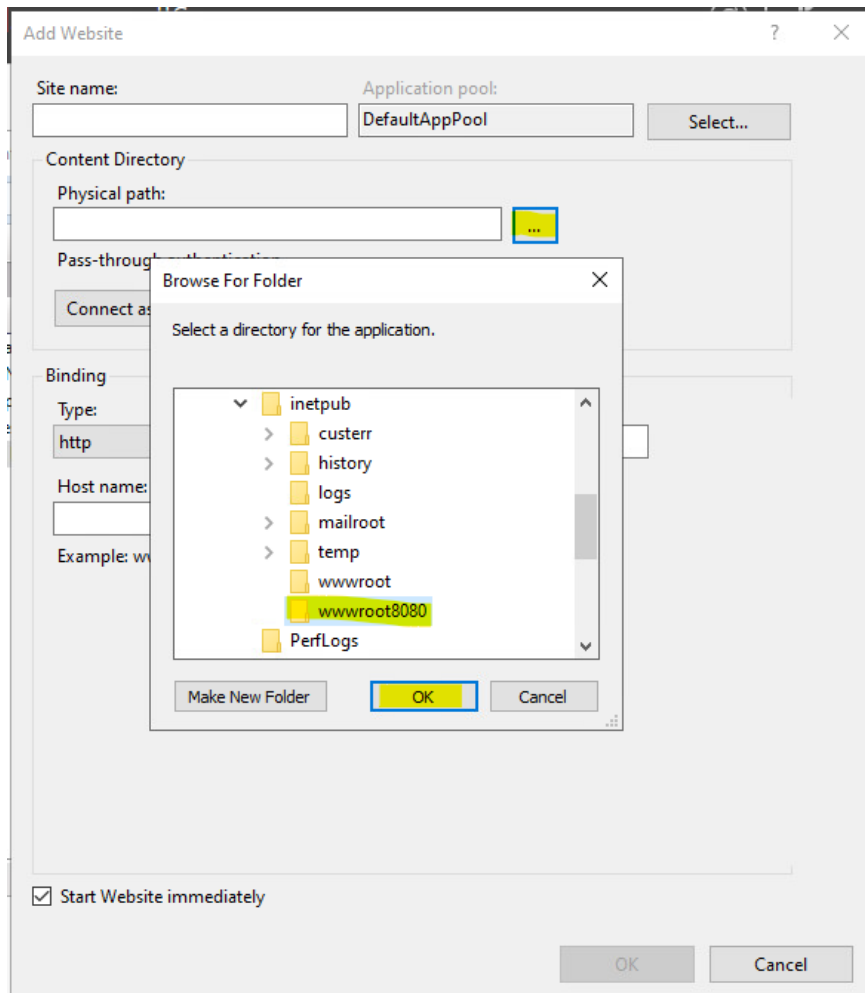
ADD website on port 8080



<https://extratruco.com/informatica-tecla-suprimir>







Add Website

Site name: Web Site 8080 Application pool: DefaultAppPool Select...

Content Directory

Physical path: C:\inetpub\wwwroot8080 ...

Pass-through authentication

Connect as... Test Settings...

Binding

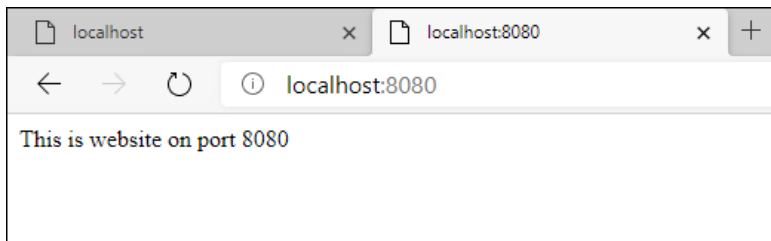
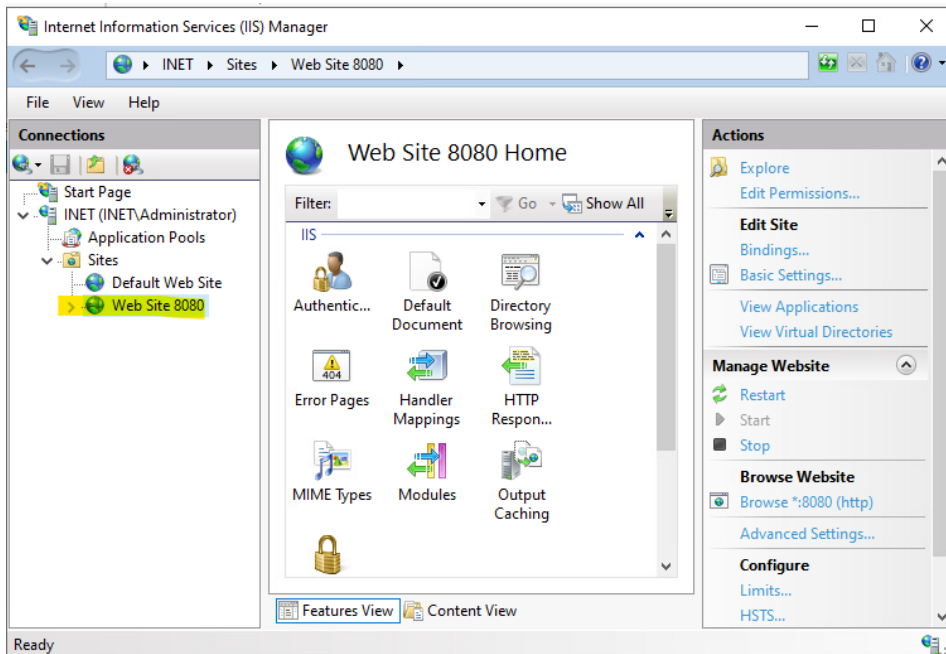
Type: http IP address: All Unassigned Port: 8080

Host name:

Example: www.contoso.com or marketing.contoso.com

☒ Start Website immediately

OK Cancel



RTR-CPH

Role: HQ Router

Tasks:

RTR-CPH - Task1: Site-to-site VPN

1. Configure IKEv2 Site-to-Site VPN with RTR-AAL

Add Roles and Features Wizard

Select server roles

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Network Policy and Acces...
Confirmation
Results

Select one or more roles to install on the selected server.

Roles	Description
☐ Active Directory Certificate Services	
☐ Active Directory Domain Services	
☐ Active Directory Federation Services	
☐ Active Directory Lightweight Directory Services	
☐ Active Directory Rights Management Services	
☐ Device Health Attestation	
☐ DHCP Server	
☐ DNS Server	
☐ Fax Server	
☒ File and Storage Services (1 of 12 installed)	
☐ Host Guardian Service	
☐ Hyper-V	
☒ Network Policy and Access Services	Network Policy and Access Services provides Network Policy Server (NPS), which helps safeguard the security of your network.
☐ Print and Document Services	
☒ Remote Access (2 of 3 installed)	
☒ DirectAccess and VPN (RAS) (Installed)	
☒ Routing (Installed)	
☐ Web Application Proxy	
☐ Remote Desktop Services	
☐ Volume Activation Services	

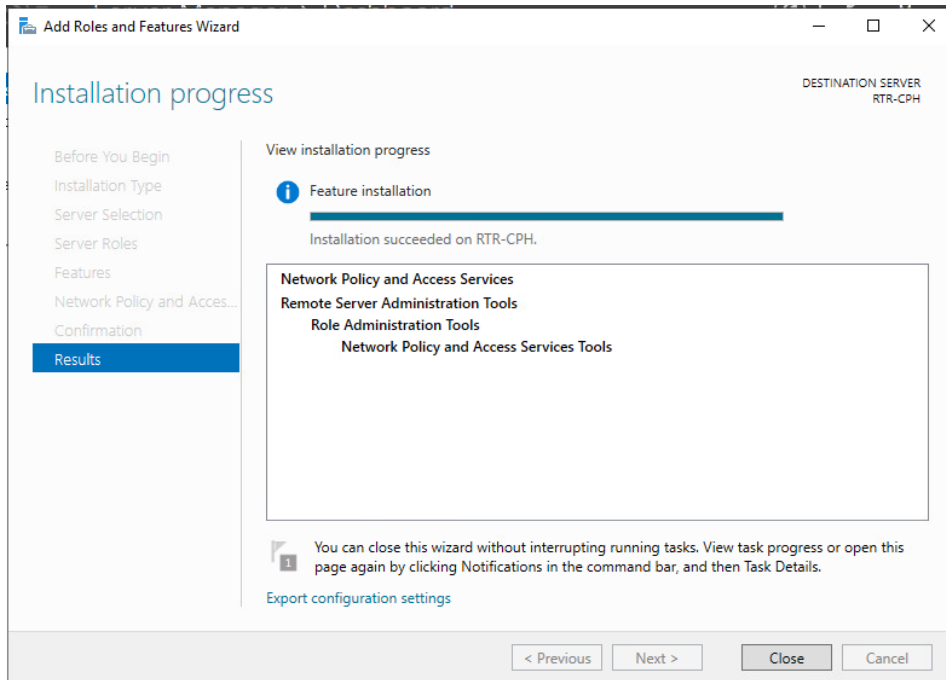
DESTINATION SERVER
RTR-CPH

< Previous
Next >
Install
Cancel

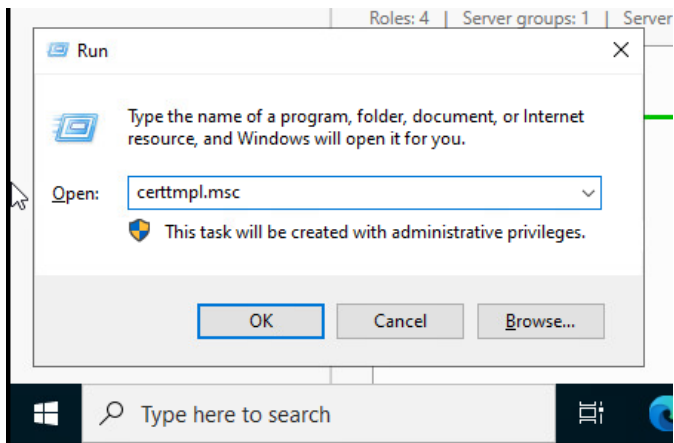
ES2025_TP_IT Network Systems Administration_39_ModuleB

Version: 1.0
Date: 07.06.25

50 of 198



- Go to DC machine (CA) to create certs



Comentado [1]: I am not able to configure S2S VPN IKEv2 using certs. Maybe it's enough using PSK.

Exchange Enrollment Agent (Offline requ...	1	4.1	
Exchange Signature Only	1	6.1	
Exchange User	1	7.1	
IPSec	1	8.1	
IPSec (Offline)		7.1	
Kerberos Aut		110.0	Client Authent
Key Recovery		105.0	Key Recovery A
OCSP Respon		101.0	OCSP Signing
RAS and IAS		101.0	Client Authent
Root Certificate		5.1	
Router (Offline request)	1	4.1	
Static Exch...	2	100.2	Client Authent

Duplicate Template

All Tasks

Properties

Help

Properties of New Template

Subject Name	Server	Issuance Requirements
Superseded Templates		
Extensions		
Security		
Compatibility	General	Request Handling
Cryptography		Key Attestation

Template display name:

Template name:

Validity period:
 years

Renewal period:
 weeks

☒ Publish certificate in Active Directory

☐ Do not automatically reenroll if a duplicate certificate exists in Active Directory

OK Cancel Apply Help

Properties of New Template ✕

Subject Name	Server	Issuance Requirements
Superseded Templates	Extensions	Security
Compatibility	General	Request Handling
	Cryptography	Key Attestation

Purpose: Signature and encryption ▾

☐ Delete revoked or expired certificates (do not archive)

☐ Include symmetric algorithms allowed by the subject

☐ Archive subject's encryption private key

☐ Authorize additional service accounts to access the private key (*)

Key Permissions...

☒ Allow private key to be exported

☐ Renew with the same key (*)

☐ For automatic renewal of smart card certificates, use the existing key if a new key cannot be created (*)

Do the following when the subject is enrolled and when the private key associated with this certificate is used:

☒ Enroll subject without requiring any user input

☐ Prompt the user during enrollment

☐ Prompt the user during enrollment and require user input when the private key is used

* Control is disabled due to [compatibility settings](#).

OK Cancel Apply Help

Properties of New Template

Subject Name Server Issuance Requirements

Compatibility General Request Handling Cryptography Key Attestation

Superseded Templates Extensions **Security**

Group or user names:

- Authenticated Users**
- Administrator
- Domain Admins (SKILLSNET\Domain Admins)
- Domain Computers (SKILLSNET\Domain Computers)
- Domain Controllers (SKILLSNET\Domain Controllers)
- Enterprise Admins (SKILLSNET\Enterprise Admins)

Add... Remove

Permissions for Authenticated Users

	Allow	Deny
Full Control	☐	☐
Read	☒	☐
Write	☐	☐
Enroll	☒	☐
Autoenroll	☒	☐

For special permissions or advanced settings, click Advanced.

Advanced

OK **Cancel** Apply Help

Properties of New Template

Subject Name Server Issuance Requirements

Compatibility General Request Handling Cryptography Key Attestation

Superseded Templates **Extensions** Security

To modify an extension, select it, and then click Edit.

Extensions included in this template:

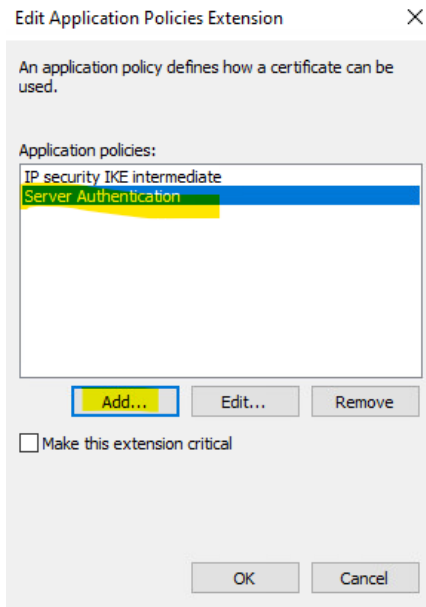
- Application Policies**
- Basic Constraints
- Certificate Template Information
- Issuance Policies
- Key Usage

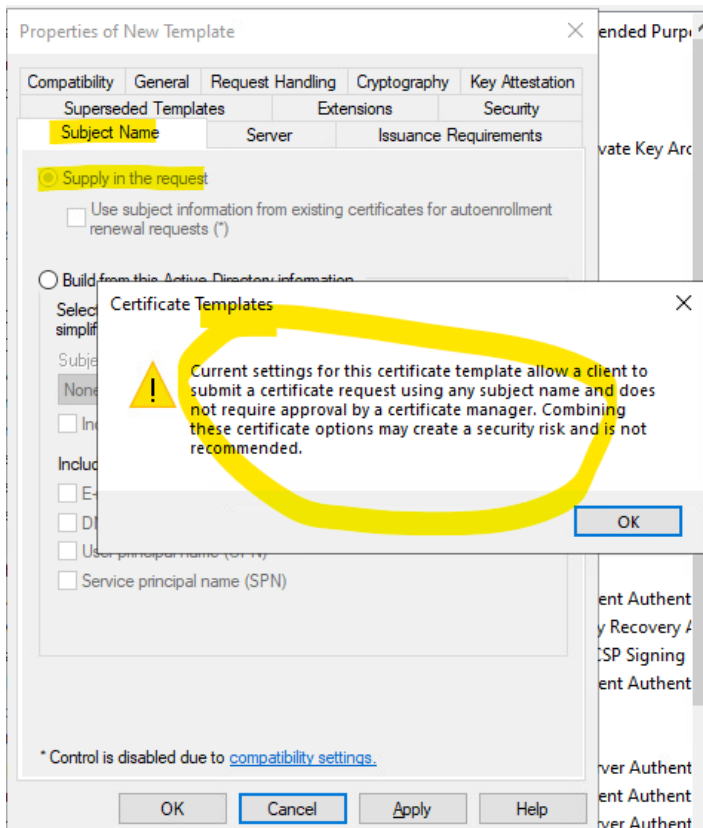
Edit...

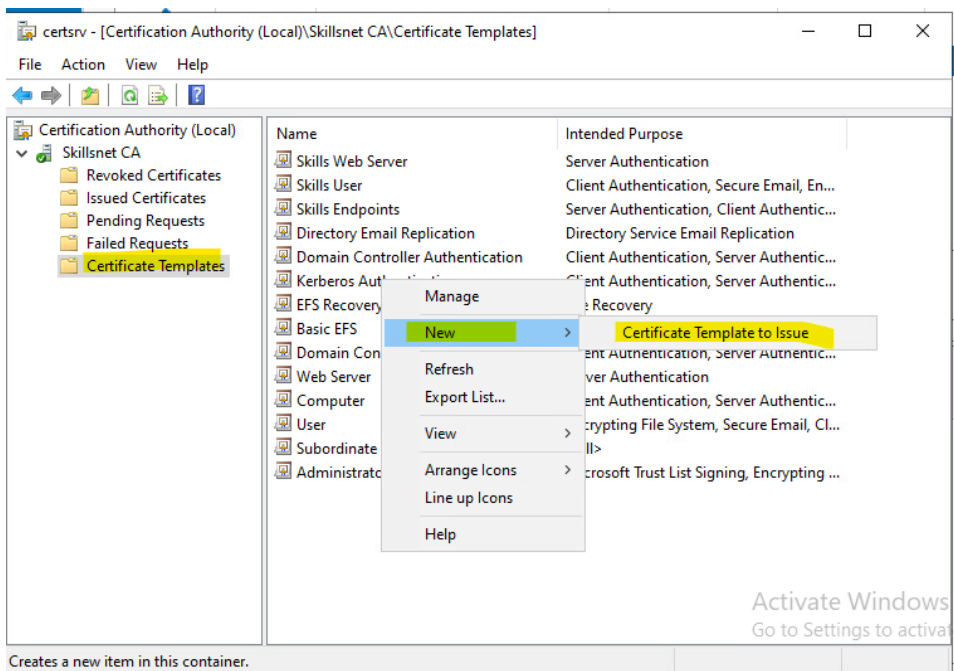
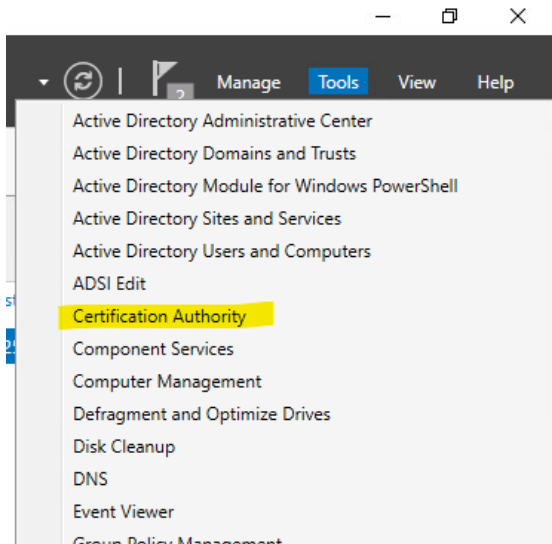
Description of Application Policies:

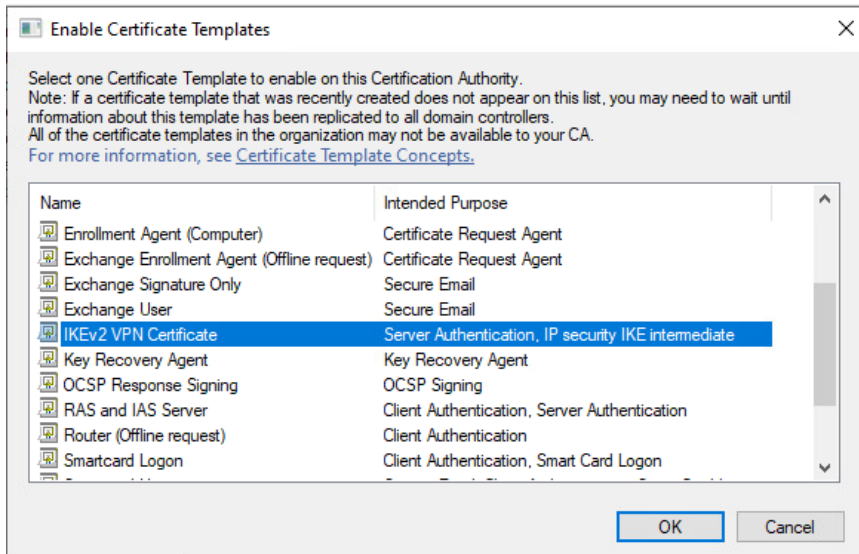
IP security IKE intermediate

OK **Cancel** Apply Help

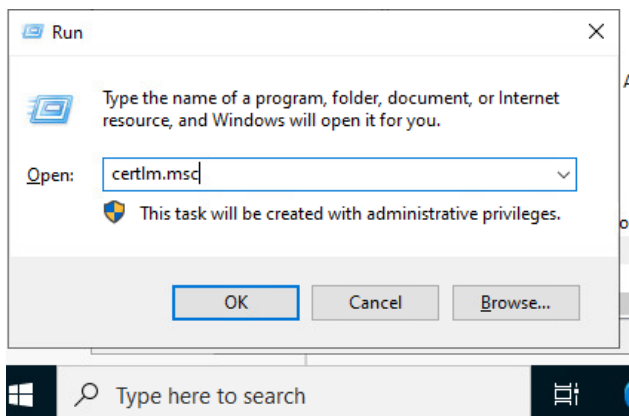


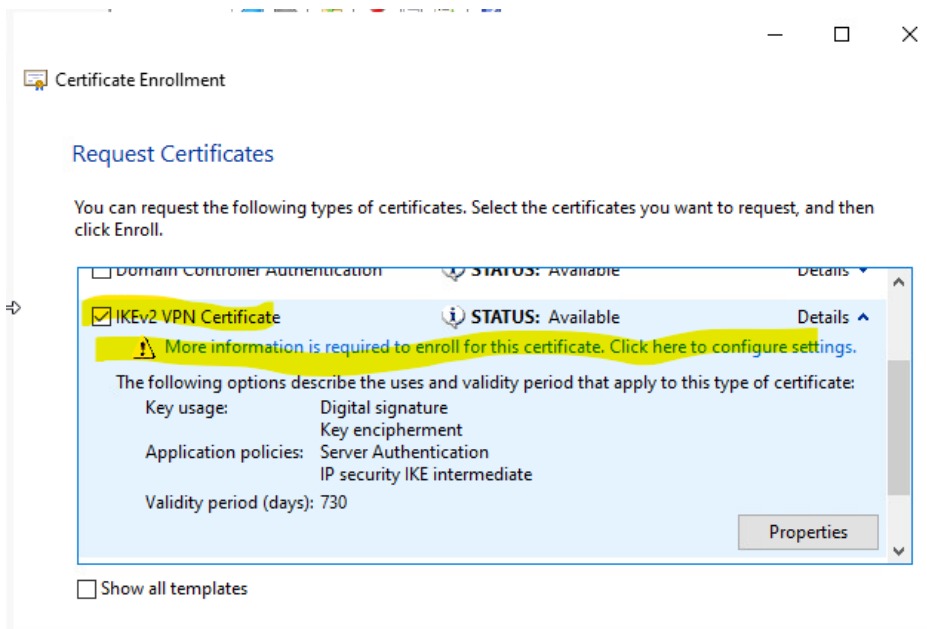
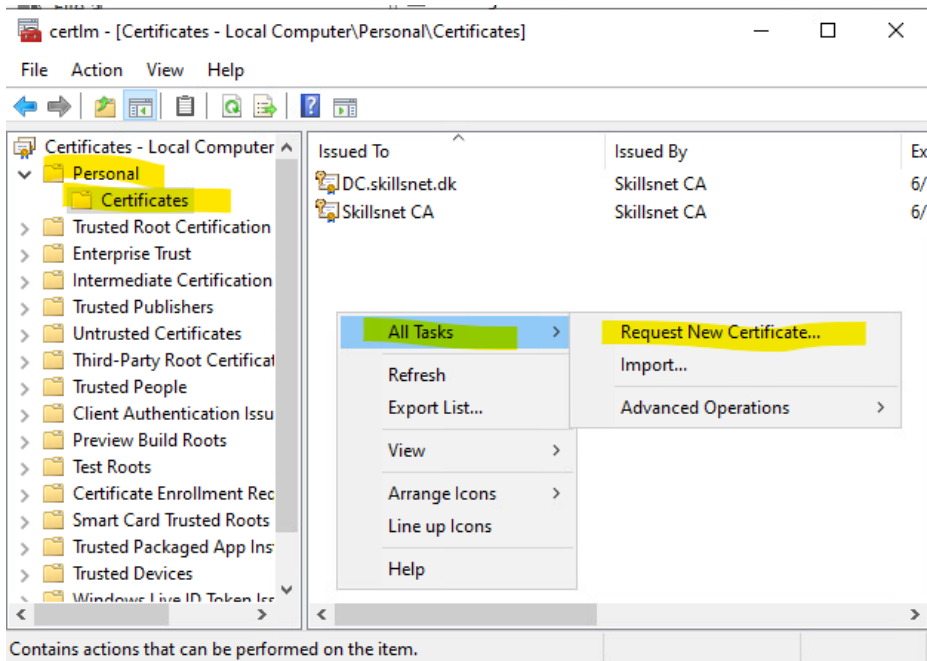






- Create certificates for RTR-CPH and RTR-AAL





Certificate Properties [X]

Subject | General | Extensions | Private Key | Certification Authority | Signature

The subject of a certificate is the user or computer to which the certificate is issued. You can enter information about the types of subject name and alternative name values that can be used in a certificate.

Subject of certificate
The user or computer that is receiving the certificate

Subject name:

Type: **Common name** [v]
Value: **RTR-CPH**

Alternative name:

Type: **Directory name** [v]
Value:

[Add >] [< Remove]

[OK] [Cancel] [Apply]

Certificate Properties [X]

Subject | General | Extensions | Private Key | Certification Authority | Signature

The subject of a certificate is the user or computer to which the certificate is issued. You can enter information about the types of subject name and alternative name values that can be used in a certificate.

Subject of certificate
The user or computer that is receiving the certificate

Subject name:

Type: **Common name** [v]
Value:

Alternative name:

Type:

CN=RTR-CPH

[Add >] [< Remove]

Certificate Enrollment

Request Certificates

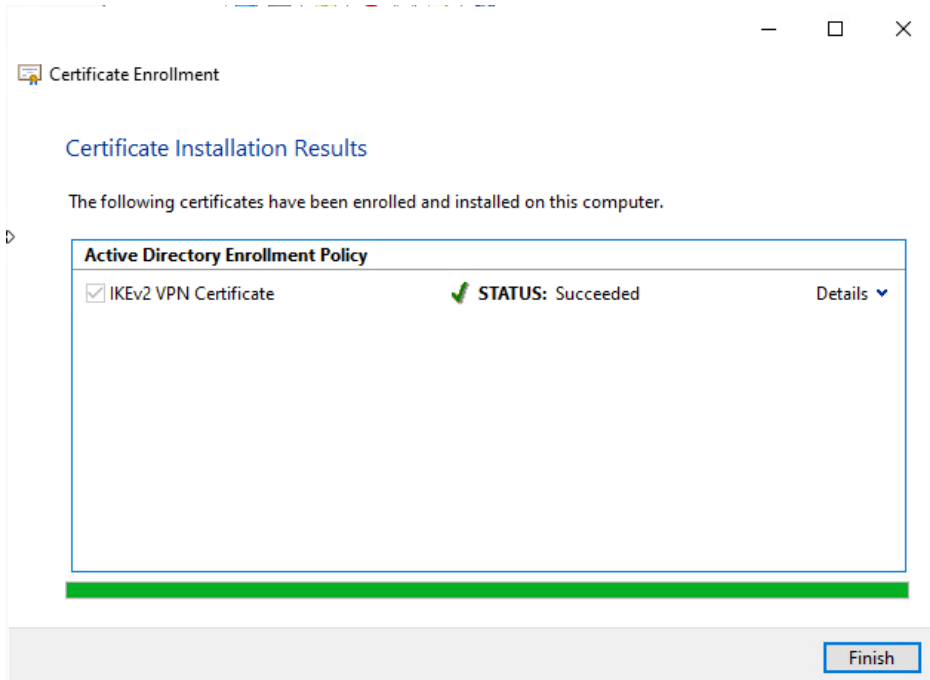
You can request the following types of certificates. Select the certificates you want to request, and then click Enroll.

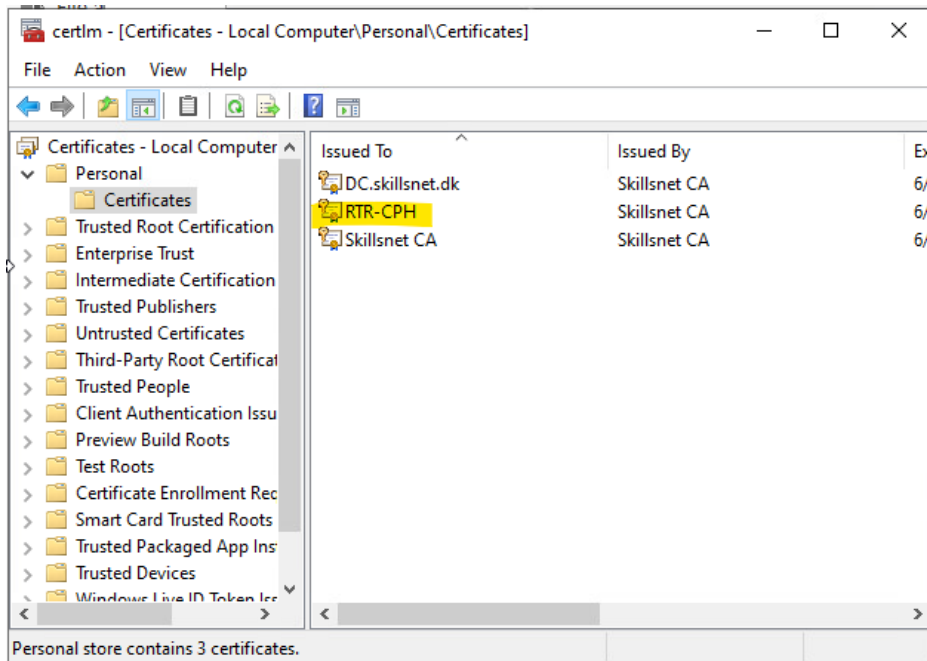
☐ Domain Controller Authentication	STATUS: Available	Details ▾										
☒ IKEv2 VPN Certificate	STATUS: Available	Details ▲										
The following options describe the uses and validity period that apply to this type of certificate: <table> <tr> <td>Key usage:</td> <td>Digital signature</td> </tr> <tr> <td></td> <td>Key encipherment</td> </tr> <tr> <td>Application policies:</td> <td>Server Authentication</td> </tr> <tr> <td></td> <td>IP security IKE intermediate</td> </tr> <tr> <td>Validity period (days):</td> <td>730</td> </tr> </table>			Key usage:	Digital signature		Key encipherment	Application policies:	Server Authentication		IP security IKE intermediate	Validity period (days):	730
Key usage:	Digital signature											
	Key encipherment											
Application policies:	Server Authentication											
	IP security IKE intermediate											
Validity period (days):	730											
☐ IPSec	STATUS: Available	Details ▾										

☐ Show all templates

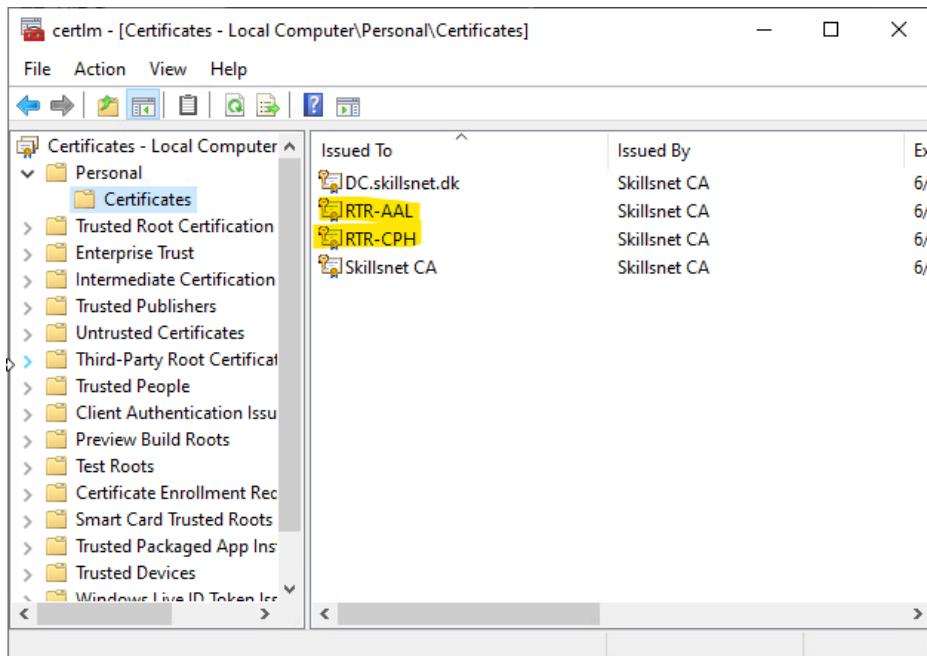
Enroll

Cancel

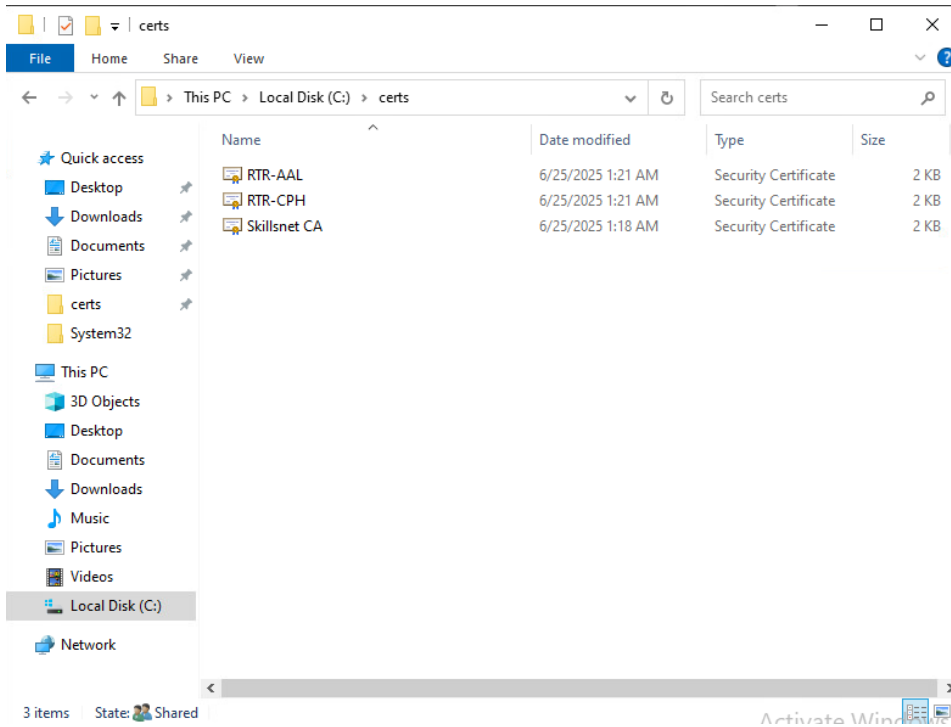




Repeat the process for RTR-AAL

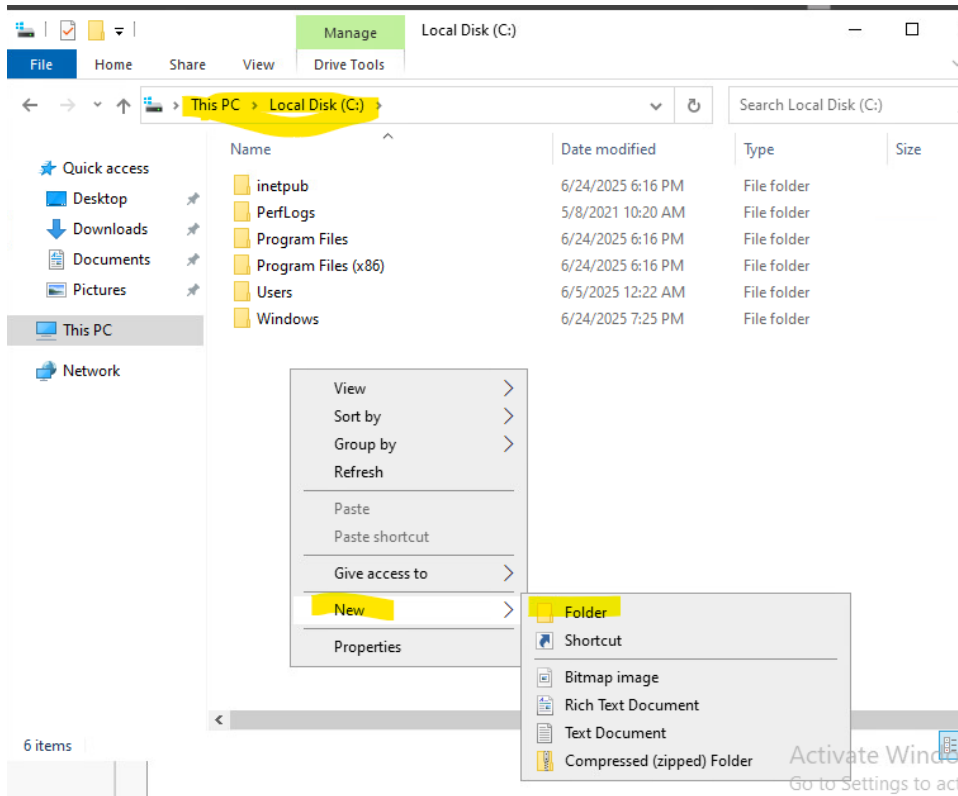


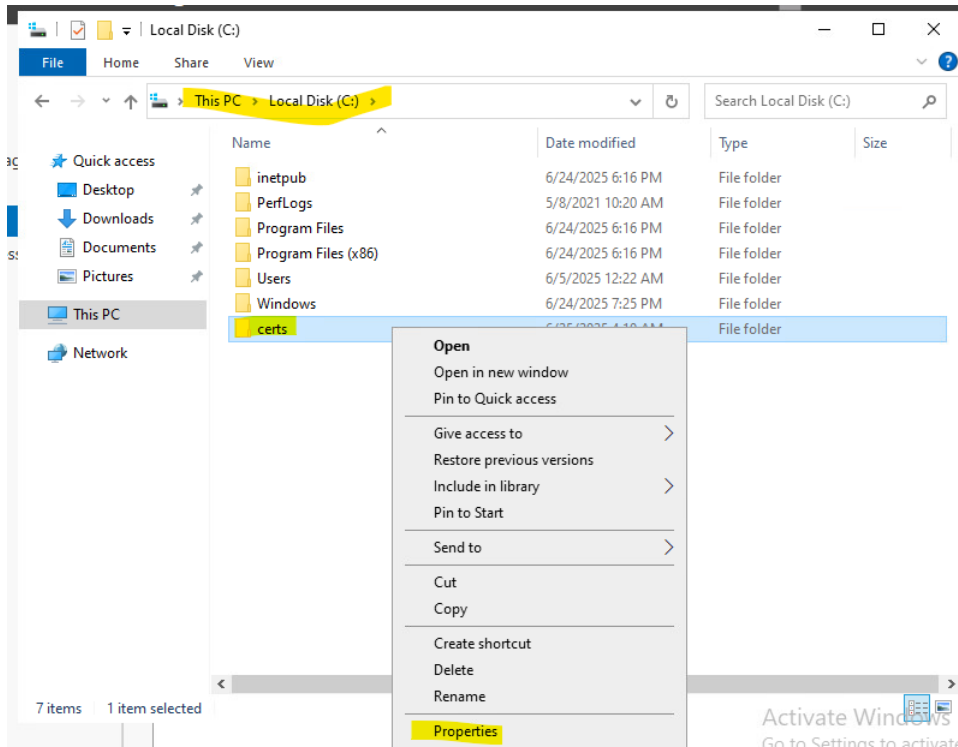
- Export certificates to shared folder (see how to in DC)

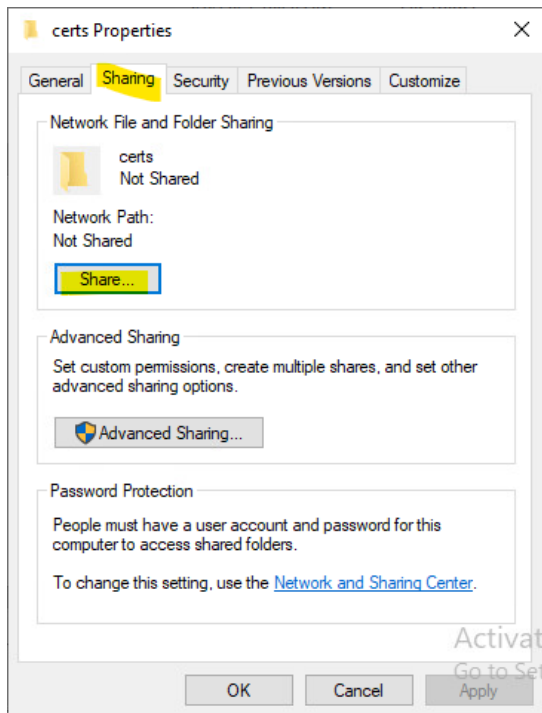


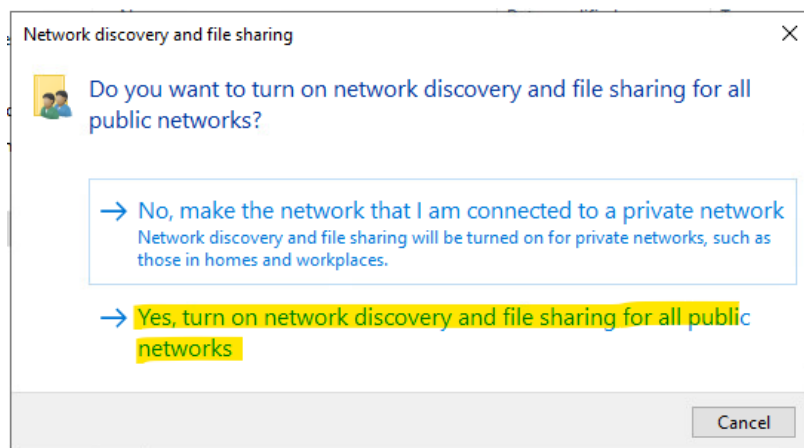
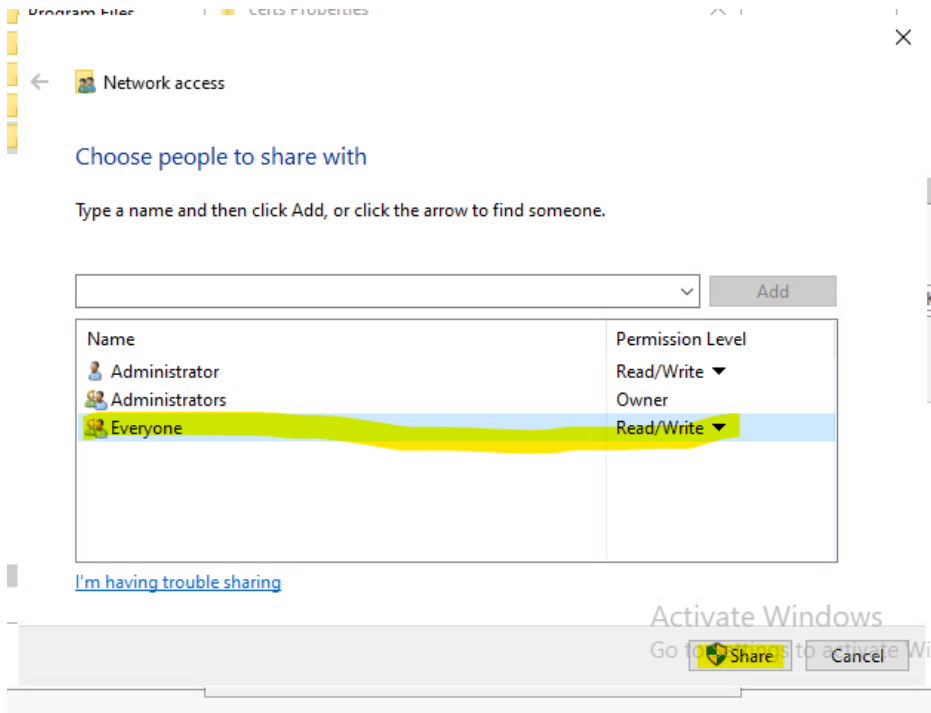
- Back to RTR-CPH

Create a shared folder to transfer certs and other things to Branch:











← Network access

Your folder is shared.

You can [e-mail](#) someone links to these shared items, or [copy](#) and paste the links into another app.

Individual Items

certs
\\RTR-CPH\certs

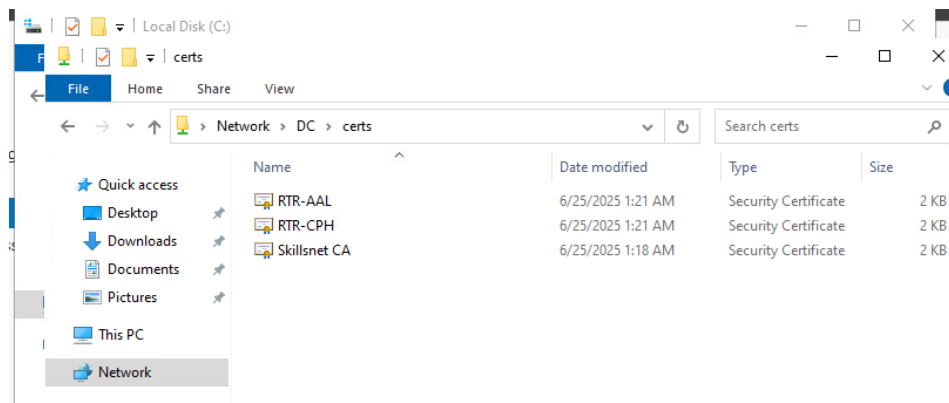
[Show me all the network shares on this computer.](#)

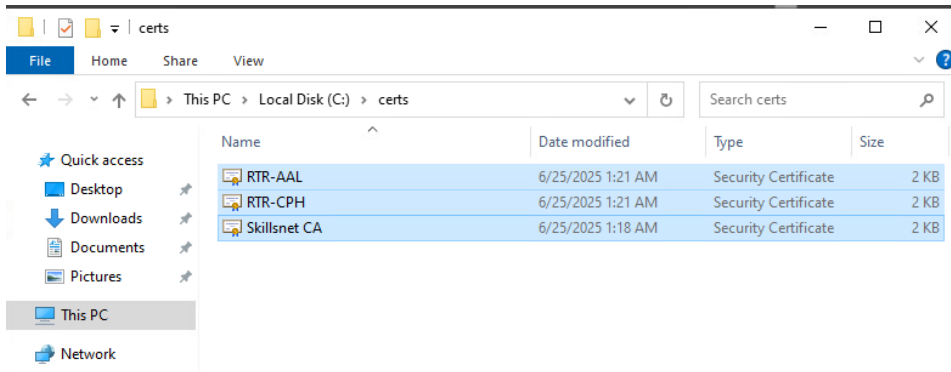
Activate Windows

Go to Settings to activate Wi

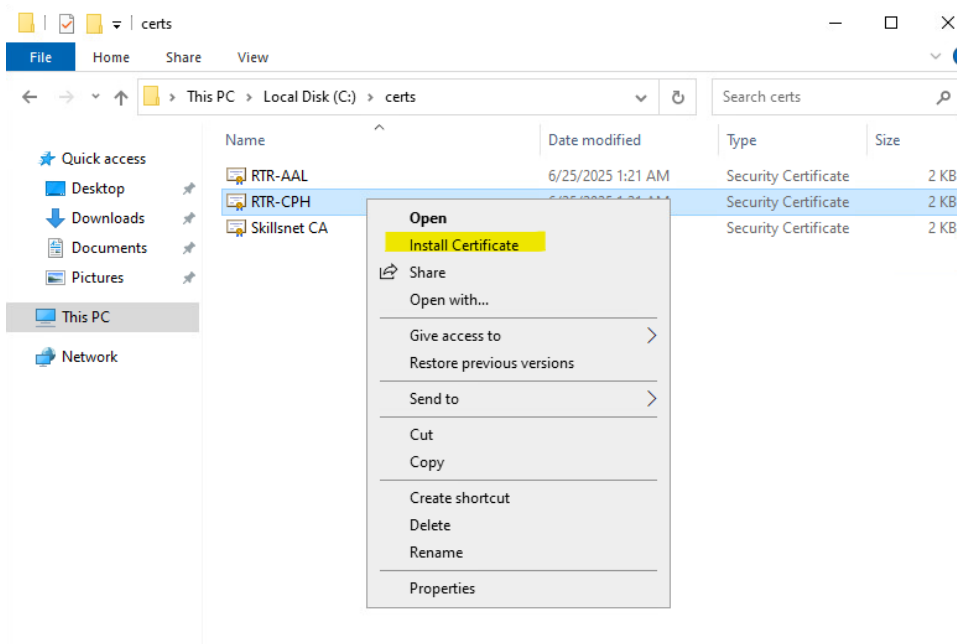
Done

Access to the shared folder in DC and copy here RTR-CPH.cer, RTR-AAL.cer and Skillsnet CA.cer:





Install RTR-CPH certificate:



 Certificate Import Wizard

Welcome to the Certificate Import Wizard

This wizard helps you copy certificates, certificate trust lists, and certificate revocation lists from your disk to a certificate store.

A certificate, which is issued by a certification authority, is a confirmation of your identity and contains information used to protect data or to establish secure network connections. A certificate store is the system area where certificates are kept.

Store Location

☐ Current User

☒ Local Machine

To continue, click Next.



Cancel



← Certificate Import Wizard

Certificate Store

Certificate stores are system areas where certificates are kept.

Windows can automatically select a certificate store, or you can specify a location for the certificate.

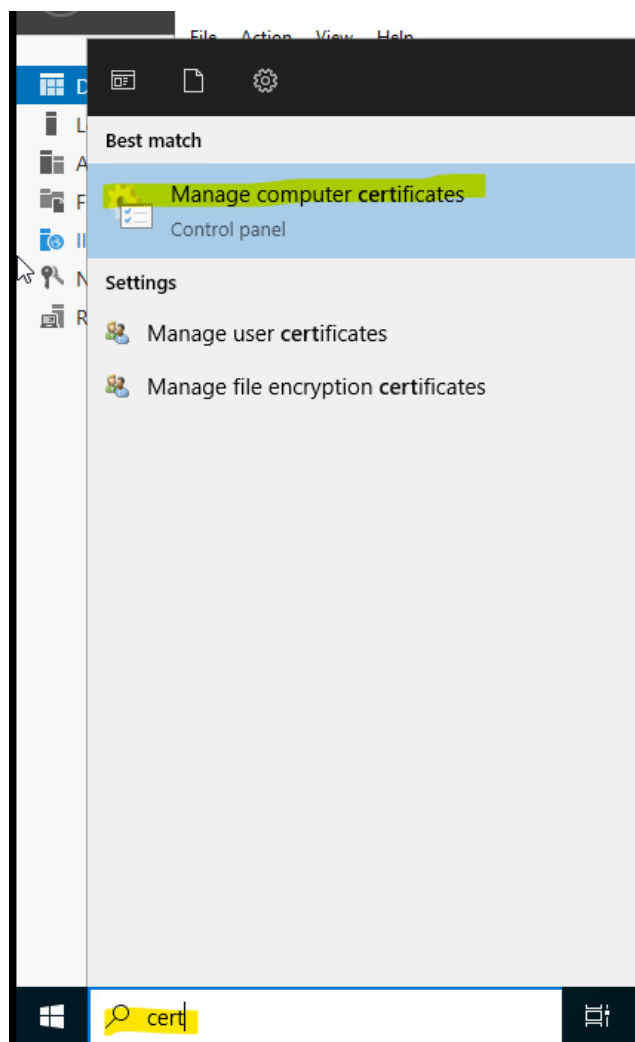
- ☒ Automatically select the certificate store based on the type of certificate
- ☐ Place all certificates in the following store

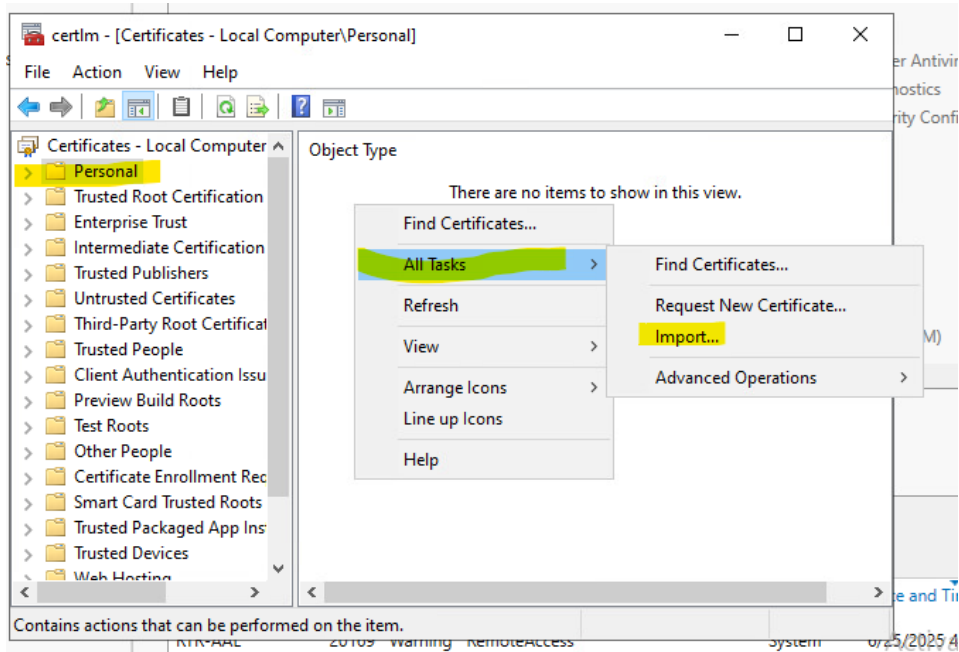
Certificate store:

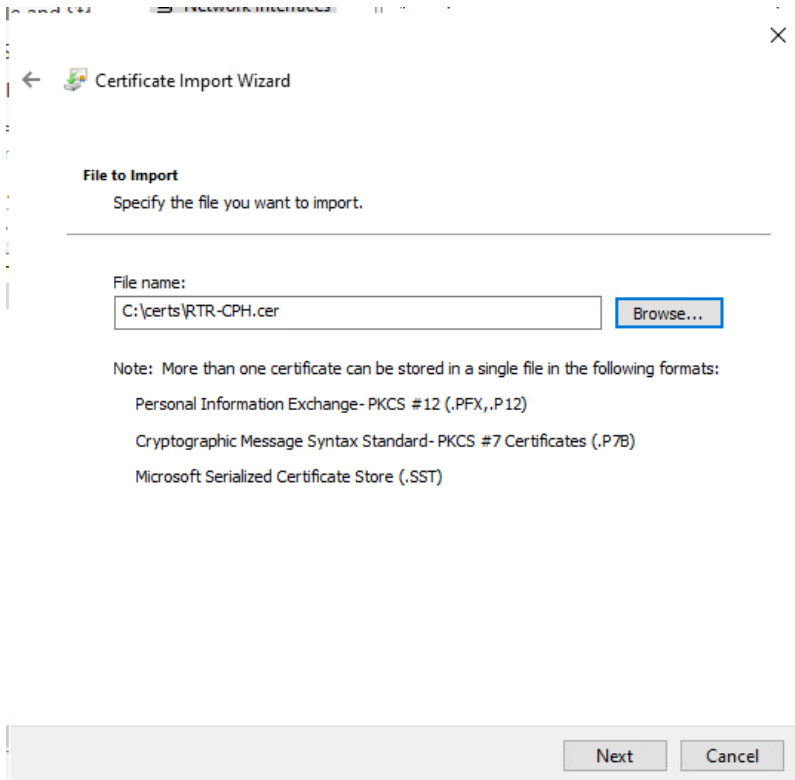
Browse...

Next

Cancel







The image shows a Windows 'Certificate Import Wizard' dialog box. At the top, there is a title bar with a back arrow, a small icon, and the text 'Certificate Import Wizard'. Below the title bar, the section 'File to Import' is displayed with the instruction 'Specify the file you want to import.' A horizontal line separates this section from the input area. In the input area, there is a label 'File name:' followed by a text box containing the path 'C:\certs\RTR-CPH.cer'. To the right of the text box is a 'Browse...' button. Below the input area, a 'Note' states: 'More than one certificate can be stored in a single file in the following formats:'. This is followed by a list of three formats: 'Personal Information Exchange- PKCS #12 (.PFX,.P12)', 'Cryptographic Message Syntax Standard- PKCS #7 Certificates (.P7B)', and 'Microsoft Serialized Certificate Store (.SST)'. At the bottom of the dialog, there are two buttons: 'Next' and 'Cancel'.

← Certificate Import Wizard

File to Import
Specify the file you want to import.

File name:
 [Browse...](#)

Note: More than one certificate can be stored in a single file in the following formats:

- Personal Information Exchange- PKCS #12 (.PFX,.P12)
- Cryptographic Message Syntax Standard- PKCS #7 Certificates (.P7B)
- Microsoft Serialized Certificate Store (.SST)

[Next](#) [Cancel](#)



Certificate Store

Certificate stores are system areas where certificates are kept.

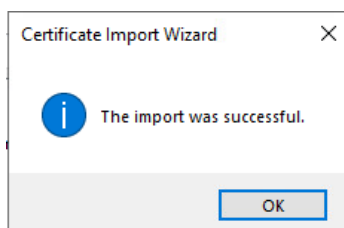
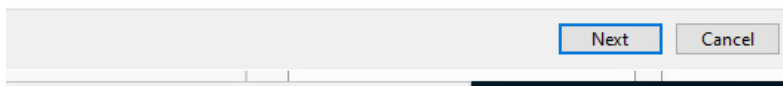
Windows can automatically select a certificate store, or you can specify a location for the certificate.

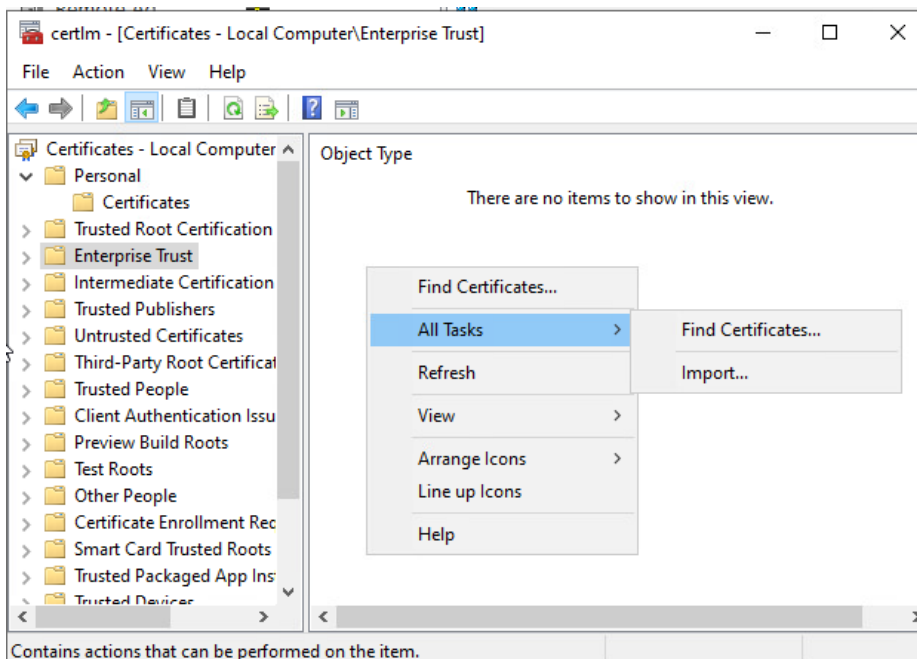
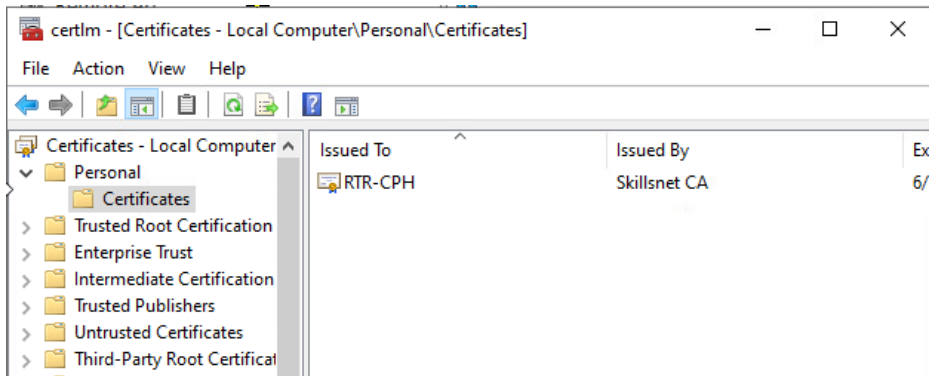
- ☐ Automatically select the certificate store based on the type of certificate
- ☒ Place all certificates in the following store

Certificate store:

Personal

Browse...





Contains actions that can be performed on the item.

←

Certificate Import Wizard

×

File to Import
Specify the file you want to import.

File name:

C:\certs\Skillsnet CA.cer

Browse...

Note: More than one certificate can be stored in a single file in the following formats:
Personal Information Exchange- PKCS #12 (.PFX,.P12)
Cryptographic Message Syntax Standard- PKCS #7 Certificates (.P7B)
Microsoft Serialized Certificate Store (.SST)

Next

Cancel

← Certificate Import Wizard

Certificate Store

Certificate stores are system areas where certificates are kept.

Windows can automatically select a certificate store, or you can specify a location for the certificate.

- ☐ Automatically select the certificate store based on the type of certificate
- ☒ Place all certificates in the following store

Certificate store:

Enterprise Trust

Browse...

Next

Cancel

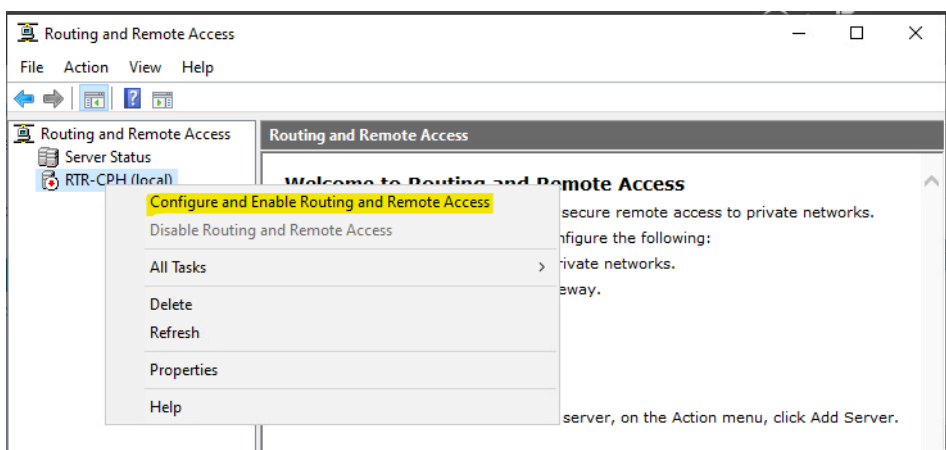
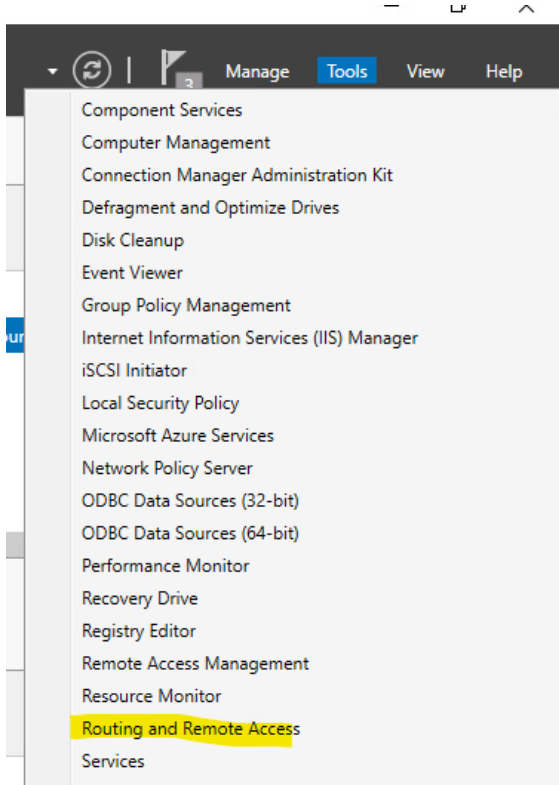
Certificate Import Wizard



The import was successful.

OK

- Site-to-site VPN configuration:



Routing and Remote Access Server Setup Wizard

Configuration

You can enable any of the following combinations of services, or you can customize this server.

- ☐ Remote access (dial-up or VPN)
Allow remote clients to connect to this server through either a dial-up connection or a secure virtual private network (VPN) Internet connection.
- ☐ Network address translation (NAT)
Allow internal clients to connect to the Internet using one public IP address.
- ☐ Virtual private network (VPN) access and NAT
Allow remote clients to connect to this server through the Internet and local clients to connect to the Internet using a single public IP address.
- ☐ Secure connection between two private networks
Connect this network to a remote network, such as a branch office.
- ☒ Custom configuration
Select any combination of the features available in Routing and Remote Access.

< Back Next > Cancel

Routing and Remote Access

File Action View Help

Routing and Remote Access

- Server Status
- RTR-CPH (local)
 - Network Interfaces
 - Remote Access Logging
 - IPv4
 - General
 - Static Routes
 - IGMP
 - NAT
 - IPv6

Network Interfaces

LAN and Demand Dial Interfaces	Type	Status	Connection State
Loopback	Loopback	Enabled	Connected
Internal	Internal	Enabled	Connected
Ethernet2	Dedicated	Enabled	Connected
Ethernet1	Dedicated	Enabled	Connected
Ethernet0	Dedicated	Enabled	Connected

New Demand-dial Interface...

Refresh

Export List...

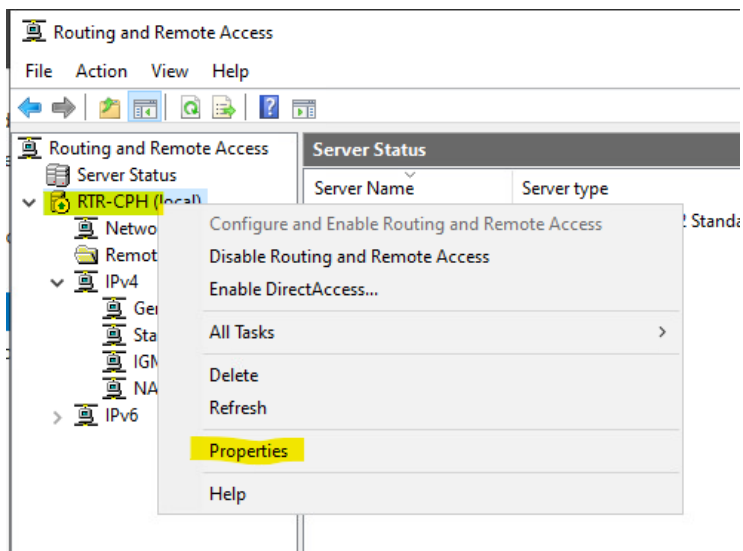
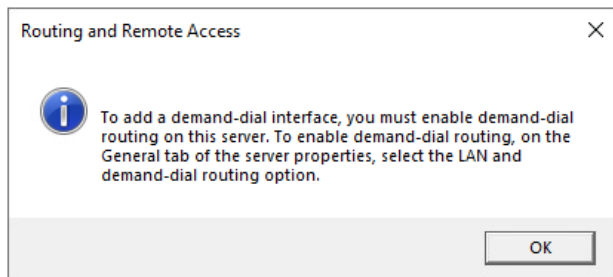
View >

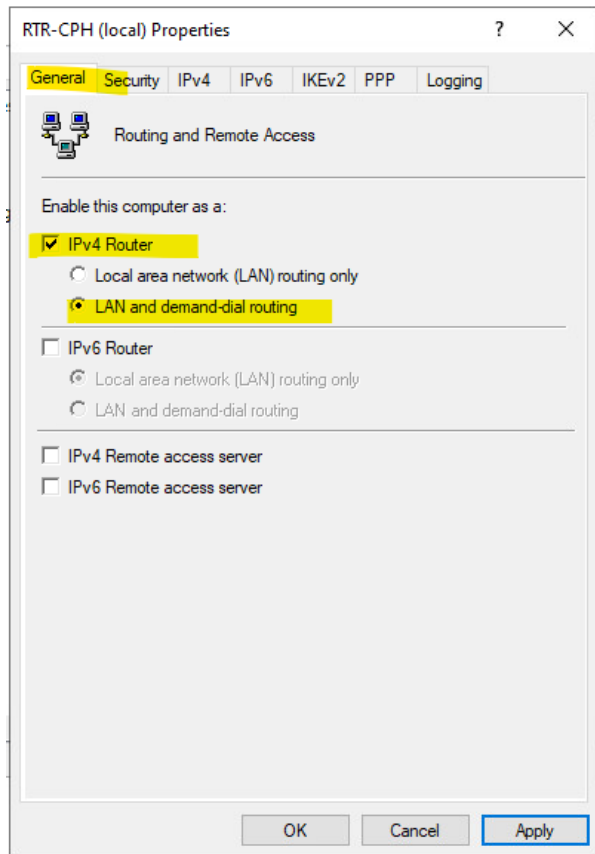
Arrange Icons >

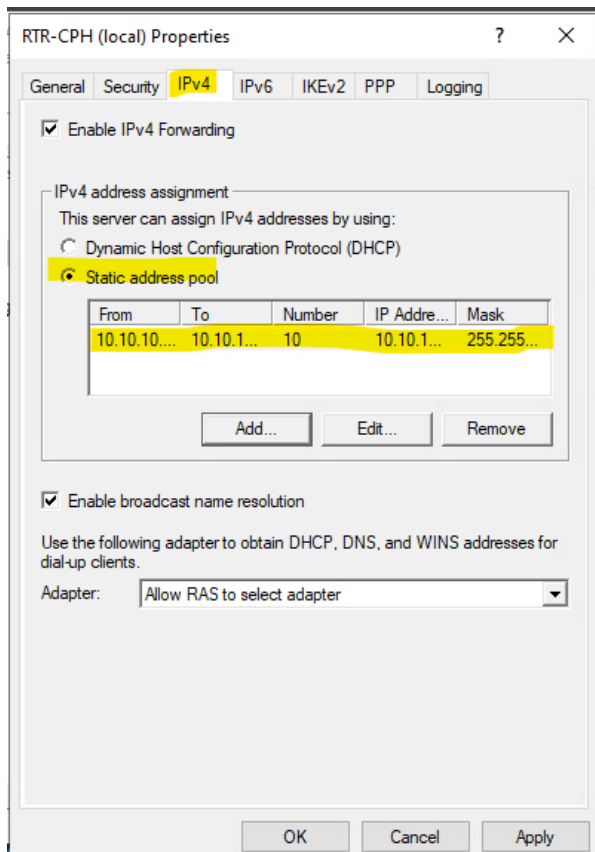
Line up Icons

Help

Create a new demand dial interface

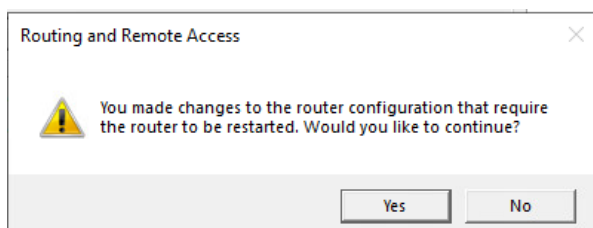




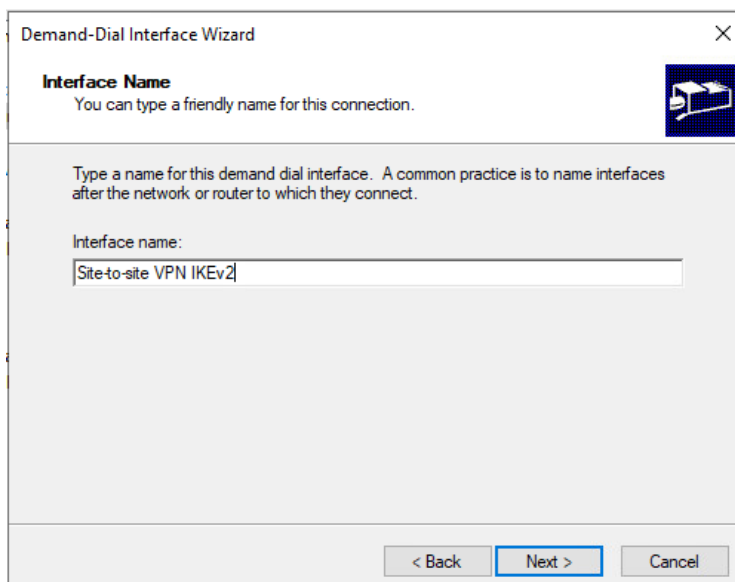
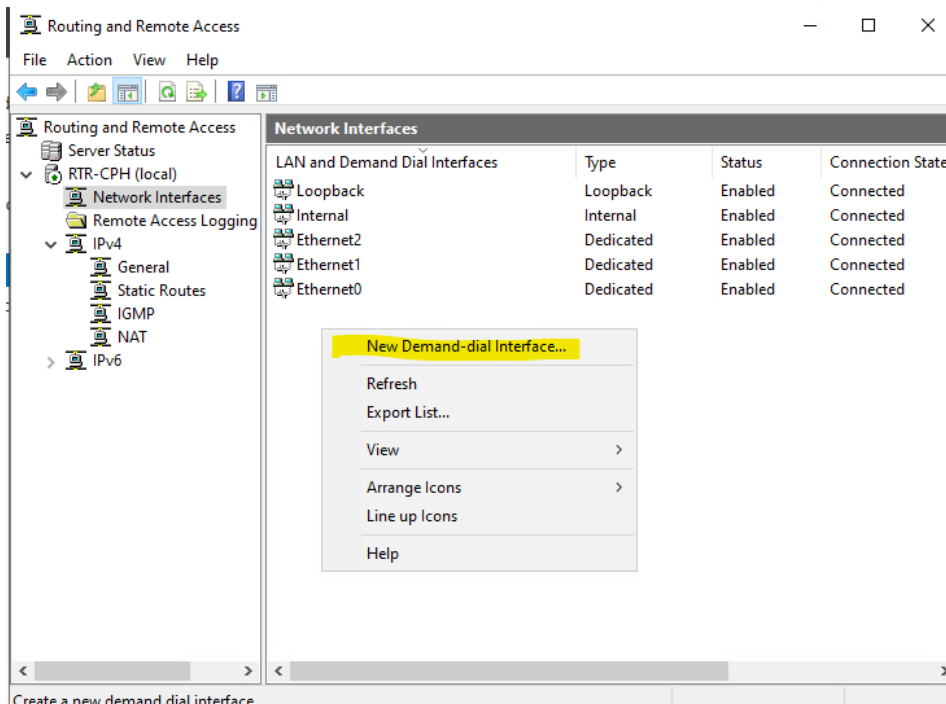


IP range: 10.10.10.11-10.10.10.20 (10 IPs)

Configuration only necessary in one extreme (for example, RTR-CPH)



Maybe you must restart the computer.



Demand-Dial Interface Wizard

Connection Type
Select the type of demand-dial interface you want to create.

☐ Connect using a modem, ISDN adapter, or other device

☒ Connect using virtual private networking (VPN)

☐ Connect using PPP over Ethernet (PPPoE)

< Back Next > Cancel

Demand-Dial Interface Wizard

VPN Type
Select the type of VPN connection you want to create.

☐ Automatic selection

☐ Point to Point Tunneling Protocol (PPTP)

☐ Layer 2 Tunneling Protocol (L2TP)

☒ IKEv2

< Back Next > Cancel

Demand-Dial Interface Wizard

Destination Address
What is the name or address of the remote router?

Enter the name or IP address of the router you are connecting to.

Host name or IP address (such as microsoft.com or 157.54.0.1 or 3ffe:1234::1111):

198.51.100.12

< Back Next > Cancel

Demand-Dial Interface Wizard

Protocols and Security
Select transports and security options for this connection.

Select all that apply:

- ☒ Route IP packets on this interface.
- ☐ Add a user account so a remote router can dial in
- ☐ Send a plain-text password if that is the only way to connect
- ☐ Use scripting to complete the connection with the remote router

< Back Next > Cancel

Demand-Dial Interface Wizard

Static Routes for Remote Networks

A static route is a manually defined, permanent route between two networks.

To activate this demand-dial connection, you must add a static route to the network. Specify the IP address of the remote networks this network will communicate with.

Static Routes:

Destination	Network Mask/Prefix length	Metric

Static Route

☒ Remote Network Support using IPv4

Destination:

 Network Mask:

 Metric:

☐ Remote Network Support using IPv6

Destination:

 Prefix Length:

 Metric:

Demand-Dial Interface Wizard
 ✕

Static Routes for Remote Networks


A static route is a manually defined, permanent route between two networks.

To activate this demand-dial connection, you must add a static route to the network. Specify the IP address of the remote networks this network will communicate with.

Static Routes:

Destination	Network Mask/Prefix length	Metric
10.2.1.0	255.255.255.0	100

Demand-Dial Interface Wizard
 ✕

Dial-Out Credentials


Supply the user name and password to be used when connecting to the remote router.

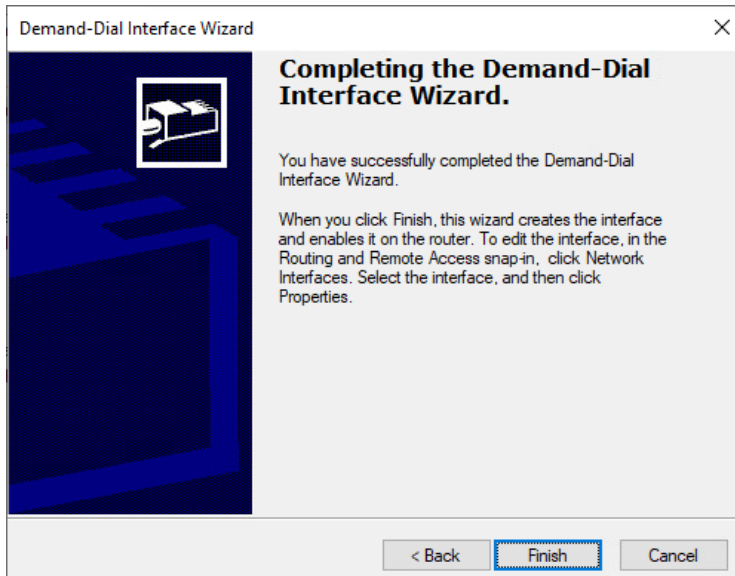
You need to set the dial out credentials that this interface will use when connecting to the remote router. These credentials must match the dial in credentials configured on the remote router.

User name:

Domain:

Password:

Confirm password:



Certificates do not seem to work properly. Use PSK instead:

Routing and Remote Access

File Action View Help

Routing and Remote Access

- Server Status
- ▼ RTR-CPH (local)
 - Network Interfaces
 - Ports
 - Remote Access Logging
 - ▼ IPv4
 - General
 - Static Routes
 - IGMP
 - NAT
 - ▼ IPv6
 - General
 - Static Routes

Network Interfaces

LAN and Demand Dial Interfaces	Type	Status	Connection State
Loopback	Loopback	Enabled	Connected
Internal	Internal	Enabled	Connected
Ethernet2	Dedicated	Enabled	Connected
Ethernet1	Dedicated	Enabled	Connected
Ethernet0	Dedicated	Enabled	Connected
Site-to-site VPN IKEv2	Dedicated	Enabled	Connected

Opens the properties dialog box for the current selection.

IP Address	Subnet Mask	Link Status	MTU
255.255.255.255	255.255.255.255	On-link	10
255.255.255.255	255.255.255.255	On-link	10

Set Credentials...

Connect

Disconnect

Enable

Disable

Unreachability Reason...

Set IP Demand-dial Filters

Set IPv6 Demand-dial Filters

Dial-out Hours

Delete

Refresh

Properties

Help

Site-to-site VPN IKEv2 Properties

General Options Security Networking

Type of VPN:
IKEv2

Advanced Settings

Data encryption:
Require encryption (disconnect if server declines)

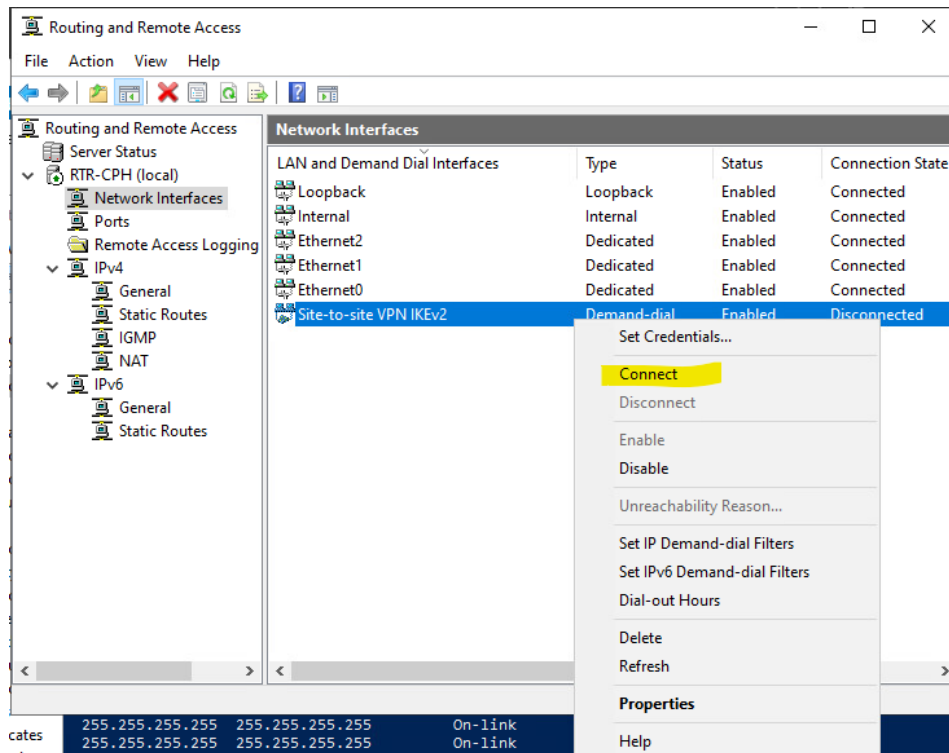
Authentication

☐ Use Extensible Authentication Protocol (EAP)
Properties

☐ Use machine certificates
☐ Verify the Name and Usage attributes of the server's certificate

☒ Use preshared key for authentication
Key: *****

OK Cancel



The screenshot shows the 'Routing and Remote Access' console window. The left pane displays the tree structure with 'Network Interfaces' selected. The right pane shows a table of network interfaces. A context menu is open for the 'Site-to-site VPN IKEv2' interface, with the 'Connect' option highlighted.

Network Interfaces			
Interface	Type	Status	Connection State
Loopback	Loopback	Enabled	Connected
Internal	Internal	Enabled	Connected
Ethernet2	Dedicated	Enabled	Connected
Ethernet1	Dedicated	Enabled	Connected
Ethernet0	Dedicated	Enabled	Connected
Site-to-site VPN IKEv2	Demand-dial	Enabled	Disconnected

Context Menu Options:

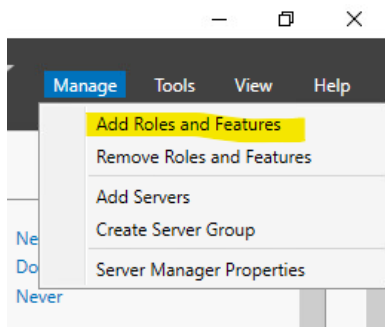
- Set Credentials...
- Connect**
- Disconnect
- Enable
- Disable
- Unreachability Reason...
- Set IP Demand-dial Filters
- Set IPv6 Demand-dial Filters
- Dial-out Hours
- Delete
- Refresh
- Properties
- Help

```

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
-----
10.1.1.0                    255.255.255.0    On-link          10.1.1.254       281
10.1.1.254                  255.255.255.255  On-link          10.1.1.254       281
10.1.1.255                  255.255.255.255  On-link          10.1.1.254       281
10.1.2.0                    255.255.255.0    On-link          10.1.2.254       281
10.1.2.254                  255.255.255.255  On-link          10.1.2.254       281
10.1.2.255                  255.255.255.255  On-link          10.1.2.254       281
10.2.1.0                    255.255.255.0    On-link          10.10.10.19      125
10.2.1.255                  255.255.255.255  On-link          10.10.10.19      281
10.10.10.11                 255.255.255.255  On-link          10.10.10.11      331
10.10.10.18                 255.255.255.255  On-link          10.10.10.19      26
10.10.10.19                 255.255.255.255  On-link          10.10.10.19      281
127.0.0.0                   255.0.0.0         On-link          127.0.0.1        331
127.0.0.1                   255.255.255.255  On-link          127.0.0.1        331
127.255.255.255             255.255.255.255  On-link          127.0.0.1        331
198.51.100.0                255.255.255.0     On-link          198.51.100.11    281
198.51.100.11               255.255.255.255  On-link          198.51.100.11    281
198.51.100.255              255.255.255.255  On-link          198.51.100.11    281
224.0.0.0                   240.0.0.0         On-link          127.0.0.1        331
224.0.0.0                   240.0.0.0         On-link          198.51.100.11    281
224.0.0.0                   240.0.0.0         On-link          10.1.2.254       281
224.0.0.0                   240.0.0.0         On-link          10.1.1.254       281
224.0.0.0                   240.0.0.0         On-link          10.10.10.11      331
224.0.0.0                   240.0.0.0         On-link          10.10.10.19      281
255.255.255.255             255.255.255.255  On-link          127.0.0.1        331
255.255.255.255             255.255.255.255  On-link          198.51.100.11    281
255.255.255.255             255.255.255.255  On-link          10.1.2.254       281
255.255.255.255             255.255.255.255  On-link          10.1.1.254       281
255.255.255.255             255.255.255.255  On-link          10.10.10.11      331
255.255.255.255             255.255.255.255  On-link          10.10.10.19      281
=====
Persistent Routes:
  
```


RTR-CPH - Task2: RRAS FW

2. Configure RRAS firewall:
 - 2.1. Block all outbound traffic to TCP port 8080
 - 2.2. Block all outbound connections to WinRM



Add Roles and Features Wizard

Select server roles

DESTINATION SERVER
RTR-CPH

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Remote Access
Role Services
Confirmation
Results

Select one or more roles to install on the selected server.

Roles	Description
☐ Active Directory Certificate Services	Remote Access provides seamless connectivity through DirectAccess, VPN, and Web Application Proxy. DirectAccess provides an Always On and Always Managed experience. RAS provides traditional VPN services, including site-to-site (branch-office or cloud-based) connectivity. Web Application Proxy enables the publishing of selected HTTP- and HTTPS-based applications from your corporate network to client devices outside of the corporate network. Routing provides traditional routing capabilities, including NAT and other connectivity options. RAS and Routing can be deployed in single-tenant or multi-tenant mode.
☐ Active Directory Domain Services	
☐ Active Directory Federation Services	
☐ Active Directory Lightweight Directory Services	
☐ Active Directory Rights Management Services	
☐ Device Health Attestation	
☐ DHCP Server	
☐ DNS Server	
☐ Fax Server	
☒ File and Storage Services (1 of 12 installed)	
☐ Host Guardian Service	
☐ Hyper-V	
☐ Network Policy and Access Services	
☐ Print and Document Services	
☒ Remote Access	
☐ Remote Desktop Services	
☐ Volume Activation Services	
☐ Web Server (IIS)	
☐ Windows Deployment Services	
☐ Windows Server Update Services	

< Previous Next > Install Cancel

Add Roles and Features Wizard

Select role services

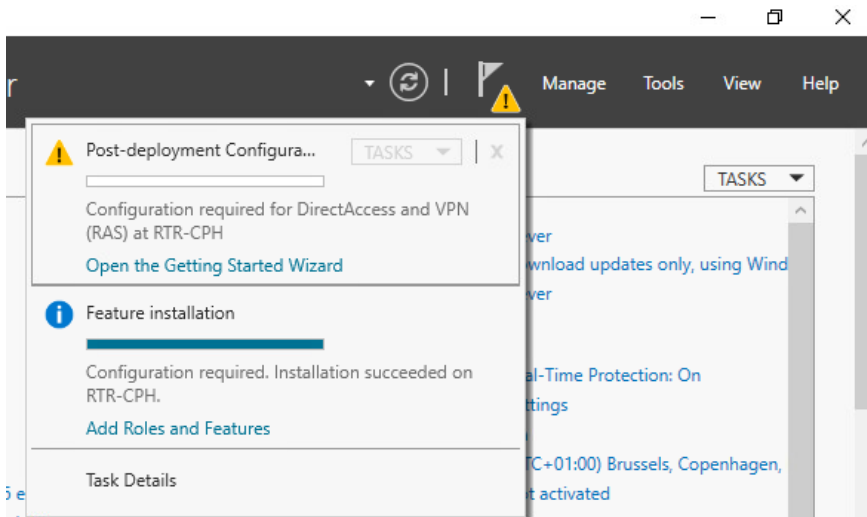
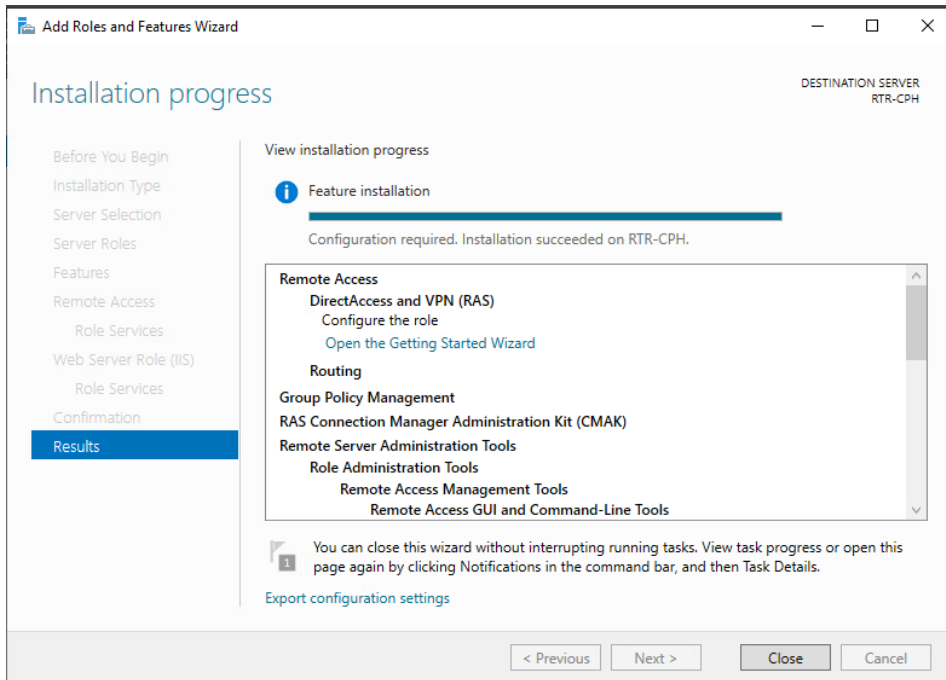
DESTINATION SERVER
RTR-CPH

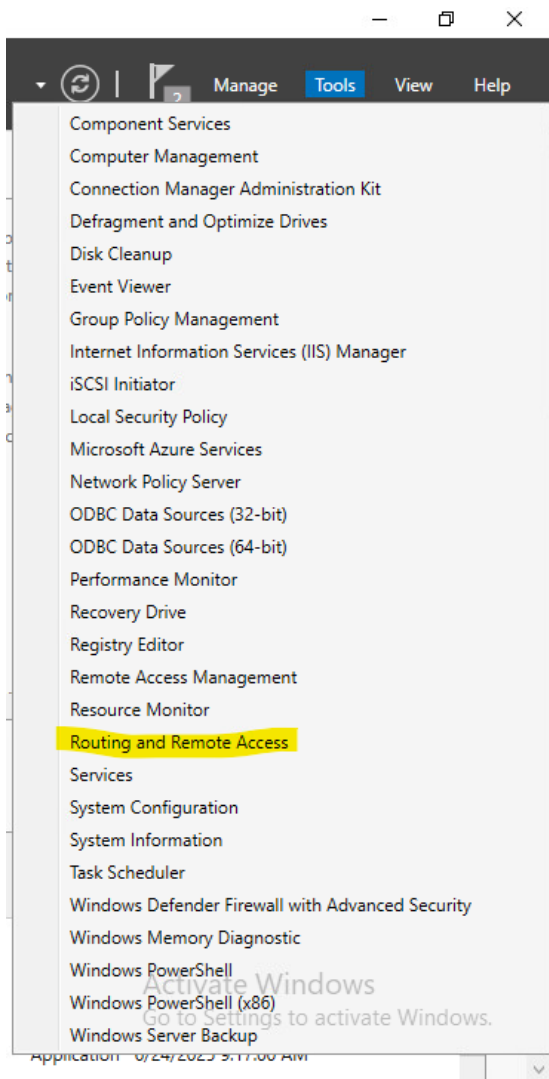
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Remote Access
Role Services
Web Server Role (IIS)
Role Services
Confirmation
Results

Select the role services to install for Remote Access

Role services	Description
☒ DirectAccess and VPN (RAS)	Routing provides support for NAT Routers, LAN Routers running BGP, RIP, and multicast capable routers (IGMP Proxy).
☒ Routing	
☐ Web Application Proxy	

< Previous Next > Install Cancel





RTR-CPH - Task3: Port forwarding

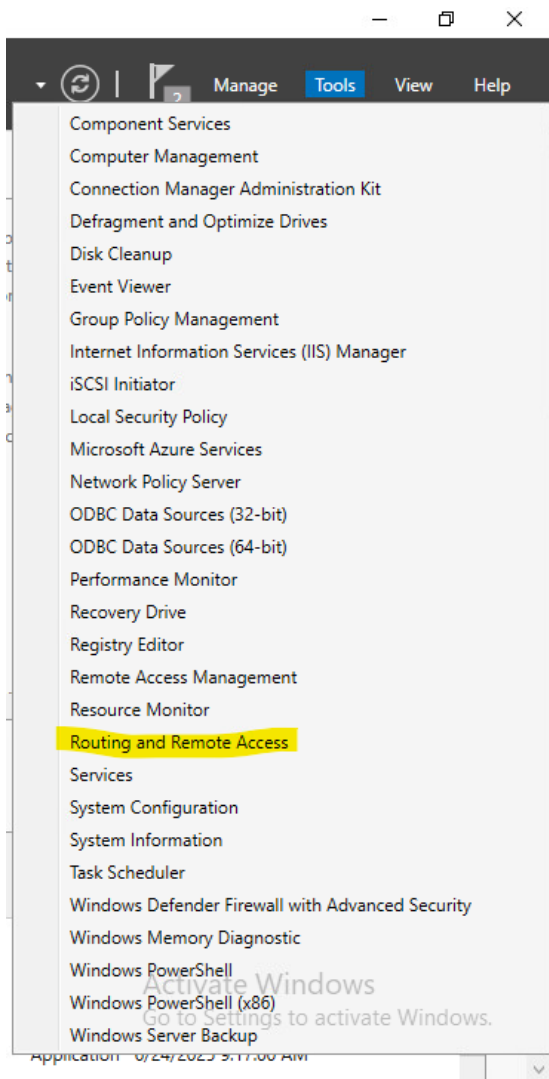
3. Port forward 198.51.100.11:8080 to DEV-SRV port 80

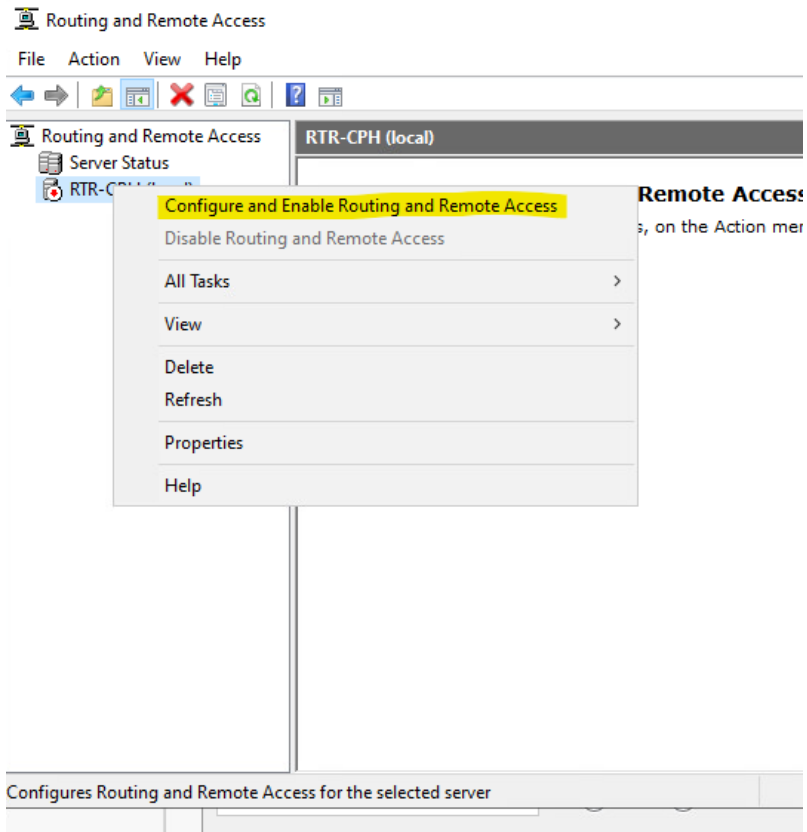
RTR-CPH - Task4: Reverse proxy

4. Configure reverse proxy against www.skillsnet.dk, host it as www.skillspublic.dk

RTR-CPH - Taskr: NAT (masquerade)

5. Configure NAT rules for outbound (external) connectivity





Routing and Remote Access Server Setup Wizard

Configuration

You can enable any of the following combinations of services, or you can customize this server.

- ☐ Remote access (dial-up or VPN)
Allow remote clients to connect to this server through either a dial-up connection or a secure virtual private network (VPN) Internet connection.
- ☒ Network address translation (NAT)
Allow internal clients to connect to the Internet using one public IP address.
- ☐ Virtual private network (VPN) access and NAT
Allow remote clients to connect to this server through the Internet and local clients to connect to the Internet using a single public IP address.
- ☐ Secure connection between two private networks
Connect this network to a remote network, such as a branch office.
- ☐ Custom configuration
Select any combination of the features available in Routing and Remote Access.

< Back

Next >

Cancel

Routing and Remote Access Server Setup Wizard

NAT Internet Connection

You can select an existing interface or create a new demand-dial interface for client computers to connect to the Internet.

- ☒ Use this public interface to connect to the Internet:

Network Interfaces:

Name	Description	IP Address
Ethernet0	Intel(R) 82574L Gigabit...	198.51.100.11
Ethernet1	Intel(R) 82574L Gigabit...	10.1.1.254
Ethernet2	Intel(R) 82574L Gigabit...	10.1.2.254

- ☐ Create a new demand-dial interface to the Internet
A demand-dial interface is activated when a client uses the Internet. Select this option if this server connects with a modem or by using the Point-to-Point Protocol over Ethernet. The Demand-Dial Interface Wizard will start at the end of this wizard.

< Back

Next >

Cancel

Routing and Remote Access Server Setup Wizard

Network Selection

You can select the network that will have shared Internet access.

Select the interface for the network that will have access to the Internet.

Network Interfaces:

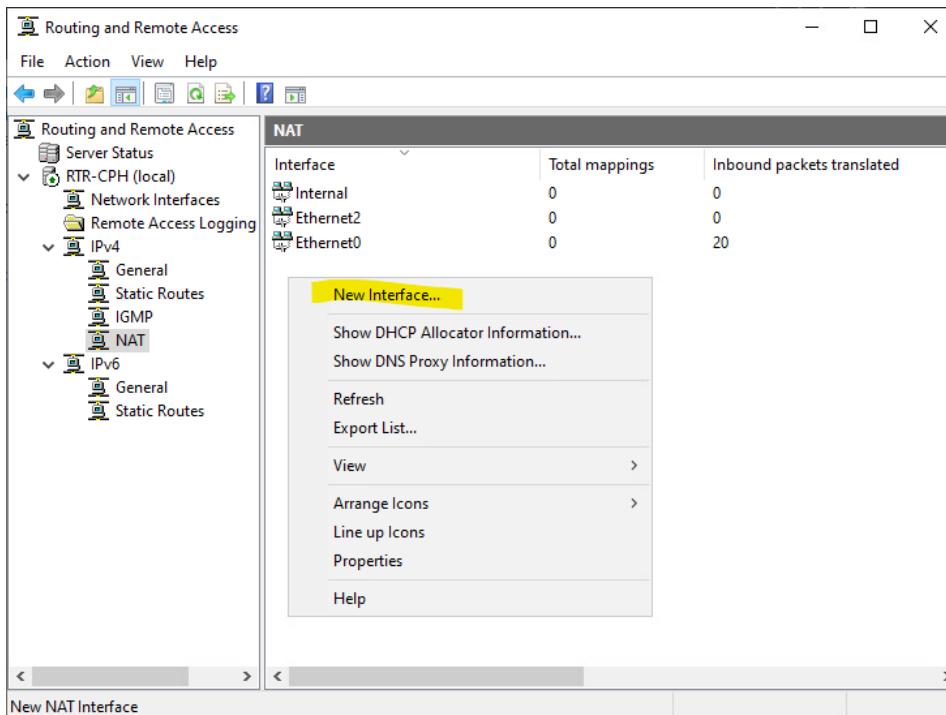
Name	Description	IP Address
Ethernet1	Intel(R) 82574L Gigabit ...	10.1.1.254
Ethernet2	Intel(R) 82574L Gigabit ...	10.1.2.254

If your network has both a NAT server and multiple private interfaces, you should configure DHCP on all private segments.

< Back

Next >

Cancel



Routing and Remote Access

File Action View Help

Routing and Remote Access

- Server Status
- ▼ RTR-CPH (local)
 - Network Interfaces
 - Remote Access Logging
 - ▼ IPv4
 - General
 - Static Routes
 - IGMP
 - NAT**
 - ▼ IPv6
 - General
 - Static Routes

NAT

Interface	Total mappings	Inbound packets translated
Internal	0	0
Ethernet2	0	0
Ethernet0	0	20

New Interface...

Show DHCP Allocator Information...

Show DNS Proxy Information...

Refresh

Export List...

View >

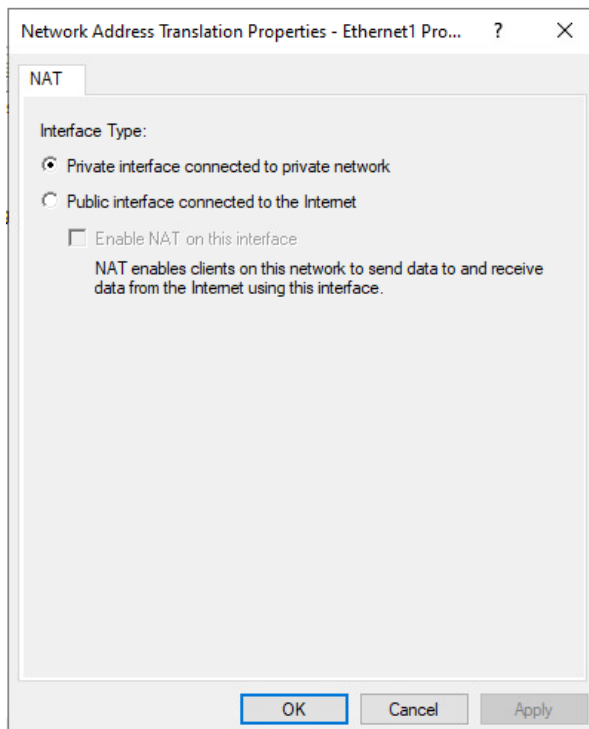
Arrange Icons >

Line up Icons

Properties

Help

New NAT Interface



RTR-AAL

Role: Branch Router

Tasks:

RTR-AAL - Task1: Site-to-site VPN

1. Configure IKEv2 Site-to-Site VPN with RTR-CPH

RTR-AAL - Task2: NAT (masquerade)

2. Configure NAT rules for outbound (external) connectivity

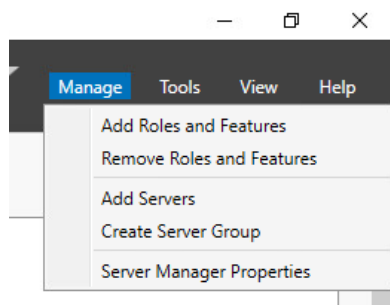
DC

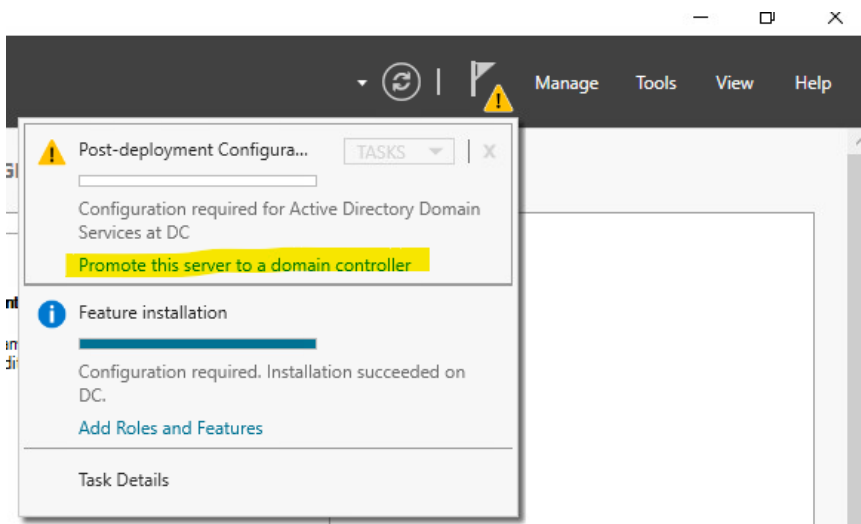
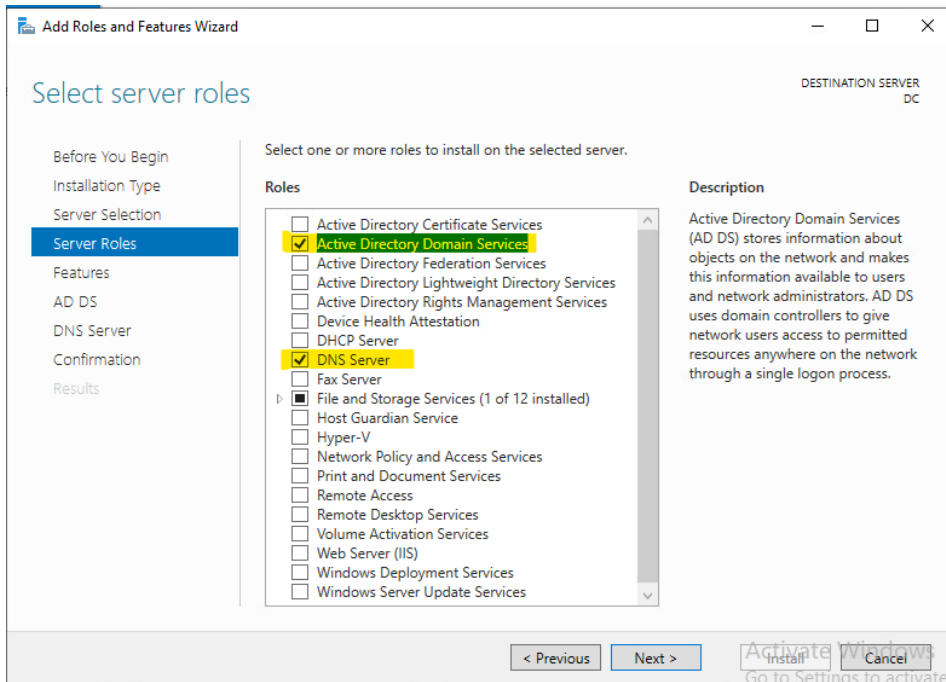
Role: Primary Domain Controller for skillsnet.dk. Provides Enterprise PKI and SAML IdP.

Active Directory & Identity Services

DC AD - Task1: Domain controller

- Don't forget rename host to DC previously.
1. Promote to DC and create domain skillsnet.dk





Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
DC

- Deployment Configuration
- Domain Controller Options
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Select the deployment operation

☐ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
DC

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

< Previous

Next >

Install

Cancel

Active Directory Domain Services Configuration Wizard

DNS Options

TARGET SERVER
DC

 A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found... [Show more](#) 

- Deployment Configuration
- Domain Controller Options
- DNS Options**
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Specify DNS delegation options

☐ Create DNS delegation

[More about DNS delegation](#)

< Previous

Next >

Install

Cancel

Active Directory Domain Services Configuration Wizard

Additional Options

TARGET SERVER
DC

- Deployment Configuration
- Domain Controller Options
- DNS Options
- Additional Options**
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name:

[More about additional options](#)

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

Review Options

- Deployment Configuration
- Domain Controller Options
 - DNS Options
- Additional Options
- Paths
- Review Options**
- Prerequisites Check
- Installation
- Results

Review your selections:

Configure this server as the first Active Directory domain controller in a new forest.

The new domain name is "skillsnet.dk". This is also the name of the new forest.

The NetBIOS name of the domain: SKILLSNET

Forest Functional Level: Windows Server 2016

Domain Functional Level: Windows Server 2016

Additional Options:

Global catalog: Yes

DNS Server: Yes

Create DNS Delegation: No

These settings can be exported to a Windows PowerShell script to automate additional installations

[View script](#)

[More about installation options](#)

< Previous

Next >

Install

Cancel

Prerequisites Check

TARGET SERVER
DC

✓ All prerequisite checks passed successfully. Click 'Install' to begin installation. [Show more](#) ×

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Prerequisites need to be validated before Active Directory Domain Services is installed on this computer

[Rerun prerequisites check](#)

⬆ View results

⚠ Windows Server 2022 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.

For more information about this setting, see Knowledge Base article 942564 (<http://go.microsoft.com/fwlink/?LinkId=104751>).

⚠ A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found or it does not run Windows DNS server. If you are integrating with an existing DNS infrastructure, you should manually create a delegation to this DNS server in the parent zone to ensure reliable name resolution from outside the domain "skillsnet.dk". Otherwise, no action is required.

⚠ If you click Install, the server automatically reboots at the end of the promotion operation.

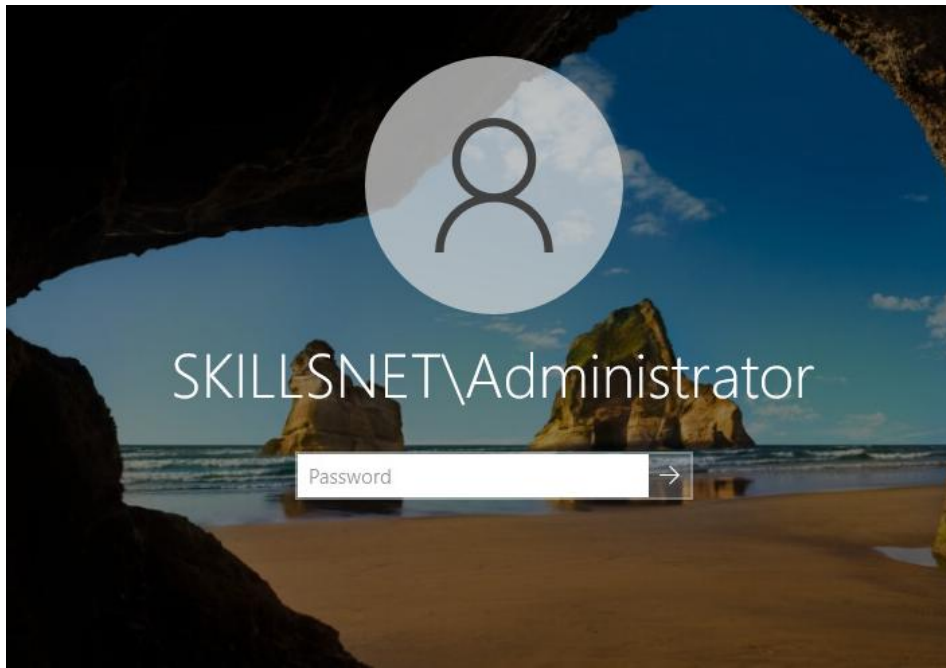
[More about prerequisites](#)

< Previous

Next >

Install

Cancel



- Join machines to domain.

Notes:

- RODC needs the VPN active to join domain.

Example with GUI: SRV1

Server Manager

Server Manager ▸ Local Server

Dashboard
Local Server
All Servers
File and Storage Services ▸

PROPERTIES
For SRV1

Computer name	SRV1	Last installed update
Workgroup	WORKGROUP	Windows Update
		Last checked for up
Microsoft Defender Firewall	Public: On	Microsoft Defender
Remote management	Enabled	Feedback & Diagno
Remote Desktop	Disabled	IE Enhanced Security
NIC Teaming	Disabled	Time zone
Ethernet0	10.1.1.2, IPv6 enabled	Product ID

System Properties

Computer Name Hardware Advanced Remote

Windows uses the following information to identify your computer on the network.

Computer description:

For example: "IIS Production Server" or "Accounting Server".

Full computer name: SRV1

Workgroup: WORKGROUP

To rename this computer or change its domain or workgroup, click Change.

Change...

OK Cancel Apply

Computer Name/Domain Changes

You can change the name and the membership of this computer. Changes might affect access to network resources.

Computer name:
SRV1

Full computer name:
SRV1

More...

Member of

☒ Domain:
skillsnet.dk

☐ Workgroup:
WORKGROUP

OK Cancel

Windows Security

Computer Name/Domain Changes

Enter the name and password of an account with permission to join the domain.

Administrator

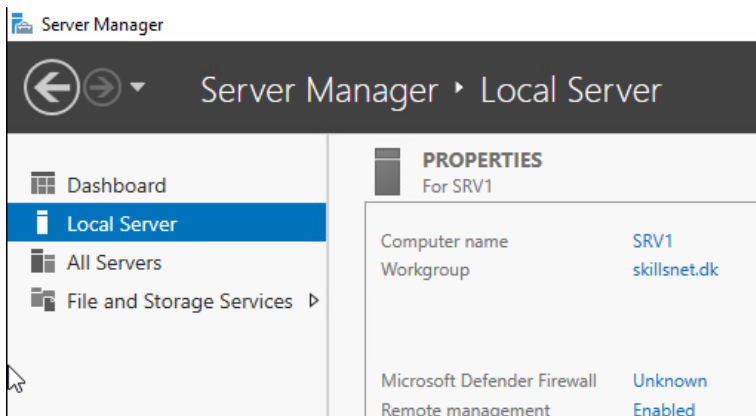
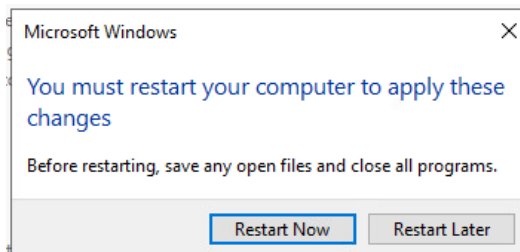
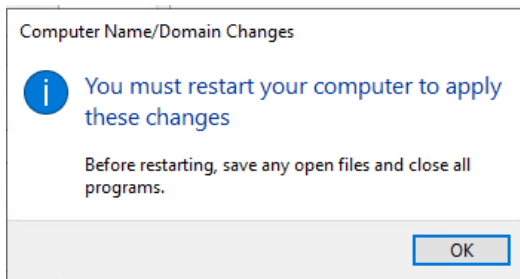
.....

OK Cancel

Computer Name/Domain Changes

Welcome to the skillsnet.dk domain.

OK



Example without GUI: SRV2

- Don't forget to fix previously dns server to 10.1.1.1 in SRV2


```
Administrator: C:\Windows\system32\cmd.exe
WARNING: To stop SConfig from launching at sign-in, type "Set-SConfig -AutoLaunch $false"

=====
                        Welcome to Windows Server 2022 Standard
=====

1) Domain/workgroup:           Workgroup: WORKGROUP
2) Computer name:              SRV2
3) Add local administrator
4) Remote management:         Enabled

5) Update setting:             Download only
6) Install updates
7) Remote desktop:            Disabled

8) Network settings
9) Date and time
10) Telemetry setting:         Required
11) Windows activation

12) Log off user
13) Restart server
14) Shut down server
15) Exit to command line (PowerShell)

Enter number to select an option: 1
```

```
Administrator: C:\Windows\system32\cmd.exe

=====
                        Change domain/workgroup membership
=====

Current workgroup: WORKGROUP

Join (D)omain or (W)orkgroup? (Blank=Cancel): D
Name of domain to join (Blank=Cancel): skillsnet.dk
Specify an authorized domain\user (Blank=Cancel): Administrator
Password for Administrator: *****
```

```
Administrator: C:\Windows\system32\cmd.exe

=====
Change domain/workgroup membership
=====

Current workgroup: WORKGROUP

Join (D)omain or (W)orkgroup? (Blank=Cancel): D
Name of domain to join (Blank=Cancel): skillsnet.dk
Specify an authorized domain\user (Blank=Cancel): Administrator
Password for Administrator: *****
Joining skillsnet.dk...
WARNING: The changes will take effect after you restart the computer SRV2.
Successfully joined domain.
Do you want to change the computer name before restarting? (Y)es or (N)o: n
Restart now? (Y)es or (N)o: y
```

```
Administrator: C:\Windows\system32\cmd.exe
WARNING: To stop SConfig from launching at sign-in, type "Set-SConfig -AutoLaunch $false"

=====
Welcome to Windows Server 2022 Standard
=====

1) Domain/workgroup:           Domain: skillsnet.dk
2) Computer name:              SRV2
3) Add local administrator
4) Remote management:         Enabled

5) Update setting:             Download only
6) Install updates
7) Remote desktop:            Disabled

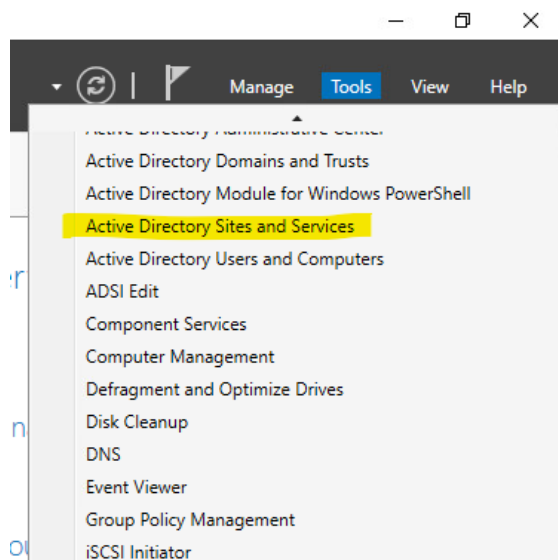
8) Network settings
9) Date and time
10) Telemetry setting:         Required
11) Windows activation

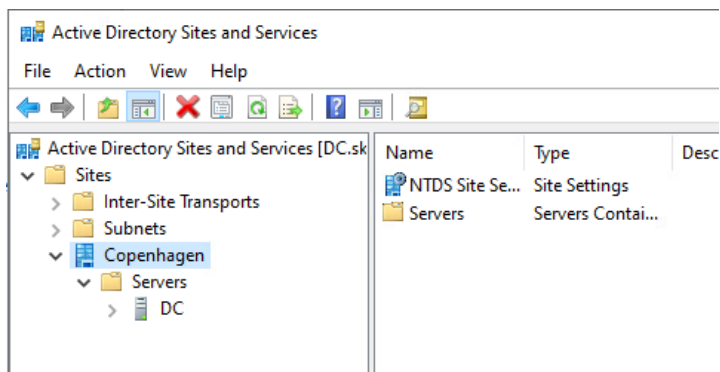
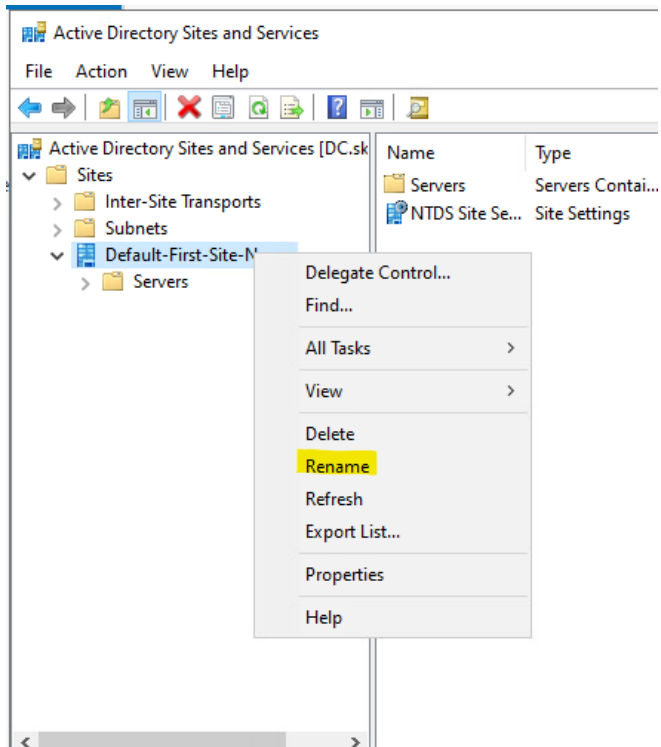
12) Log off user
13) Restart server
14) Shut down server
15) Exit to command line (PowerShell)

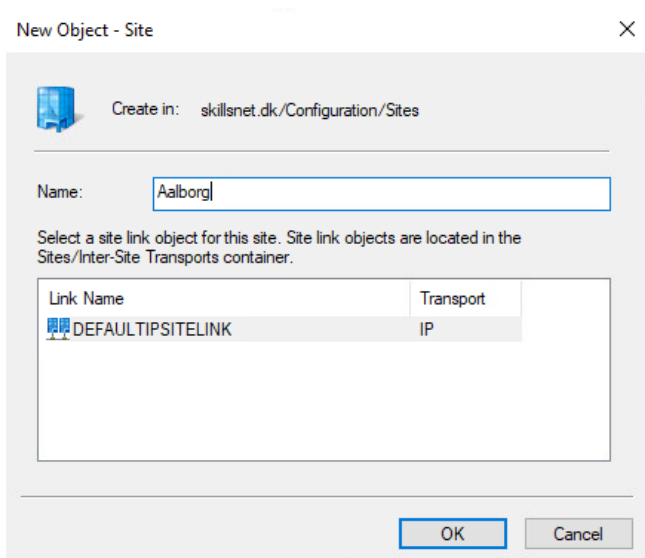
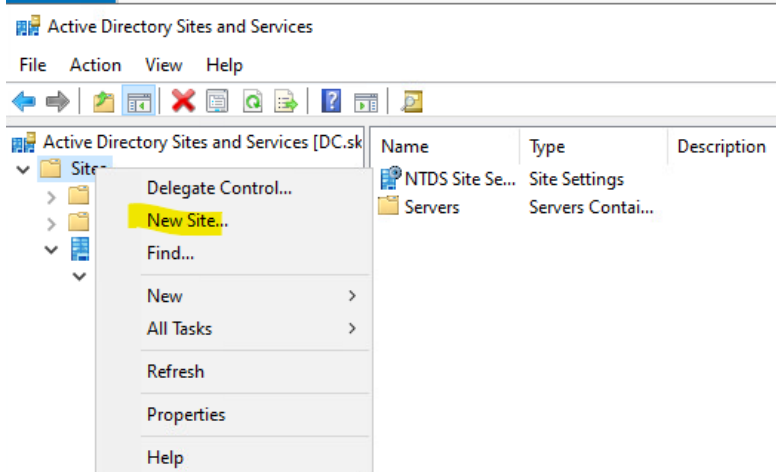
Enter number to select an option: _
```

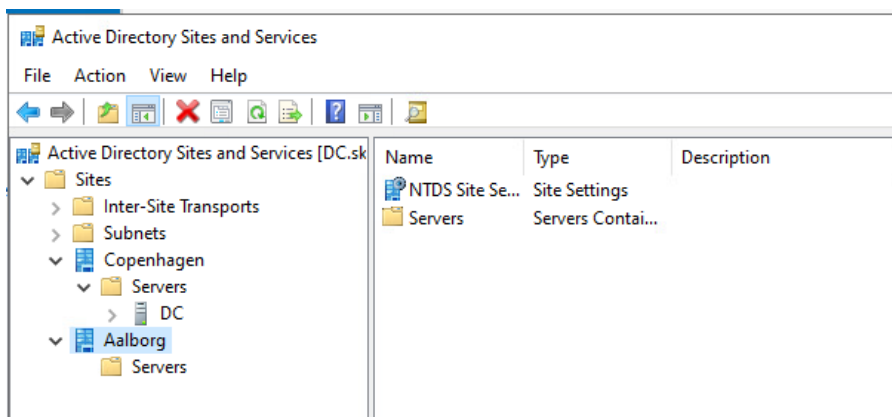
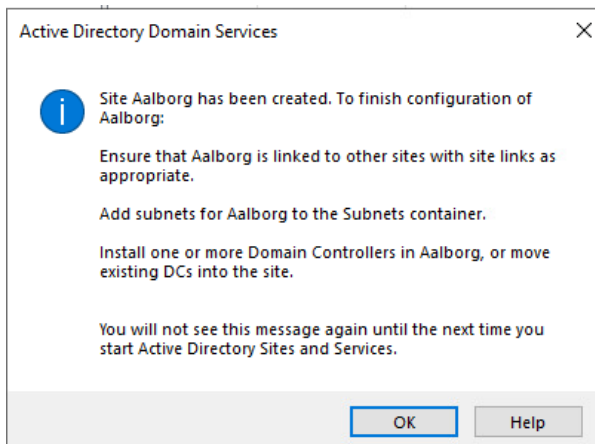
DC AD- Task2: Configure sites

2. Configure Sites: **Copenhagen** and **Aalborg**; assign DC/RODC accordingly









DC AD- Task3: Create OU structure

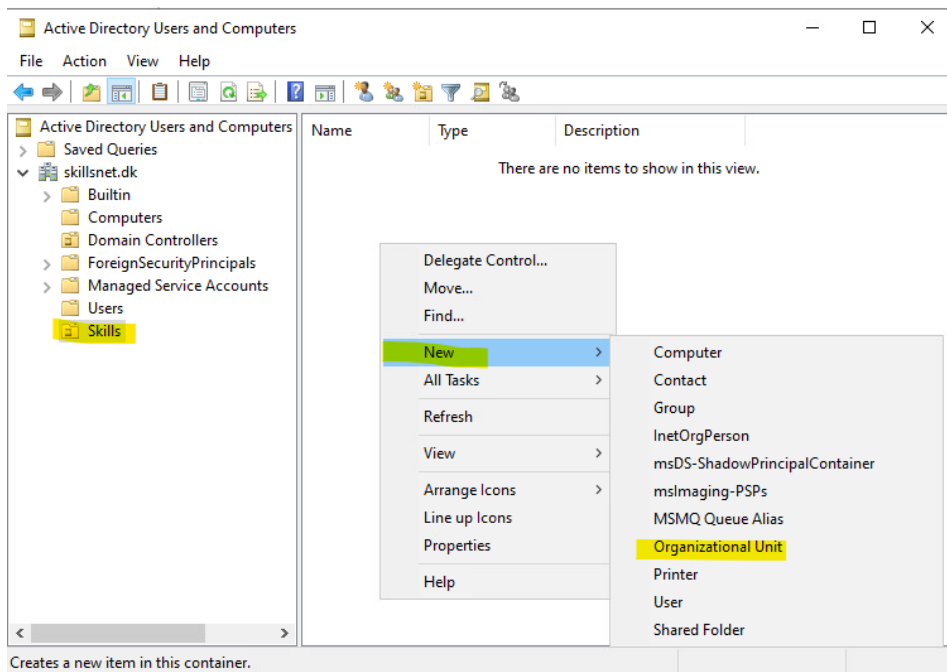
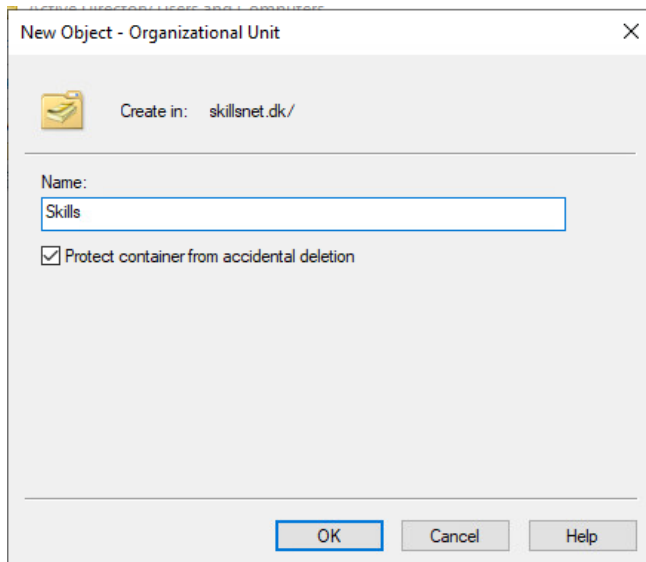
3. Create OU structure according to the **Appendix C**

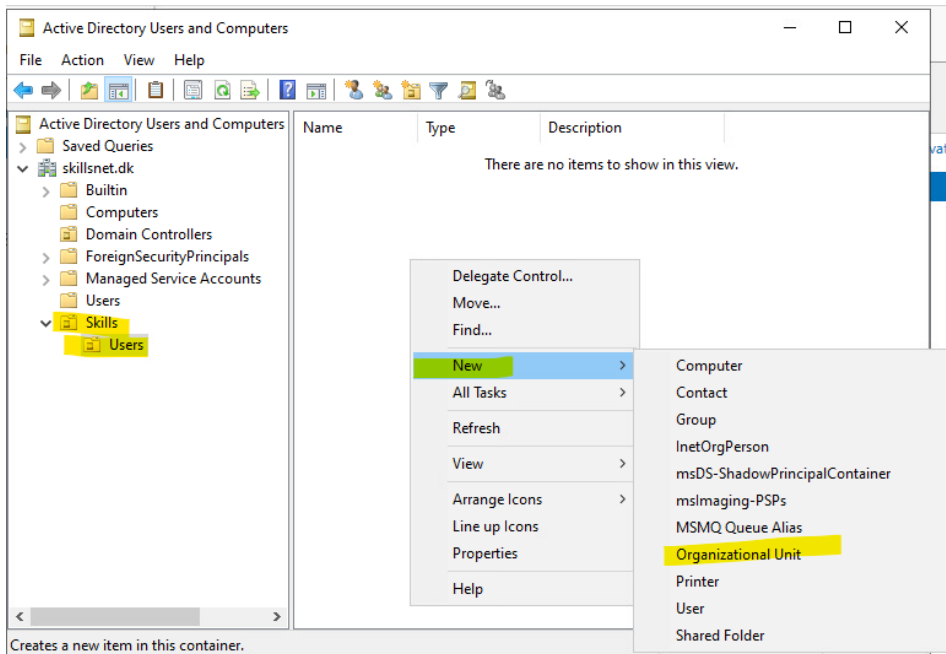
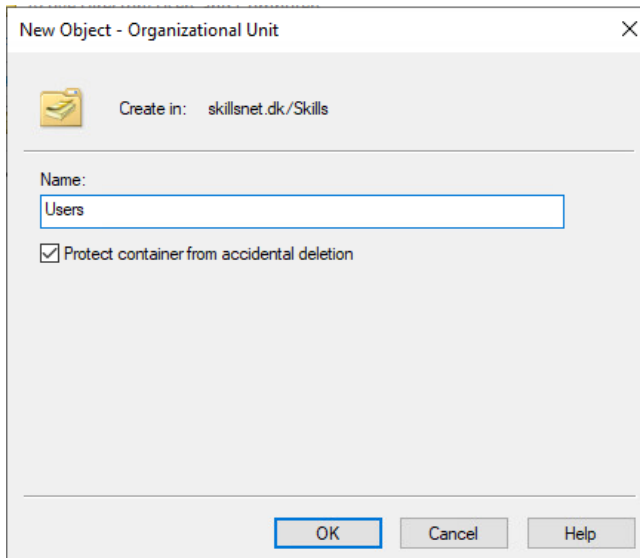
The screenshot shows the 'Active Directory Users and Computers' console. The left pane shows the tree structure with 'skillsnet.dk' selected. The right pane shows a list of containers. A context menu is open over the 'Users' container, and the 'New' option is selected, opening a sub-menu where 'Organizational Unit' is highlighted.

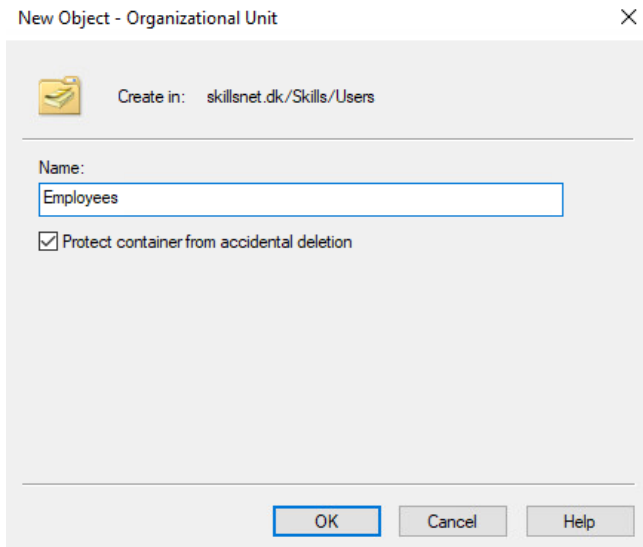
Name	Type	Description
Builtin	builtinDomain	
Computers	Container	Default container for up...
Domain Con...	Organizational...	Default container for do...
ForeignSecu...	Container	Default container for sec...
Managed Se...	Container	Default container for ma...
Users	Container	Default container for up...

Creates a new item in this container.

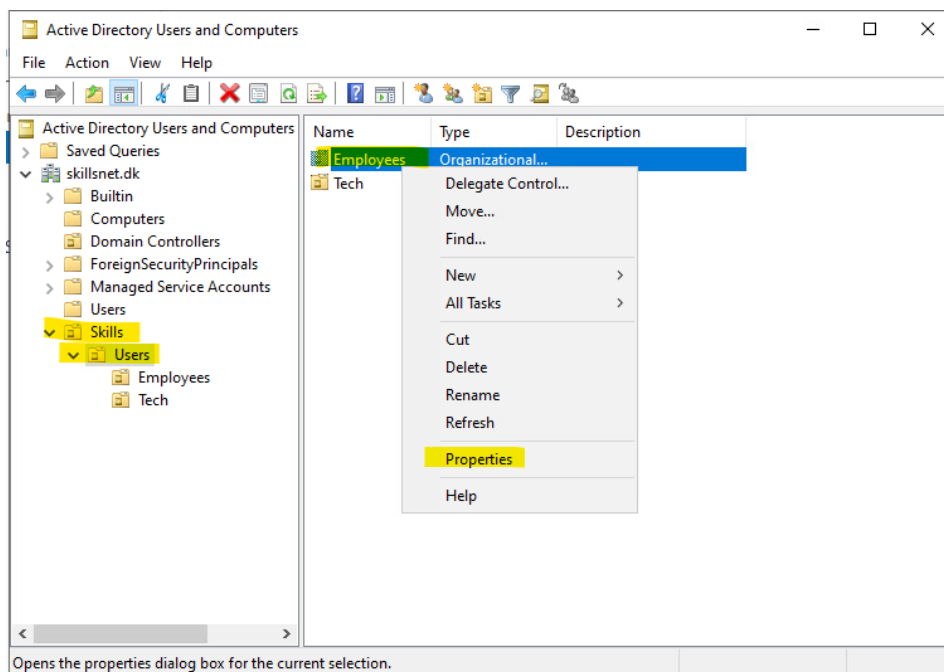
Activate Wi
Go to Settings







Don't forget to change the Description:



Employees Properties

General Managed By COM+

Employees

Description: All regular non-technical users

Street:

City:

State/province:

Zip/Postal Code:

Country/region:

OK Cancel Apply

Repeat the process for all the OU structure. The result should be this:

Active Directory Users and Computers

File Action View Help

Active Directory Users and Computers

- Saved Queries
- skillsnet.dk
 - Builtin
 - Computers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - Managed Service Accounts
 - Skills
 - Desktops
 - Groups
 - Servers
 - Users
 - Contractors
 - Development
 - Employees
 - Finance
 - Sales
 - Tech
 - Users

Name	Type	Description
Desktops	Organizational...	Computer objects for client PCs
Groups	Organizational...	All security and distribution groups
Servers	Organizational...	Computer objects for servers
Users	Organizational...	

Active Directory Users and Computers

File Action View Help

Active Directory Users and Computers

- Saved Queries
- skillsnet.dk
 - Builtin
 - Computers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - Managed Service Accounts
 - Skills
 - Desktops
 - Groups
 - Servers
 - Users
 - Contractors
 - Development
 - Employees
 - Finance
 - Sales
 - Tech
 - Users

Name	Type	Description
Contractors	Organizational...	External/temporary workers
Development	Organizational...	Software/dev team members
Employees	Organizational...	All regular non-technical users
Finance	Organizational...	Finance team users
Sales	Organizational...	Sales department staff
Tech	Organizational...	IT personnel and admins

DC AD- Task4: Password policies

4. Create password policies according to the **Appendix D**

DC AD- Task5: ADFS

5. Deploy ADFS at **sso.skillsnet.dk** using UPN logins

DC AD- Task6: Import users

6. Import users from **C:\Resources\ES2025_TP39_ModuleB_Users_Skillsnet.csv**
 - 6.1. Ensure all users are activated and have UPN defined
 - 6.2. Based on Department value, create unique groups
 - 6.3. Assign users to role groups and OUs based on their Department value
 - 6.4. Usernames must follow the format: **Firstname.Lastname**

Group Policies

Use the **first bullet point** in each policy block as the **GPO name**. The **subsequent bullets** define the **settings required** in that GPO.

Group Policies need to be mapped and configured according to best practices.

DC GPO - Task1: Baseline Security Policy for Servers

1. Baseline Security Policy for Servers
 - 1.1. Send NTLMv2 responses only;

- 1.2. Allow only AES256_HBA_SHA1 and future encryption types for Kerberos;
- 1.3. Audit successful and failed logons;
- 1.4. Do not store LM hash value on next password change;
- 1.5. Disallow WinRM Server and Client Basic authentication and unencrypted traffic;
- 1.6. Set high level encryption and secure RPC communication on Remote Desktop Services Host;

DC GPO - Task2: Baseline Security Policy for Desktops

2. Baseline Security Policy for Desktops
 - 2.1. Send NTLMv2 responses only;
 - 2.2. Allow only AES256_HBA_SHA1 and future encryption types for Kerberos;
 - 2.3. Set high level encryption and secure RPC communication on Remote Desktop Services Host;
 - 2.4. Do not store LM hash value on next password change;
 - 2.5. Disallow WinRM Server and Client Basic authentication and unencrypted traffic;
 - 2.6. Turn off Microsoft consumer experiences;
 - 2.7. Turn off Autoplay on all drives.

DC GPO - Task3: Administrative accounts

3. Administrative Accounts
 - 3.1. Grant administrative rights on desktops only to Tech users

DC GPO - Task4: Users – Restrict Tools

4. Users – Restrict Tools
 - 4.1. Applicable only for Sales, Finance and Contractors;
 - 4.2. Disable ability to use Registry editing tools, PowerShell and Command Prompt;

DC GPO - Task5: Contractors – Customization

5. Contractors – Customization
 - 5.1. Login banner that reads:
Welcome to Skillsnet, you're a contractor and all your activities are recorded. You will be legally responsible for any damage that you cause.

DC GPO - Task6: Folder Redirection

6. Folder Redirection
 - 6.1. Applicable only for Finance, Sales
 - 6.2. Redirect all user folders to `\\srv1.skillsnet.dk\Users\<username>`

DC GPO - Task7: Disk Encryption

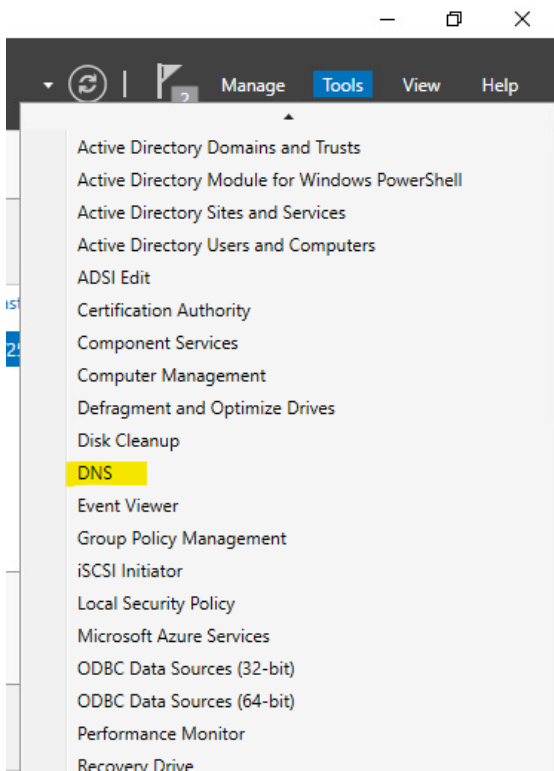
7. Disk Encryption
 - 7.1. Store all recovery keys for domain-joined systems in Active Directory

Name Resolution

DC DNS - Task1: A & AAAA records

1. Create **A** and **AAAA** records for all relevant hosts based on the **Appendix A**

DNS server already installed with ADDS



¿Susceptibles DNS Forward Records A and AAAA for skillsnet.dk?:

1. skillsnet.dk 10.1.1.1
2. dc.skillsnet.dk 10.1.1.1
3. srv1.skillsnet.dk 10.1.1.2
4. srv2.skillsnet.dk 10.1.1.3
5. rodc.skillsnet.dk (Aalborg) 10.2.1.1
6. client.skillsnet.dk (Aalborg) 10.2.1.10
7. www.skillsnet.dk 10.1.1.3 ¿Cname de srv2?
8. sso.skillsnet.dk 10.1.1.1 ¿Cname de dc?
9. cdp.skillsnet.dk 10.1.1.3 ¿Cname de srv2?
10. crl.skillsnet.dk 10.1.1.3 ¿Cname de srv2?
11. aia.skillsnet.dk 10.1.1.3 ¿Cname de srv2?
12. cacerts.skillsnet.dk 10.1.1.3 ¿Cname de srv2?
13. ocsp.skillsnet.dk 10.1.1.1 ¿Cname de dc?
14. intra.skillsnet.dk 10.1.1.3 ¿Cname de srv2?
15. app.skillsnet.dk 10.1.1.3 ¿Cname de srv2?

DC DNS - Task2: Reverse lookup zones

2. Create reverse lookup zones for both IPv4 and IPv6 subnets. Enable dynamic updates for domain-joined machines

DC DNS - Task3: Authority delegation

3. Delegate authority for the **skillsdev.dk** to the **DEV-SRV**

DC DNS - Task4: DNS forwarding

4. Configure DNS forwarding so that all unresolved queries are forwarded to the INET server

Certificate services

DC Certs - Task1: PKI and Root CA

1. Deploy an Enterprise PKI. The root Certification Authority must be named **"Skillsnet CA"**

Add Roles and Features Wizard

Select server roles

DESTINATION SERVER
DC.skillsnet.dk

Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD CS
 Role Services
Confirmation
Results

Select one or more roles to install on the selected server.

Roles

- ☒ **Active Directory Certificate Services**
- ☒ Active Directory Domain Services (Installed)
- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☐ DHCP Server
- ☒ DNS Server (Installed)
- ☐ Fax Server
- ☒ File and Storage Services (2 of 12 installed)
 - ☐ Host Guardian Service
 - ☐ Hyper-V
 - ☐ Network Policy and Access Services
 - ☐ Print and Document Services
 - ☐ Remote Access
 - ☐ Remote Desktop Services
 - ☐ Volume Activation Services
 - ☐ Web Server (IIS)
 - ☐ Windows Deployment Services
 - ☐ Windows Server Update Services

Description

Active Directory Certificate Services (AD CS) is used to create certification authorities and related role services that allow you to issue and manage certificates used in a variety of applications.

< Previous

Next >

Install

Cancel

Add Roles and Features Wizard

Select role services

DESTINATION SERVER
DC.skillsnet.dk

Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD CS
Role Services
Confirmation
Results

Select the role services to install for Active Directory Certificate Services

Role services	Description
☒ Certification Authority	Certification Authority (CA) is used to issue and manage certificates. Multiple CAs can be linked to form a public key infrastructure.
☐ Certificate Enrollment Policy Web Service	
☐ Certificate Enrollment Web Service	
☐ Certification Authority Web Enrollment	
☐ Network Device Enrollment Service	
☐ Online Responder	

< Previous Next > Install Cancel

Add Roles and Features Wizard

Installation progress

DESTINATION SERVER
DC.skillsnet.dk

- Before You Begin
- Installation Type
- Server Selection
- Server Roles
- Features
- AD CS
- Role Services
- Confirmation
- Results**

View installation progress

Feature installation

Configuration required. Installation succeeded on DC.skillsnet.dk.

Active Directory Certificate Services

Additional steps are required to configure Active Directory Certificate Services on the destination server

[Configure Active Directory Certificate Services on the destination server](#)

Certification Authority

Remote Server Administration Tools

Role Administration Tools

Active Directory Certificate Services Tools

Certification Authority Management Tools

You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.

[Export configuration settings](#)

< Previous Next > Close Cancel

Post-deployment Configuration

Configuration required for Active Directory Certificate Services at DC

[Configure Active Directory Certificate Services on th...](#)

Feature installation TASKS X

Configuration required. Installation succeeded on DC.skillsnet.dk.

[Add Roles and Features](#)

Task Details

2

3

4

5 [Connect this server to cloud services](#)

AD CS Configuration

Credentials

DESTINATION SERVER
DC.skillsnet.dk

Specify credentials to configure role services

To install the following role services you must belong to the local Administrators group:

- Standalone certification authority
- Certification Authority Web Enrollment
- Online Responder

To install the following role services you must belong to the Enterprise Admins group:

- Enterprise certification authority
- Certificate Enrollment Policy Web Service
- Certificate Enrollment Web Service
- Network Device Enrollment Service

Credentials:

[More about AD CS Server Roles](#)

< Previous **Next >** Configure Cancel

AD CS Configuration

Role Services

DESTINATION SERVER
DC.skillsnet.dk

Select Role Services to configure

- ☒ Certification Authority
- ☐ Certification Authority Web Enrollment
- ☐ Online Responder
- ☐ Network Device Enrollment Service
- ☐ Certificate Enrollment Web Service
- ☐ Certificate Enrollment Policy Web Service

[More about AD CS Server Roles](#)

< Previous Next > Configure Cancel


AD CS Configuration

DESTINATION SERVER
DC.skillsnet.dk

Setup Type

Credentials
 Role Services
Setup Type
 CA Type
 Private Key
 Cryptography
 CA Name
 Validity Period
 Certificate Database
 Confirmation
 Progress
 Results

Specify the setup type of the CA

Enterprise certification authorities (CAs) can use Active Directory Domain Services (AD DS) to simplify the management of certificates. Standalone CAs do not use AD DS to issue or manage certificates.

☒ Enterprise CA

Enterprise CAs must be domain members and are typically online to issue certificates or certificate policies.

☐ Standalone CA

Standalone CAs can be members or a workgroup or domain. Standalone CAs do not require AD DS and can be used without a network connection (offline).

[More about Setup Type](#)

< Previous
Next >
Configure
Cancel


 AD CS Configuration

DESTINATION SERVER
DC.skillsnet.dk

CA Type

Credentials
 Role Services
 Setup Type
CA Type
 Private Key
 Cryptography
 CA Name
 Validity Period
 Certificate Database
 Confirmation
 Progress
 Results

Specify the type of the CA

When you install Active Directory Certificate Services (AD CS), you are creating or extending a public key infrastructure (PKI) hierarchy. A root CA is at the top of the PKI hierarchy and issues its own self-signed certificate. A subordinate CA receives a certificate from the CA above it in the PKI hierarchy.

☒ Root CA
 Root CAs are the first and may be the only CAs configured in a PKI hierarchy.

☐ Subordinate CA
 Subordinate CAs require an established PKI hierarchy and are authorized to issue certificates by the CA above them in the hierarchy.

[More about CA Type](#)

< Previous

Next >

Configure

Cancel

Private Key

 DESTINATION SERVER
DC.skillsnet.dk

- Credentials
- Role Services
- Setup Type
- CA Type
- Private Key**
- Cryptography
- CA Name
- Validity Period
- Certificate Database
- Confirmation
- Progress
- Results

Specify the type of the private key

To generate and issue certificates to clients, a certification authority (CA) must have a private key.

- ☒ **Create a new private key**
Use this option if you do not have a private key or want to create a new private key.
- ☐ **Use existing private key**
Use this option to ensure continuity with previously issued certificates when reinstalling a CA.
 - ☐ **Select a certificate and use its associated private key**
Select this option if you have an existing certificate on this computer or if you want to import a certificate and use its associated private key.
 - ☐ **Select an existing private key on this computer**
Select this option if you have retained private keys from a previous installation or want to use a private key from an alternate source.

[More about Private Key](#)

< Previous

Next >

Configure

Cancel

Cryptography for CA

DESTINATION SERVER
DC.skillsnet.dk

- Credentials
- Role Services
- Setup Type
- CA Type
- Private Key
- Cryptography**
- CA Name
- Validity Period
- Certificate Database
- Confirmation
- Progress
- Results

Specify the cryptographic options

Select a cryptographic provider:

RSA#Microsoft Software Key Storage Provider

Key length:

2048

Select the hash algorithm for signing certificates issued by this CA:

- SHA256
- SHA384
- SHA512
- SHA1
- MD5

☐ Allow administrator interaction when the private key is accessed by the CA.

[More about Cryptography](#)

< Previous

Next >

Configure

Cancel

AD CS Configuration

CA Name

DESTINATION SERVER
DC.skillsnet.dk

Credentials
Role Services
Setup Type
CA Type
Private Key
Cryptography
CA Name
Validity Period
Certificate Database
Confirmation
Progress
Results

Specify the name of the CA

Type a common name to identify this certification authority (CA). This name is added to all certificates issued by the CA. Distinguished name suffix values are automatically generated but can be modified.

Common name for this CA:

Distinguished name suffix:

Preview of distinguished name:

[More about CA Name](#)

< Previous Next > Configure Cancel

AD CS Configuration

Validity Period

DESTINATION SERVER
DC.skillsnet.dk

Credentials
Role Services
Setup Type
CA Type
Private Key
Cryptography
CA Name
Validity Period
Certificate Database
Confirmation
Progress
Results

Specify the validity period

Select the validity period for the certificate generated for this certification authority (CA):

5 Years

CA expiration Date: 6/24/2030 8:18:00 PM

The validity period configured for this CA certificate should exceed the validity period for the certificates it will issue.

[More about Validity Period](#)

< Previous Next > Configure Cancel

AD CS Configuration

CA Database

- Credentials
- Role Services
- Setup Type
- CA Type
- Private Key
 - Cryptography
 - CA Name
 - Validity Period
- Certificate Database**
- Confirmation
- Progress
- Results

Specify the database locations

Certificate database location:

Certificate database log location:

[More about CA Database](#)

DESTINATION SERVER
DC.skillsnet.dk

< Previous

Next >

Configure

Cancel


AD CS Configuration
— □ ×

Confirmation

Credentials
 Role Services
 Setup Type
 CA Type
 Private Key
 Cryptography
 CA Name
 Validity Period
 Certificate Database
Confirmation
 Progress
 Results

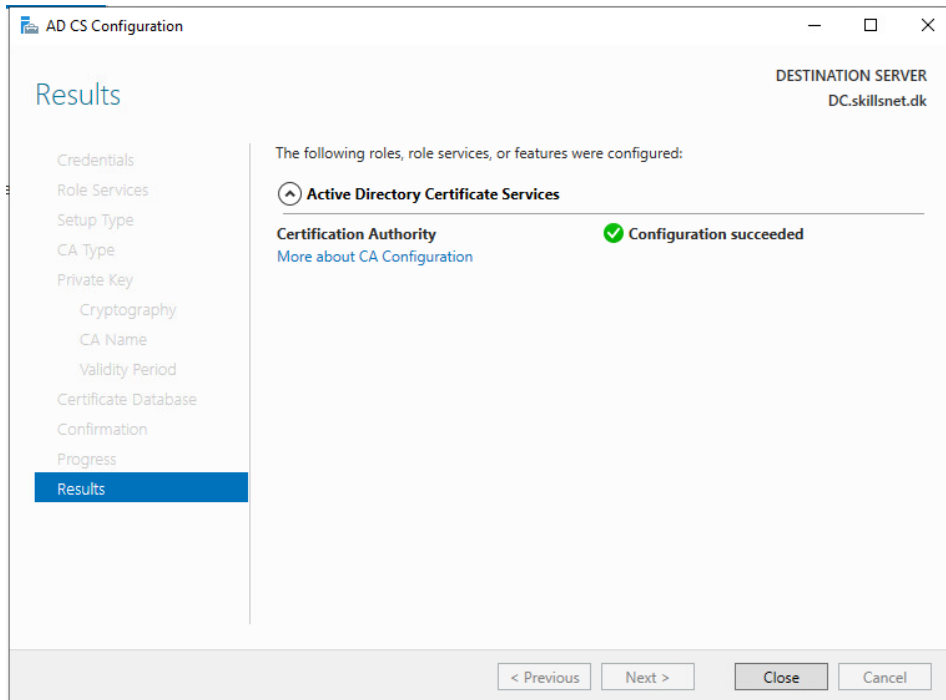
DESTINATION SERVER
 DC.skillsnet.dk

To configure the following roles, role services, or features, click Configure.

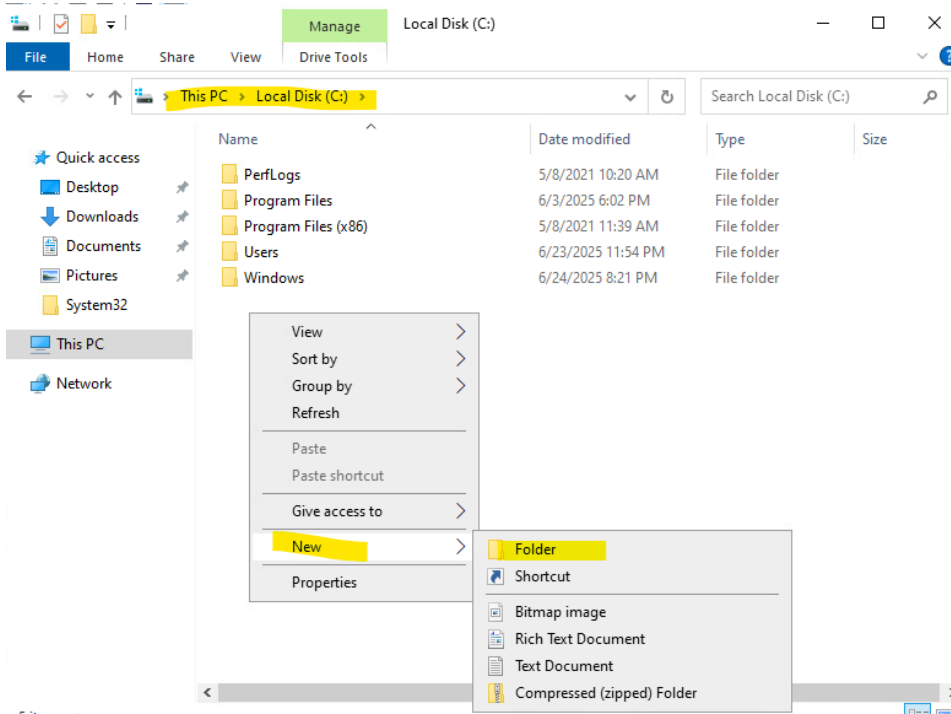
^ **Active Directory Certificate Services**

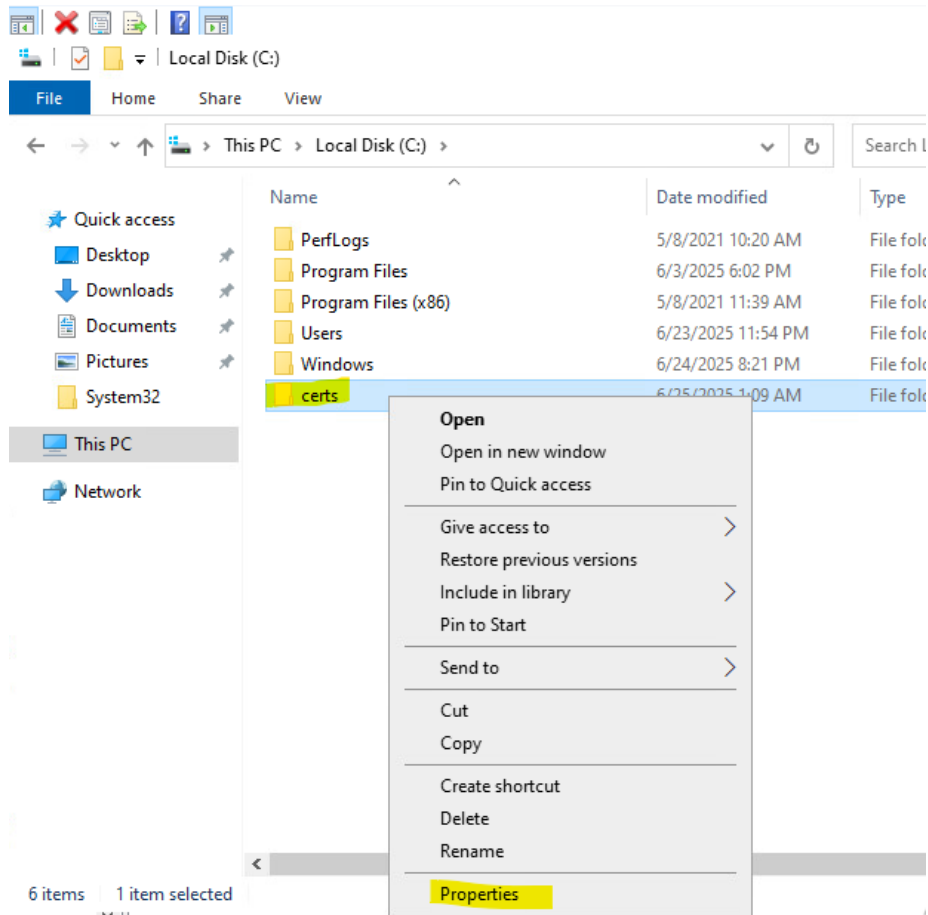
Certification Authority
 CA Type: Enterprise Root
 Cryptographic provider: RSA#Microsoft Software Key Storage Provider
 Hash Algorithm: SHA256
 Key Length: 2048
 Allow Administrator Interaction: Disabled
 Certificate Validity Period: 6/24/2030 8:18:00 PM
 Distinguished Name: CN=Skillsnet CA,DC=skillsnet,DC=dk
 Certificate Database Location: C:\Windows\system32\CertLog
 Certificate Database Log Location: C:\Windows\system32\CertLog

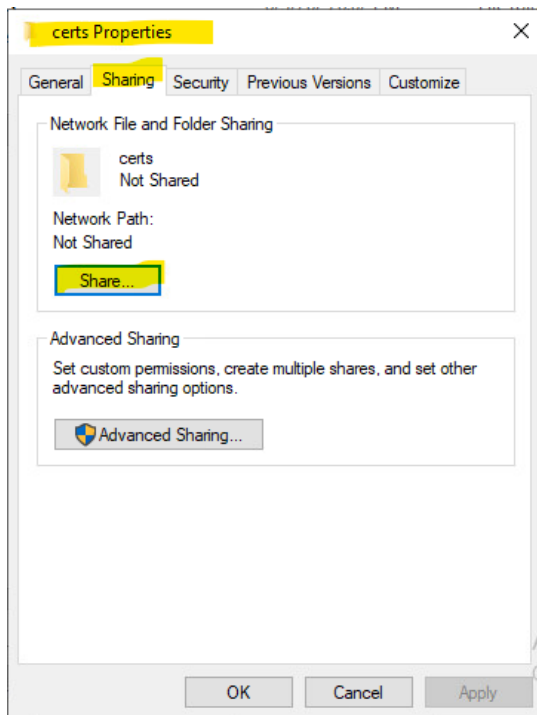
< Previous
Next >
Configure
Cancel



- Create a shared folder for sharing certificates







>

←  Network access

Choose people on your network to share with

Type a name and then click Add, or click the arrow to find someone.

▼
Add

Name	Permission Level
 Administrator	Read/Write ▼
 Administrators	Owner
 Everyone	Read/Write ▼

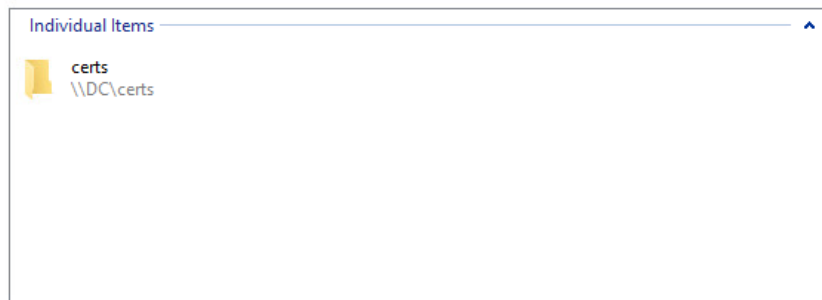
[I'm having trouble sharing](#)

Activate Windows

Network access

Your folder is shared.

You can [e-mail](#) someone links to these shared items, or [copy](#) and paste the links into another app.

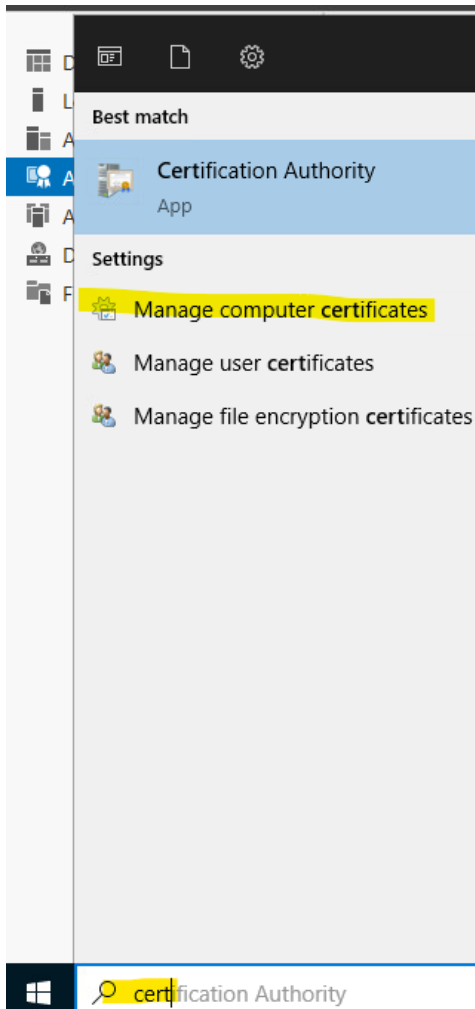


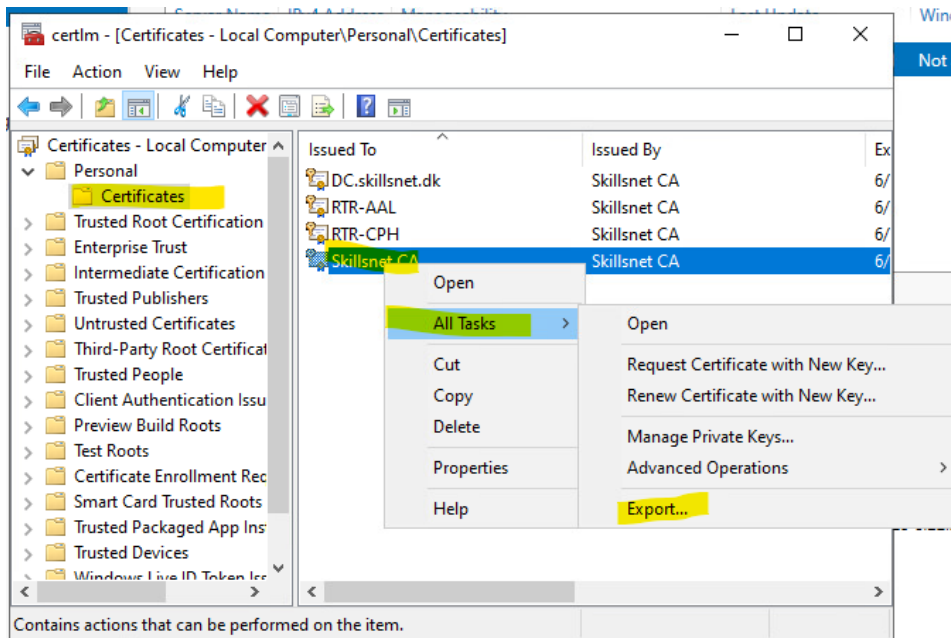
[Show me all the network shares on this computer.](#)

Activate Windows
Go to Settings to activate Windows.

Done

- Share the Skillsnet CA







←  Certificate Export Wizard

Export Private Key

You can choose to export the private key with the certificate.

Private keys are password protected. If you want to export the private key with the certificate, you must type a password on a later page.

Do you want to export the private key with the certificate?

- ☐ Yes, export the private key
- ☒ No, do not export the private key

Next

Cancel

← Certificate Export Wizard

Export File Format

Certificates can be exported in a variety of file formats.

Select the format you want to use:

- ☐ DER encoded binary X.509 (.CER)
- ☒ Base-64 encoded X.509 (.CER)
- ☐ Cryptographic Message Syntax Standard - PKCS #7 Certificates (.P7B)
- ☐ Include all certificates in the certification path if possible
- ☐ Personal Information Exchange - PKCS #12 (.PFX)
- ☐ Include all certificates in the certification path if possible
- ☐ Delete the private key if the export is successful
- ☐ Export all extended properties
- ☐ Enable certificate privacy
- ☐ Microsoft Serialized Certificate Store (.SST)

Next

Cancel

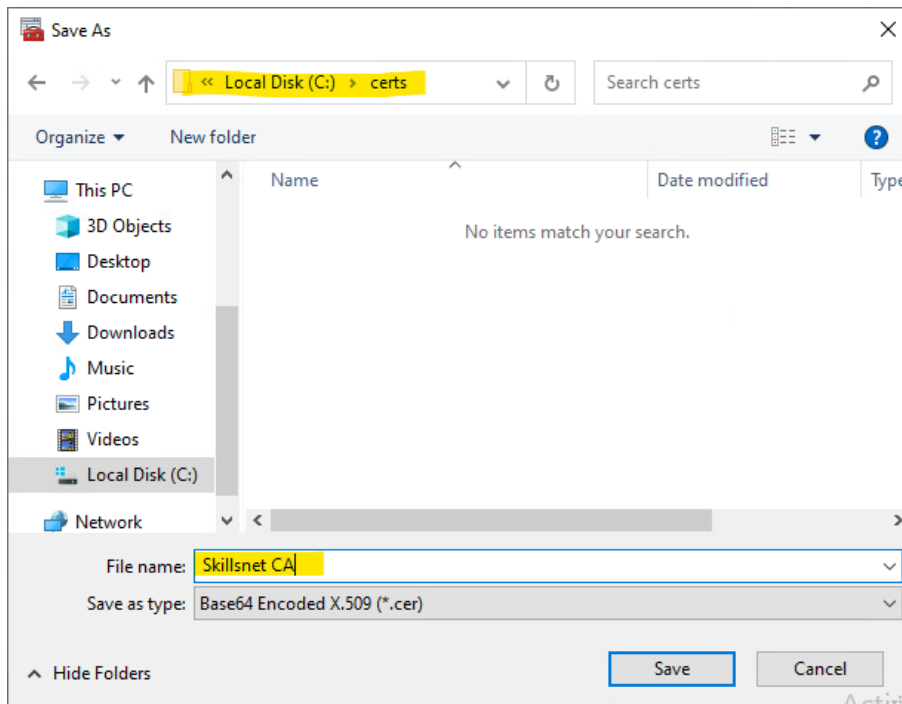
← Certificate Export Wizard

File to Export

Specify the name of the file you want to export

File name:

Browse...



X

← Certificate Export Wizard

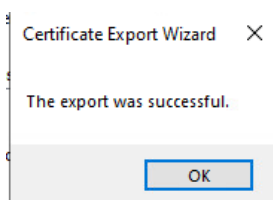
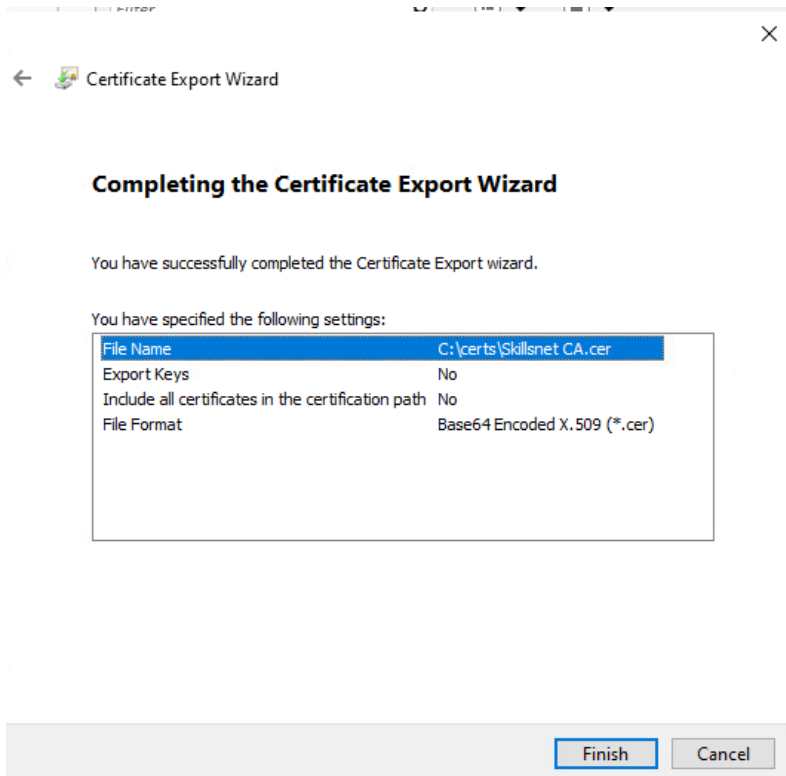
File to Export

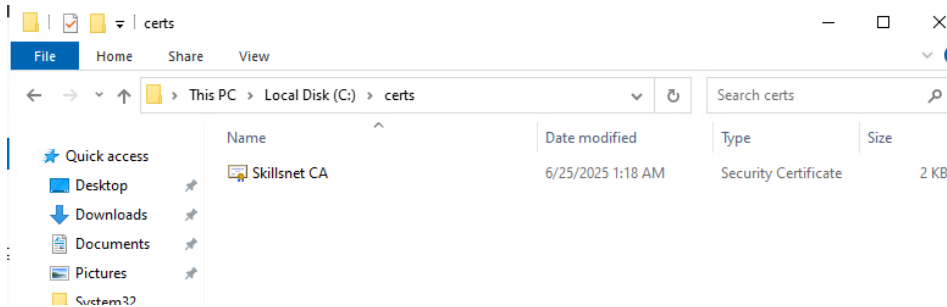
Specify the name of the file you want to export

File name:

C:\certs\Skillsnet CA.cer

Browse...





DC Certs - Task2: CDP

2. Configure the **CDP** endpoint at <http://crl.skillsnet.dk/SkillsnetCA.crl>

DC Certs - Task3: AIA

3. Configure the **AIA** endpoint at <http://cacerts.skillsnet.dk/SkillsnetCA.crt>

DC Certs - Task4: OCSP

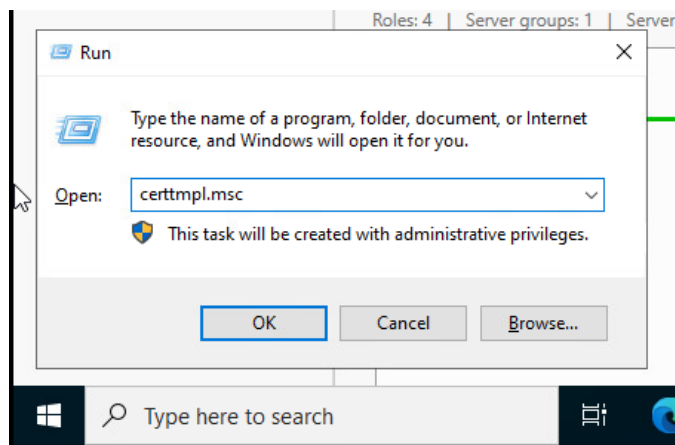
4. Configure the **OCSP** responder endpoint at <http://ocsp.skillsnet.dk/>

DC Certs - Task5: CDP and AIA on SRV2

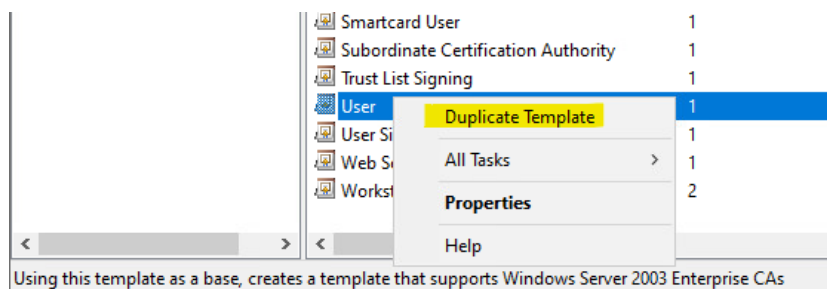
5. CDP and AIA certificate endpoints need to be **hosted on SRV2**. OCSP responder is **hosted on DC**.

DC Certs - Task6: certificate templates

6. Create following certificate templates:
 - 6.1. **Skills Users** – For user certificates; certificate needs to be auto-enrolled for all users
 - 6.2. **Skills Endpoints** – For computer certificates; certificate needs to be auto-enrolled for all endpoints
 - 6.3. **Skills Web Server** – For web server certificates



6.1. Skills Users



Properties of New Template X

Superseded Templates		Extensions		Security	
Subject Name		Server		Issuance Requirements	
Compatibility	General	Request Handling	Cryptography	Key Attestation	
Template display name: Skills User					
Template name: SkillsUser					
Validity period: 1 years		Renewal period: 6 weeks			
☒ Publish certificate in Active Directory					
☐ Do not automatically reenroll if a duplicate certificate exists in Active Directory					

OK Cancel Apply Help

Properties of New Template

Subject Name Server Issuance Requirements

Compatibility General Request Handling Cryptography Key Attestation

Superseded Templates Extensions **Security**

Group or user names:

- Authenticated Users**
- Administrator
- Domain Admins (SKILLSNET\Domain Admins)
- Domain Users (SKILLSNET\Domain Users)
- Enterprise Admins (SKILLSNET\Enterprise Admins)

Add... Remove

Permissions for Authenticated Users

	Allow	Deny
Full Control	☐	☐
Read	☒	☐
Write	☐	☐
Enroll	☒	☐
Autoenroll	☒	☐

For special permissions or advanced settings, click Advanced.

Advanced

OK **Cancel** Apply Help

6.2 Skills Endpoints

Template Display Name	Schema Version	Version	Intended Pur
Code Signing	1	3.1	
Computer	1	5.1	
Cross Certification A		105.0	
Directory Email Repli		115.0	Directory Sen
Domain Controller		4.1	
Domain Controller A		110.0	Client Auther
EFS Recovery Agent		6.1	
Enrollment Agent		4.1	
Enrollment Agent (Computer)	1	5.1	
Exchange Enrollment Agent (Offline requ...	1	4.1	

Duplicate Template
 All Tasks >
 Properties
 Help

Properties of New Template

Subject Name	Server	Issuance Requirements
Superseded Templates	Extensions	Security
Compatibility	General	Request Handling
	Cryptography	Key Attestation

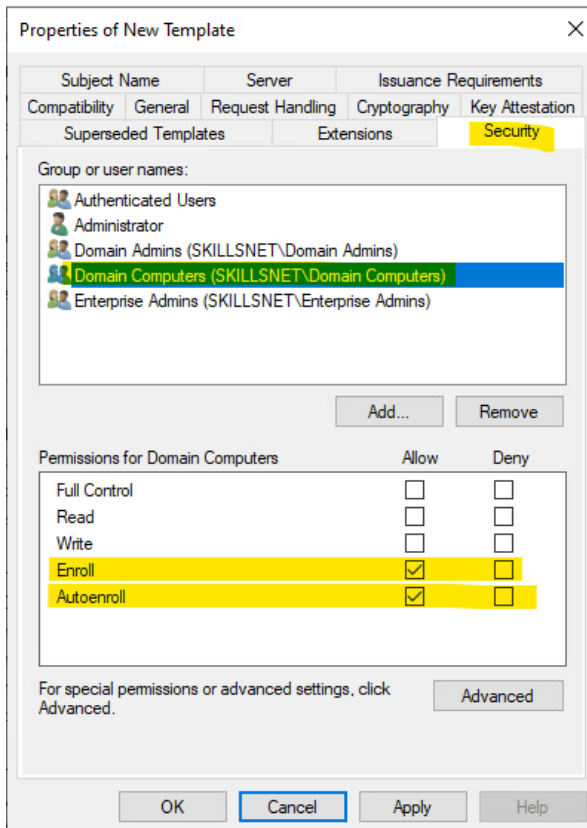
Template display name:
Skills Endpoints

Template name:
SkillsEndpoints

Validity period: 1 years
Renewal period: 6 weeks

☒ Publish certificate in Active Directory
☐ Do not automatically reenroll if a duplicate certificate exists in Active Directory

OK Cancel Apply Help



6.3. Skills Web Servers

Properties of New Template

Subject Name	Server	Issuance Requirements
Superseded Templates	Extensions	Security
Compatibility	General	Request Handling
	Cryptography	Key Attestation

Template display name:
Skills Web Server

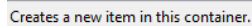
Template name:
SkillsWebServer

Validity period: 2 years
Renewal period: 6 weeks

☒ Publish certificate in Active Directory
☐ Do not automatically reenroll if a duplicate certificate exists in Active Directory

OK Cancel Apply Help

PUBLISH



Enable Certificate Templates

Select one Certificate Template to enable on this Certification Authority.
Note: If a certificate template that was recently created does not appear on this list, you may need to wait until information about this template has been replicated to all domain controllers.
All of the certificate templates in the organization may not be available to your CA.
For more information, see [Certificate Template Concepts](#).

Name	Intended Purpose
Key Recovery Agent	Key Recovery Agent
OCSP Response Signing	OCSP Signing
RAS and IAS Server	Client Authentication, Server Authentication
Router (Offline request)	Client Authentication
Skills Endpoints	Server Authentication, Client Authentication
Skills User	Client Authentication, Secure Email, Encrypting File System
Skills Web Server	Server Authentication
Smartcard Logon	Client Authentication, Smart Card Logon
Smartcard User	Secure Email, Client Authentication, Smart Card Logon
Trust List Signing	Microsoft Trust List Signing

OK

Cancel

certsrv - [Certification Authority (Local)\Skillsnet CA\Certificate Templates]

File Action View Help

- Certification Authority (Local)
 - Skillsnet CA
 - Revoked Certificates
 - Issued Certificates
 - Pending Requests
 - Failed Requests
 - Certificate Templates

Name	Intended Purpose
Skills Web Server	Server Authentication
Skills User	Client Authentication, Secure Email, En...
Skills Endpoints	Server Authentication, Client Authentic...
Directory Email Replication	Directory Service Email Replication
Domain Controller Authentication	Client Authentication, Server Authentic...
Kerberos Authentication	Client Authentication, Server Authentic...
EFS Recovery Agent	File Recovery
Basic EFS	Encrypting File System
Domain Controller	Client Authentication, Server Authentic...
Web Server	Server Authentication
Computer	Client Authentication, Server Authentic...
User	Encrypting File System, Secure Email, Cl...
Subordinate Certification Authority	<All>
Administrator	Microsoft Trust List Signing, Encrypting ...

Part 6: Enable Autoenrollment via Group Policy

1. **Open GPMC** on DC.
2. Create/Edit a GPO linked to the domain.

Navigate:

pgsql

CopiarEditor

User Configuration > Policies > Windows Settings > Security Settings > Public Key Policies

3.
 - o Enable **Autoenrollment** for users.
4. Repeat under **Computer Configuration** for computers.

Backups

DC Backups - Task1: Backup script

1. Create a PowerShell script and save it as **C:\Scripts\Backup.ps1**
2. The script must back up the following items:
 - 2.1. **C:\Backups\Users.csv** – A CSV file containing the full OU path and all key user attributes (e.g samAccountName, UPN, first name, last name, etc.)
 - 2.2. **C:\Backups\GPOs** - A directory containing exports of all existing Group Policy Objects
 - 2.3. **C:\Backups\Web\<site>** - A directory containing web root folders served by the IIS web servers on skillsnet.dk, organized using the corresponding DNS name of each site.
3. Implement email notification functionality within the script. It must send a success or failure report to: support@nordicbackup.net (You may choose the message format)

4. Configure Windows Server Backup to perform scheduled backups of the entire C:\Backups directory to an iSCSI target hosted on SRV2
5. Ensure the backup is scheduled to run **daily at 02:00**

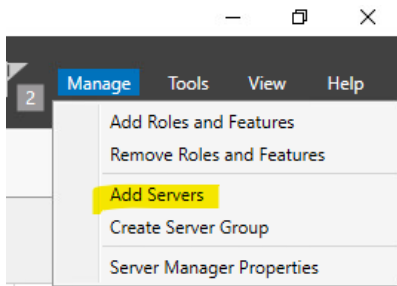
RODC

Role: Read-Only Domain Controller to provide authentication to Aalborg site.

RODC- Task1: Promote as RODC

Task: Promote as RODC on domain **skillsnet.dk**

Add the management of this server to DC. In DC machine:



Add Servers

Active Directory | DNS | Import

Location: skillsnet

Operating System: All

Name (CN): *Name, or beginning of name*

Find Now

Name	Operating System
DC	Windows Server 2022 Standard
SRV1	Windows Server 2022 Standard
SRV2	Windows Server 2022 Standard
RODC	Windows Server 2022 Standard

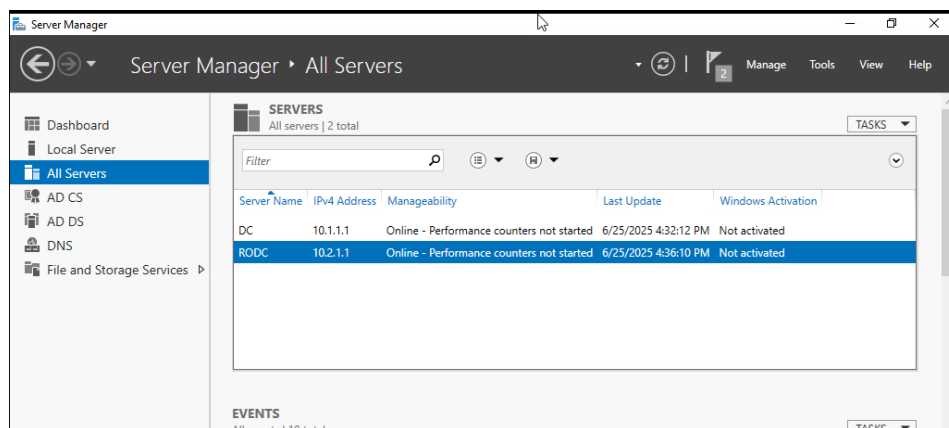
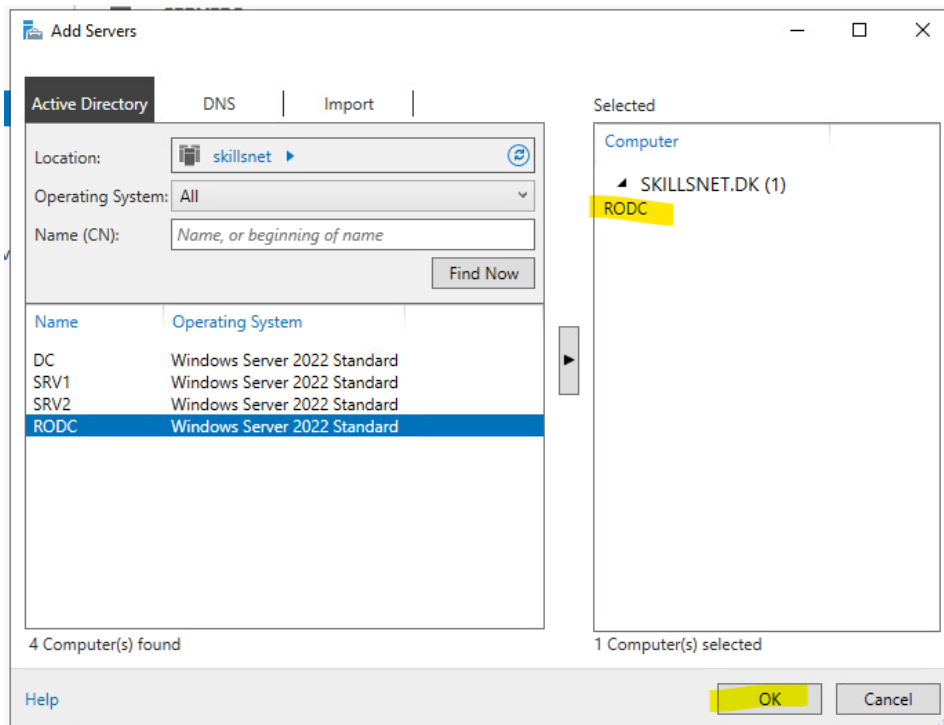
4 Computer(s) found

Selected

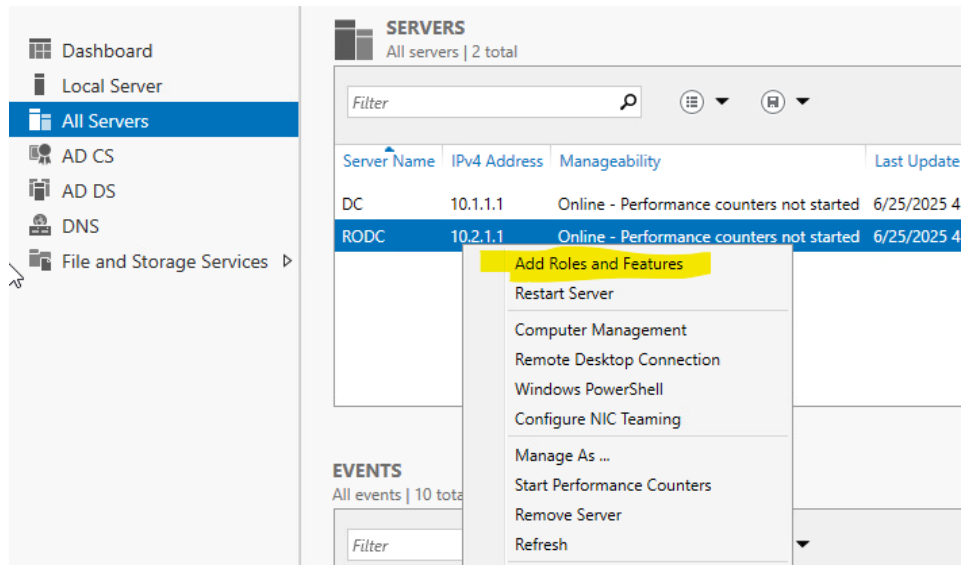
Computer

0 Computer(s) selected

Help **OK** **Cancel**



Install and configure ADDS



SERVICES
All servers | 2 total

Filter

Server Name	IPv4 Address	Manageability	Last Update
DC	10.1.1.1	Online - Performance counters not started	6/25/2025 4
RODC	10.2.1.1	Online - Performance counters not started	6/25/2025 4

EVENTS
All events | 10 total

Filter

- Add Roles and Features
- Restart Server
- Computer Management
- Remote Desktop Connection
- Windows PowerShell
- Configure NIC Teaming
- Manage As ...
- Start Performance Counters
- Remove Server
- Refresh

Select destination server

DESTINATION SERVER
RODC.skillsnet.dk

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool

☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
DC.skillsnet.dk	10.1.1.1	Microsoft Windows Server 2022 Standard
RODC.skillsnet.dk	10.2.1.1	Microsoft Windows Server 2022 Standard

2 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous

Next >

Install

Cancel

Add Roles and Features Wizard

Select server roles

DESTINATION SERVER
RODC.skillsnet.dk

Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
DNS Server
Confirmation
Results

Select one or more roles to install on the selected server.

Roles	Description
☐ Active Directory Certificate Services	Active Directory Domain Services (AD DS) stores information about objects on the network and makes this information available to users and network administrators. AD DS uses domain controllers to give network users access to permitted resources anywhere on the network through a single logon process.
☒ Active Directory Domain Services	
☐ Active Directory Federation Services	
☐ Active Directory Lightweight Directory Services	
☐ Active Directory Rights Management Services	
☐ Device Health Attestation	
☐ DHCP Server	
☒ DNS Server	
☒ File and Storage Services (1 of 12 installed)	
☐ Host Guardian Service	
☐ Hyper-V	
☐ Print and Document Services	
☐ Remote Access	
☐ Remote Desktop Services	
☐ Volume Activation Services	
☐ Web Server (IIS)	
☐ Windows Deployment Services	
☐ Windows Server Update Services	

< Previous Next > Install Cancel

Add Roles and Features Wizard

Installation progress

DESTINATION SERVER
RODC.skillsnet.dk

- Before You Begin
- Installation Type
- Server Selection
- Server Roles
- Features
- AD DS
- DNS Server
- Confirmation
- Results**

View installation progress

Feature installation

Configuration required. Installation succeeded on RODC.skillsnet.dk.

Active Directory Domain Services
Additional steps are required to make this machine a domain controller.
[Promote this server to a domain controller](#)

DNS Server

Group Policy Management

Remote Server Administration Tools
Role Administration Tools
AD DS and AD LDS Tools
Active Directory module for Windows PowerShell

 You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.

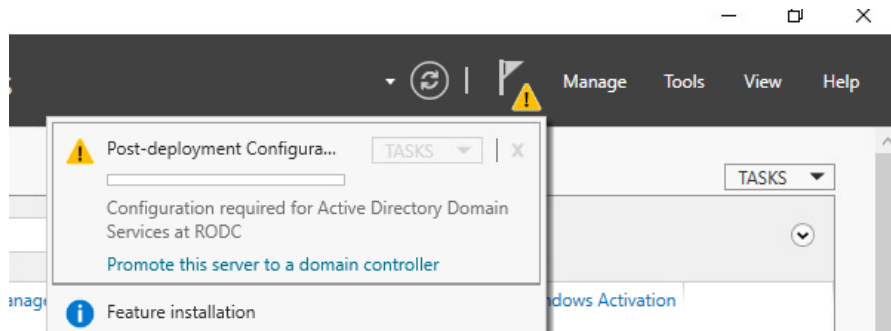
[Export configuration settings](#)

< Previous

Next >

Close

Cancel



Deployment Configuration

TARGET SERVER
RODC.skillsnet.dk

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

- ☒ Add a domain controller to an existing domain
☐ Add a new domain to an existing forest
☐ Add a new forest

Specify the domain information for this operation

Domain:

Supply the credentials to perform this operation

<No credentials provided>

[More about deployment configurations](#)

Activate Windows

Go to Settings to activate Windows.

< Previous

Next >

Install

Cancel

Windows Security

Credentials for deployment operation

Supply credentials for the deployment operation

Domain: SKILLSNET

OK

Cancel

Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
RODC.skillsnet.dk

Deployment Configuration

- Domain Controller Options
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Select the deployment operation

☒ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☐ Add a new forest

Specify the domain information for this operation

Domain:

Supply the credentials to perform this operation

[More about deployment configurations](#)

Activate Windows
Go to Settings to activate Windows.

< Previous Next >

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
RODC.skillsnet.dk

Deployment Configuration

Domain Controller Options

RODC Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Specify domain controller capabilities and site information

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☒ Read only domain controller (RODC)

Site name: Aalborg

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

Activate Windows
Go to Settings to activate Windows.

[Install](#) [Cancel](#)

[< Previous](#) [Next >](#)

RODC Options

TARGET SERVER
RODC.skillsnet.dk

- Deployment Configuration
- Domain Controller Options
- RODC Options**
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Delegated administrator account

<Not provided>

Select...

Accounts that are allowed to replicate passwords to the RODC

SKILLSNET\Allowed RODC Password Replication Group

Add...

Remove

Accounts that are denied from replicating passwords to the RODC

BUILTIN\Administrators
BUILTIN\Server Operators
BUILTIN\Backup Operators

Add...

Remove

If the same account is both allowed and denied, denied takes precedence.

[More about RODC options](#)

Activate Windows

Go to Settings to activate Windows.

< Previous

Next >

Install

Cancel

Additional Options

TARGET SERVER
RODC.skillsnet.dk

Deployment Configuration
Domain Controller Options
RODC Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Specify Install From Media (IFM) Options

☐ Install from media

Specify additional replication options

Replicate from: Any domain controller

[More about additional options](#)

Activate Windows
Go to Settings to activate
Install Cancel

< Previous

Next >

Active Directory Domain Services Configuration Wizard

Paths

TARGET SERVER
RODC.skillsnet.dk

Deployment Configuration
Domain Controller Options
 RODC Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Specify the location of the AD DS database, log files, and SYSVOL

Database folder:

Log files folder:

SYSVOL folder:

[More about Active Directory paths](#)

Activate Windows
Go to Settings to activate Windows.

< Previous Next > Install Cancel

Review Options

TARGET SERVER
RODC.skillsnet.dk

- Deployment Configuration
- Domain Controller Options
 - RODC Options
- Additional Options
- Paths
- Review Options**
- Prerequisites Check
- Installation
- Results

Review your selections:

Configure this server as an additional Active Directory domain controller for the domain "skillsnet.dk".

Site Name: Aalborg

Additional Options:

Read-only domain controller: Yes

Global catalog: Yes

DNS Server: Yes

Update DNS Delegation: No

Source domain controller: any writable domain controller

These settings can be exported to a Windows PowerShell script to automate additional installations

[View script](#)[More about installation options](#)

Activate Windows

Go to Settings to activate Windows.

< Previous

Next >

Install

Cancel

Active Directory Domain Services Configuration Wizard

Prerequisites Check

TARGET SERVER
RODC.skillsnet.dk

✓ All prerequisite checks passed successfully. Click 'Install' to begin installation. [Show more](#)

Deployment Configuration
Domain Controller Options
RODC Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Prerequisites need to be validated before Active Directory Domain Services is installed on this computer

[Rerun prerequisites check](#)

View results

⚠ Windows Server 2022 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.

For more information about this setting, see Knowledge Base article 942564 (<http://go.microsoft.com/fwlink/?LinkId=104751>).

ℹ Prerequisites Check Completed

✓ All prerequisite checks passed successfully. Click 'Install' to begin installation.

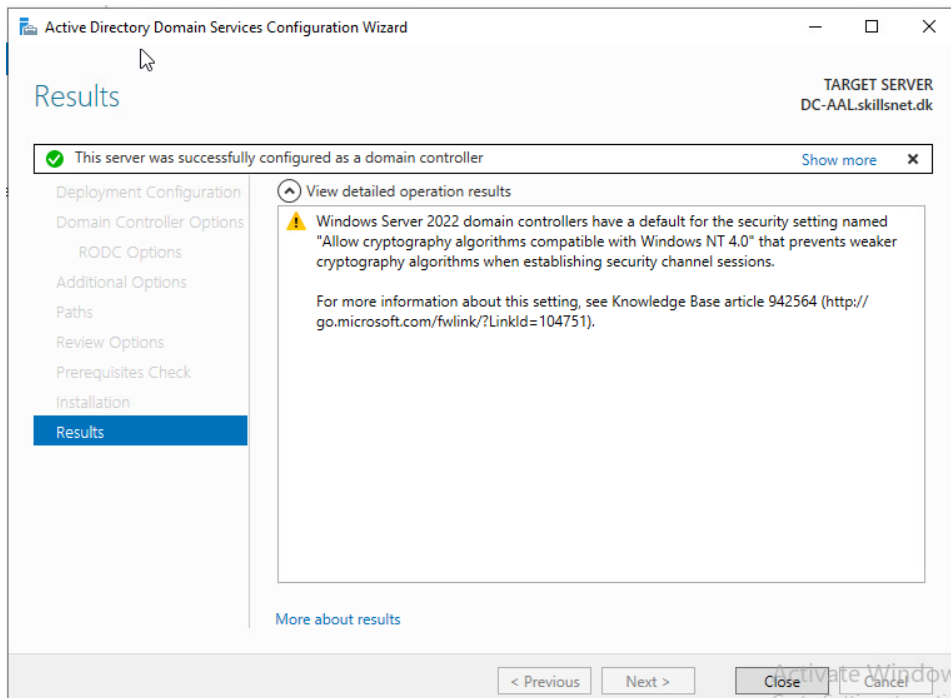
⚠ If you click Install, the server automatically reboots at the end of the promotion operation.

[More about prerequisites](#)

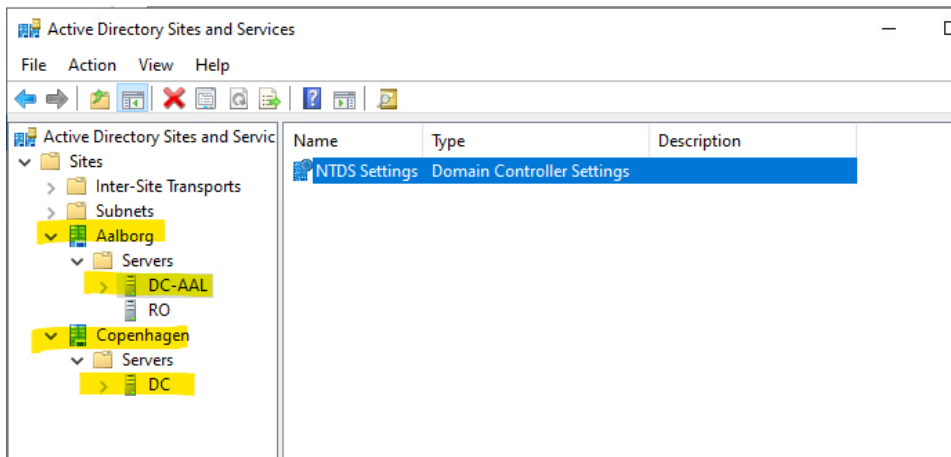
Activate Windows
Go to Settings to activate Windows

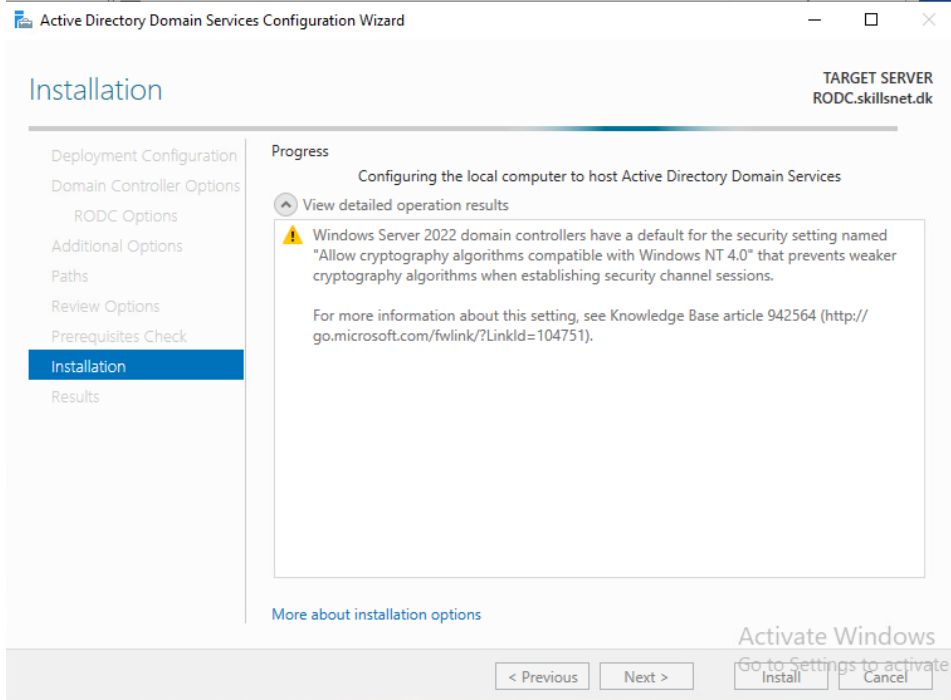
< Previous Next > Install Cancel

Problems with RODC, many errors. Reinstalled as DC-AAL



Active Directory Sites and Services on DC:





SRV2

Role: Hosts public and internal web services, provides certificate-based authentication and SAML integration, and serves as a centralized iSCSI backup target and WEF log collector.

Tasks:

SRV2 - Task1: public web server

1. Host the **public-facing** website at www.skillsnet.dk

SRV2 - Task2: intranet

2. Configure **certificate-based authentication** for the internal site intra.skillsnet.dk

SRV2 - Task3: SAML based auth app

3. Host **SAML-based authentication** application site under app.skillsnet.dk (expected result at Appendix G)
 - 3.1. Web App code is available at C:\Resources\ClaimsApp.zip
 - 3.2. Application is preconfigured with ADFS authentication against sso.skillsnet.dk
 - 3.3. ADFS relying party identifier and endpoint is <https://app.skillsnet.dk>
 - 3.4. Site webroot must be C:\inetpub\approot\
 - 3.5. ADFS needs to provide following claims:
 - 3.5.1. Attribute: **User-Principal-Name**, Outgoing Claim: **Name ID**
 - 3.5.2. Attribute: **Surname**, Outgoing Claim: **Surname**
 - 3.5.3. Attribute: **Display-Name**, Outgoing Claim: **Name**

SRV2 - Task4: web server certificates

4. Ensure all hosted web services use **valid certificates** issued by Skillsnet CA

SRV2 - Task5: iSCSI Target

5. Deploy and configure an iSCSI Target to be used for centralized backups:
 - 5.1. Add new disk of 12 GB for storing the backups
 - 5.2. Initialize and format the disk using the **ReFS file system**, then mount it as the B: drive
 - 5.3. Create a **10 GB iSCSI virtual disk** at the following path **B:\iSCSIVirtualDisks\Skillsnet-Backup.vhdx**
 - 5.4. Configure the virtual disk as an iSCSI Target and allow access from authorized initiator DC server

SRV2 - Task6: Windows Event Forwarding

6. Configure Windows Event Forwarding
 - 6.1. Use **Source-Initiated** subscription mode

- 6.2. Add exceptions to existing hardening guidelines only for SRV2 to make WEF work
- 6.3. Ensure all domain-joined computers are forwarding **Security** event logs to SRV2

SRV1 & SRV2

Role: Provide shared file services, enforce secure data storage policies, support DFS-based replication, and deliver high-availability DHCP services for internal networks.

Tasks:

SRV1 & SRV2 - Task1: SMB encryption

1. Enable **SMB encryption** on all file shares to secure data in transit

SRV1 & SRV2 - Task2: DFS shares

2. Use separate 10 GB disk on each server:
 - 2.1. Initialize and format the disk using NTFS, and mount it as **D:**
 - 2.2. This disk will serve as the data volume for all DFS shares

SRV1 & SRV2 - Task3: Bitlocker encryption

3. Configure BitLocker encryption on the D:\ volume using PowerShell, ensuring it uses TPM-based protection

SRV1 & SRV2 - Task4: Prevent writing

4. Prevent writing the following file types to shared folders: .exe, .com, .vbs, .msi

SRV1 & SRV2 - Task5: Quota for roaming profiles

5. Configure user profile folders such that each user is assigned **512 MB** quota for their roaming profile directory

SRV1 & SRV2 - Task6: DFS replication

6. Set up DFS replication for <\\skillsnet.dk\DFS\Users>

SRV1 & SRV2 - Task7: HA DHCP

7. Deploy a High-Availability DHCP failover pair that assigns IP addresses within the skillsdev.dk network range

SRV1 & SRV2 - Task8: DHCP reservation

8. Configure DEV-PC to obtain its IP address via DHCP, matching its currently assigned static configuration

DEV-PC

Role: Serves as the automation control node for the DEV environment; used to configure and deploy the skillsdev.dk domain and associated services via Ansible on DEV-SRV.

Tasks:

Ansible - Task1: hostname

1. **1-hostname.yaml:** Configure the **hostname** to **SRV**

Ansible - Task2: ADDS

2. **2-adds.yaml:** Set up an **AD DS environment** with the domain skillsdev.dk

Ansible - Task3: Users

3. **3-users.yaml:** Import the predefined OU structure (Appendix C), users and groups into the domain
 - 3.1. Source file `/ansible/resources/ES2025_TP39_ModuleB_Users_Skillsdev.json`
 - 3.2. Ensure all users are activated and have UPN defined
 - 3.3. Based on Department value, create unique groups
 - 3.4. Assign users to role groups and OUs based on their Department value
 - 3.5. Usernames must follow the format: **Firstname.Lastname**

Ansible - Task4: Web server

4. **4-web.yaml:** Deploy an IIS web server with static HTML homepage accessible at www.skillsdev.dk.
 - 4.1. Homepage must have large message "Skills Development"

Ansible - Task5: Shares

5. **5-shares.yaml**: Configure File Server based on /ansible/resources/ES2025_TP39_ModuleB_Shares.yaml

Ansible configuration notes:

1. Ansible is **preconfigured** on DEV-PC with an inventory file located at /etc/ansible/hosts
2. Connections to DEV-SRV are available via SSH using a preinstalled key
3. All required playbooks must be created and saved in the /ansible directory

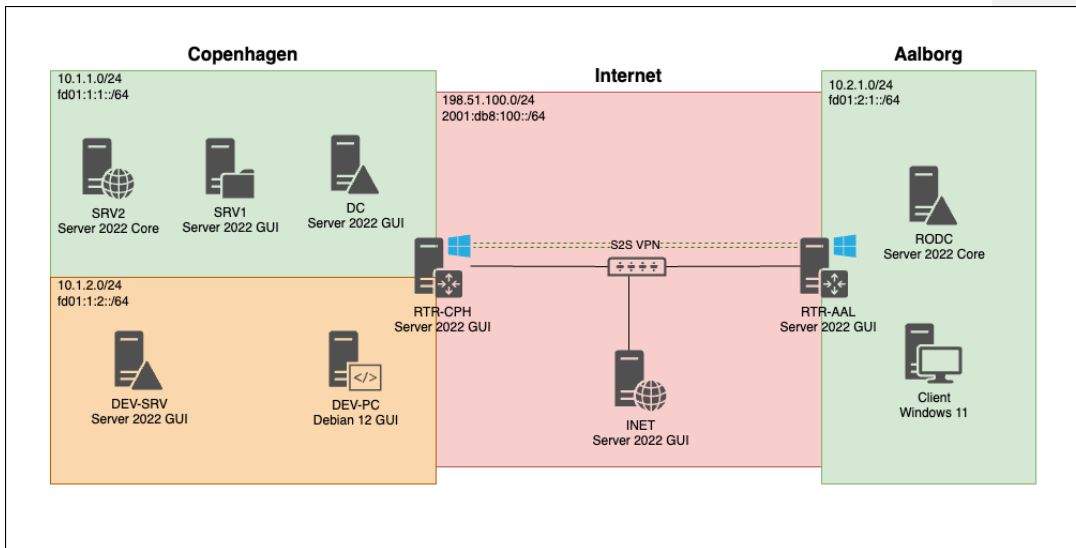
Assessment notes:

1. DEV-SRV will be restored to “Start” snapshot
2. Playbooks will be executed in sequence from the **/ansible** directory

Appendix A: Configuration table

Hostname	Domain	IPv4 Address (/24)	IPv6 Address (/64)
INET	WORKGROUP	198.51.100.1	2001:db8:100::1
RTR-CPH	WORKGROUP	INET: 198.51.100.11 INT: 10.1.1.254 DEV: 10.1.2.254	INET: 2001:db8:100::11 INT: fd01:1:1::254 DEV: fd01:1:2::254
RTR-AAL	WORKGROUP	INET: 198.51.100.12 INT: 10.2.1.254	INET: 2001:db8:100::12 INT: fd01:2:1::254
DC	SKILLSNET.DK	10.1.1.1	fd01:1:1::1
RODC	SKILLSNET.DK	10.2.1.1	fd01:2:1::1
SRV1	SKILLSNET.DK	10.1.1.2	fd01:1:1::2
SRV2	SKILLSNET.DK	10.1.1.3	fd01:1:1::3
DEV-PC	NOT APPLICABLE	10.1.2.10	fd01:1:2::10
DEV-SRV	SKILLSDEV.DK	10.1.2.1	fd01:1:2::1
CLIENT	SKILLSNET.DK	10.2.1.10	fd01:2:1::10

Appendix B: Network topology



Appendix C: OU Structure

OU Path	Description
Skills\Users\Employees	All regular non-technical users
Skills\Users\Tech	IT personnel and admins
Skills\Users\Sales	Sales department staff
Skills\Users\Finance	Finance team users
Skills\Users\Development	Software/dev team members
Skills\Users\Contractors	External/temporary workers
Skills\Groups	All security and distribution groups
Skills\Desktops	Computer objects for client PCs
Skills\Servers	Computer objects for server

Appendix D: Password Policies

Policy Name	Target	Min Length	Max Password Age	Complexity	Lockout Threshold
FGPP-Users	Regular users	12 chars	730 days	Enabled	5 attempts, 5 minutes

Policy Name	Target	Min Length	Max Password Age	Complexity	Lockout Threshold
FGPP-Tech	Tech users	20 chars	365 days	Enabled	3 attempts, 15 minutes
FGPP-Contractors	Contractors	16 chars	90 days	Enabled	3 attempts, 30 minutes

Appendix E: INET DNS records

Record Type	Name	Value
A	www.skillspublic.dk	198.51.100.11
AAAA	www.skillspublic.dk	2001:db8:100::11
A	mail.nordicbackup.net	198.51.100.1
AAAA	mail.nordicbackup.net	2001:db8:100::1
MX	nordicbackup.net	mail.nordicbackup.net

Appendix F: DNS Authoritative Servers

DNS Zone	Authoritative Server(s)
skillspublic.dk	INET
nordicbackup.net	INET
skillsnet.dk	DC, RODC
skillsdev.dk	DEV-SRV

Appendix G: ADFS successful integration

Home Page - My ASP.NET Application x +

← → 🔍 ☆ 📌 InPrivate ...

Application name Home Hello, John Doe! Sign out

Claims Display App

Claim Type: http://schemas.xmlsoap.org/ws/2005/05/identity/claims/nameidentifier
Claim Value: john.doe@skillsnet.dk

Claim Type: http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname
Claim Value: Doe

Claim Type: http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name
Claim Value: John Doe

Claim Type: http://schemas.microsoft.com/ws/2008/06/identity/claims/authenticationmethod
Claim Value: urn:oasis:names:tc:SAML:2.0:ac:classes:PasswordProtectedTransport

Claim Type: http://schemas.microsoft.com/ws/2008/06/identity/claims/authenticationinstant
Claim Value: 2025-06-05T13:10:30.488Z
